

VERTEX ALGEBRAS

github.com/danimalabares/vertex-algebras

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1. CARTAN SUBALGEBRA, CARTAN MATRIX AND SERRE RELATIONS

Kac-Moody algebras are Lie algebras, whose definition is motivated by the structure of finite-dimensional simple Lie algebras over $\mathbb{C}$.

Let $\mathfrak{g}$ be a finite-dimensional semisimple Lie algebra over $\mathbb{C}$. Then $\mathfrak{g}$ has a *Cartan subalgebra* $\mathfrak{h} \subset \mathfrak{g}$ (abelian + ...). Fixing $\mathfrak{h} \subset \mathfrak{g}$ gives a *root space decomposition*

$$\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Delta} \mathfrak{g}_{\alpha}$$

where $\Delta \subset \mathfrak{h}^*$ linear dual, and, by definition

$$\mathfrak{g}_{\alpha} = \{X \in \mathfrak{g} \mid [H, X] = \alpha(H)X \ \forall H \in \mathfrak{h}\}$$

Turns out the $\mathfrak{g}_\alpha$ are all 1-dimensional, though this property is lost when we go to Kac-Moody algebras.

$$[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subset \mathfrak{g}_{\alpha+\beta}$$

The Killing form $\kappa : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{C}$, $\kappa(x, y) = \text{Tr}_{\mathfrak{g}} \text{ad}(x)\text{ad}(y)$ is nondegenerate. “This is kind of the definition of semisimple.” (Think of $\mathfrak{h}$ as $\mathfrak{g}_0$, btw.)

$\kappa|_{\mathfrak{g}_\alpha \times \mathfrak{g}_\beta} \neq 0$ only when $\beta = -\alpha$. $\kappa|_{\mathfrak{h} \times \mathfrak{h}}$ is non-degenerate. This gives a linear isomorphism $\mathfrak{h} \xrightarrow{\nu} \mathfrak{h}$ via $\nu(H)(H') = \kappa(H, H')$.

So, $\mathfrak{h}^*$ comes with a non-degenerate bilinear form.

The *reflection* $r_\alpha : \mathfrak{h} \rightarrow \mathfrak{h}^*$ in $\alpha \in \mathfrak{h}^*$ (usually a root) is $r_\alpha(\lambda) = \lambda - 2 \frac{(\lambda, \alpha)}{(\alpha, \alpha)} \cdot \alpha$.

“Classify root systems [...] classify semisimple Lie algebras” It is a fact that $r_\alpha(\Delta) = \Delta$ for all $\alpha \in \Delta$, which motivates the definition of *root system* and permits classification.

Example 1.1. $\mathfrak{g} = \mathfrak{sl}_2$, $\mathfrak{h} =$ diagonal matrices

$$\begin{pmatrix} 1 & & \\ & -1 & \\ & & 0 \end{pmatrix} \quad \begin{pmatrix} 0 & & \\ & 1 & \\ & & -1 \end{pmatrix}$$

is a basis of $\mathfrak{h}$. There are 6 roots vectors

$$E_{12} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

E_{23}, E_{13} , etc.

Exercise 1.2. $[H_1, E_{12}] = 2E_{12}$, $[H_2, E_{12}] = -E_{12}$, $\alpha_{12} = (2, -1)$.

[Drawing of roots]

Notions of *positive roots* and *simple roots* (set of rank $\mathfrak{g}$ simple roots has ℓ elements, where $\ell = \dim(\mathfrak{h}^*)$. This will also fail for Kac-Moody algebras more generally). Next write the *Cartan matrix*

$$A = (a_{ij}), \quad a_{ij} = 2 \frac{(\alpha_i, \alpha_j)}{(\alpha_i, \alpha_i)}$$

for $1 \leq i, j \leq \ell$.

Example 1.3. $\mathfrak{sl}_3$. [Picture, hexagonal pattern]. $(\alpha_1, \alpha_1) = (\alpha_2, \alpha_2) = 2$, $(\alpha_1, \alpha_2) = -1$, so

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

Example 1.4. $\mathfrak{sl}_5$. [Picture, square pattern]. $|\alpha_2| = 1$, $|\alpha_1| = 2$, $(\alpha_1, \alpha_2) = -2$, so

$$A = \begin{pmatrix} 2 & -1 \\ -2 & 2 \end{pmatrix}$$

Remark 1.5. Since $\mathfrak{g}_\alpha$ is 1-dimensional, set $\mathfrak{g}_\alpha = \mathbb{C}E_\alpha$ and $E_i = E_{\alpha_i}$, $i = 1, 2, \dots, \ell$ (simple root vectors). It turns out that

$$\text{ad}(E_i)^{1-a_{ij}} E_j = 0.$$

This is called a *Serre relation*.

2. SOME INFINITE DIMENSIONAL LIE ALGEBRAS

Let $\mathfrak{g}$ be a finite-dimensional semisimple Lie algebra, and define the *loop algebra*

$$\begin{aligned} L\mathfrak{g} &= \mathfrak{g}[t, t^{-1}], \text{ (with basis } at^m | a \in \text{a basis of } \mathfrak{g} \text{ } \\ &\quad m \in \mathbb{Z} \text{)} \\ &= \mathfrak{g} \otimes_{\mathbb{C}} \mathbb{C}[t, t^{-1}] \end{aligned}$$

with the Lie bracket

$$[at^m, bt^n] = [a, b]t^{m+n}.$$

“This construction is absurdly general — we don’t need $\mathfrak{g}$ to be semisimple [...]”

Take $\mathfrak{g} = \mathfrak{sl}_2$. Recall that

$$E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$$

[Picture with $F, H, E, Ft, Ht, Et, Et^2 \dots$] E was a root vector, corresponding to the unique root in $\mathfrak{sl}_2$, call it α_1 . We seem to have a second simple root α_0 , corresponding to Ft .

This looks like it wants to have a Cartan matrix

$$A = \begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}$$

We will indeed recover (a variant of) $L\mathfrak{g}$ as a Lie algebra “built from” $A = \begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}$, a Kac-Moody algebra. But note first, $\mathfrak{h} = \mathbb{C}H$ is too small. “Problem with α_0 and α_1 being linearly independent ...”

Exercise 2.1. Consider $L\mathfrak{g} \oplus \mathbb{C}d$, and set $[d, at^m] = mat^m$, $[d, d] = 0$. Check this defines a Lie algebra.

Proof. Skew-commutativity, i.e. for all $x \in L\mathfrak{g} \oplus \mathbb{C}d$,

$$(2.1.1) \quad [x, x] = 0,$$

is immediate from skew commutativity in $L\mathfrak{g}$ and the hypothesis that $[d, d] = 0$.

To confirm Jacobi identity, i.e. that for all $x, y, z \in L\mathfrak{g} \oplus \mathbb{C}d$

$$(2.1.2) \quad [x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0,$$

notice that since this is a cyclic sum on x, y, z we only need to consider three elements in $L\mathfrak{g} \oplus \mathbb{C}d$ up to cyclic permutation. The cases in which the three elements are either in $L\mathfrak{g}$ or in $\mathbb{C}d$ are obvious, so that there are only two interesting possibilities:

$$(2.1.3) \quad x = d, \quad y = at^m, \quad z = bt^n$$

$$(2.1.4) \quad x = d, \quad y = d, \quad z = at^n$$

Case 2.1.3 gives

$$\begin{aligned} &[d, [at^m, bt^n]] + [at^m, [bt^n, d]] + [bt^n, [d, at^m]] \\ &= [d, [a, b]t^{m+n}] + [at^m, -nbt^n] + [bt^n, mat^m] \\ &= (m+n)[a, b]t^{m+n} - n[a, b]t^{m+n} + m[b, a]t^{m+n} \\ &= (m+n)[a, b]t^{m+n} - n[a, b]t^{m+n} - m[a, b]t^{m+n} \\ &= (m+n)[a, b]t^{m+n} - (m+n)[a, b]t^{m+n} = 0. \end{aligned}$$

Case 2.1.4 gives

$$\begin{aligned} & [d, [d, at^m]] + [d, [at^m, d]] + [at^m, [d, d]] \\ &= [d, mat^m] + [d, -mat^m] = 0. \end{aligned}$$

□

The Kac-Moody algebra turns out to be, not quite this, but slightly larger still.

Recall that an *invariant bilinear form* $(\cdot, \cdot)$ on a Lie algebra $\mathfrak{g}$ is a bilinear form such that

$$(2.1.5) \quad ([a, b], c) = (a, [b, c]) \quad \forall a, b, c \in \mathfrak{g}.$$

Exercise 2.2. Prove that an invariant bilinear form on a simple Lie algebra must in fact be symmetric.

Proof. It's enough to show that $\mathfrak{g}$ is *perfect*, i.e. that $[\mathfrak{g}, \mathfrak{g}] = \mathfrak{g}$. In this case, let $a, b \in \mathfrak{g}$ and suppose that $b = [x, y]$. Then

$$\begin{aligned} (a, b) &= (a, [x, y]) = (a, -[y, x]) = (-[a, y], x) = ([y, a], x) \\ &= (y, [a, x]) = (y, -[x, a]) = (-[y, x], a) = ([x, y], a) = (b, a) \end{aligned}$$

To confirm that $\mathfrak{g}$ is perfect just observe that $[\mathfrak{g}, \mathfrak{g}]$ is a nontrivial ideal of $\mathfrak{g}$. □

Definition 2.3. Given $\mathfrak{g}$ simple, with $(\cdot, \cdot) : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{C}$ invariant bilinear form, the *affine Lie algebra* is

$$\hat{\mathfrak{g}} = L\mathfrak{g} \oplus \mathbb{C}K,$$

with $[K, \hat{\mathfrak{g}}] = 0$, and $[at^m, bt^n] = [a, b]t^{m+n} + m(a, b)\delta_{m, -n}K$.

“For the construction to work it doesn't actually have to be nondegenerate.”

Exercise 2.4. Check that the affine Lie algebra $\hat{\mathfrak{g}}$ is a Lie algebra.

Proof. (Skew-commutativity.) Since $[K, \hat{\mathfrak{g}}] = 0$ and $K \in \hat{\mathfrak{g}}$, it is immediate that $[K, K] = 0$. For the case of an element in $L\mathfrak{g}$, we see that $[at^m, at^m] = 0$ by skew-commutativity of the bracket in $\mathfrak{g}$ and the Kronecker delta.

(Jacobi identity.) As in Exercise 2.1, any choice of x, y, z involving K is immediate by $[K, \hat{\mathfrak{g}}] = 0$. Thus the only interesting case is for Jacobi identity consider the cases

$$\begin{aligned} & [at^m, [bt^n, ct^\ell]] + [bt^n, [ct^\ell, at^m]] + [ct^\ell, [at^m, bt^n]] \\ &= [at^m, [b, c]t^{n+\ell} + n(b, c)\delta_{n, -\ell}K] \\ &+ [bt^n, [c, a]t^{\ell+m} + \ell(c, a)\delta_{\ell, -m}K] \\ &+ [ct^\ell, [a, b]t^{m+n} + m(a, b)\delta_{m, -n}K] \\ &= [at^m, [b, c]t^{n+\ell}] + [at^m, n(b, c)\delta_{n, -\ell}K] \\ &+ [bt^n, [c, a]t^{\ell+m}] + [bt^n, \ell(c, a)\delta_{\ell, -m}K] \\ &+ [ct^\ell, [a, b]t^{m+n}] + [ct^\ell, m(a, b)\delta_{m, -n}K] \\ &= [a, [b, c]]t^{m+(n+\ell)} + m(a, [b, c])\delta_{m, -(n+\ell)}K \\ &+ [b, [c, a]]t^{n+(\ell+m)} + n(b, [c, a])\delta_{n, -(\ell+m)}K \\ &+ [c, [a, b]]t^{\ell+(m+n)} + \ell(c, [a, b])\delta_{\ell, -(m+n)}K = 0 \end{aligned}$$

It is clear that we obtain a Jacobi equation on $\mathfrak{g}$. To see that the remaining terms vanish, notice that the condition on the Kronecker delta in its three appearances is the same, namely, $m + n + \ell = 0$. In this case, we only need to check that $(a, [b, c]) = (b, [c, a]) = (c, [a, b])$ to conclude. This follows from the invariance of $(\cdot, \cdot)$ and the fact that $\mathfrak{g}$ simple using Exercise 2.2. $\square$

We also have

Definition 2.5. The *extended affine Lie algebra* is

$$\tilde{\mathfrak{g}} = L\mathfrak{g} \oplus \mathbb{C}K \oplus \mathbb{C}d,$$

with $[d, at^m] = mat^m$ as before, and $[K, d] = 0$.

The extended affine Lie algebra is an example of a Kac-Moody algebra.

Exercise 2.6 (For those who like geometry). Let $R = \mathbb{C}[t, t^{-1}]$. If $D \in \text{Der}(R)$, then $L\mathfrak{g} \oplus \mathbb{C}d$ is a Lie algebra with $[d, a \otimes r] = a \otimes D(r)$. Is $(\mathfrak{g} \otimes R) \oplus \text{Der}(R)$ a Lie algebra? (The Lie algebra $L\mathfrak{g} \oplus \mathbb{C}d$ from Exercise 2.1 is a particular case, for $D = t \frac{d}{dt}$.)

Proof. Checking that $L\mathfrak{g} \oplus \mathbb{C}d$ is a Lie algebra with $[d, a \otimes r] = a \otimes D(r)$ is similar to Exercise 2.1: skew-commutativity is immediate from skew-commutativity in each of the components, while Jacobi identity is verified in two cases. For $x = y = d$ and $z = a \otimes r$ we quickly obtain

$$\begin{aligned} & [x, [y, z]] + [y, [z, x]] + [z, [x, y]] \\ &= [d, [d, a \otimes r]] + [d, [a \otimes r, d]] + [a \otimes r, [d, d]] \\ &= [d, a \otimes D(r)] + [d, -a \otimes D(r)] = 0. \end{aligned}$$

And for $x = d$, $y = a \otimes r$ and $z = b \otimes s$, we get

$$\begin{aligned} & [x, [y, z]] + [y, [z, x]] + [z, [x, y]] \\ (2.6.1) \quad &= [d, [a \otimes r, b \otimes s]] + [a \otimes r, [b \otimes s, d]] + [b \otimes s, [d, a \otimes r]] \\ &= [d, [a, b] \otimes rs] + [a \otimes r, -b \otimes D(s)] + [b \otimes s, a \otimes D(r)] \\ &= [a, b] \otimes D(rs) - [a, b] \otimes rD(s) + [b, a] \otimes sD(r) = 0. \end{aligned}$$

To check whether $(\mathfrak{g} \otimes R) \oplus \text{Der}(R)$ is a Lie algebra first put the Lie bracket on $\text{Der}(R)$ as $[D, D_1] = DD_1 - D_1D$. It is clear that this bracket is skew-commutative. Jacobi identity reads

$$\begin{aligned} & [D, [D_1, D_2]] + [D_1, [D_2, D]] + [D_2, [D, D_1]] \\ &= [D, D_1D_2 - D_2D_1] + [D_1, D_2D - DD_2] + [D_2, DD_1 - D_1D] \\ &= D(D_1D_2 - D_2D_1) - (D_1D_2 - D_2D_1)D + D_1(D_2D - DD_2) \\ &\quad - (D_2D - DD_2)D_1 + D_2(DD_1 - D_1D) - (DD_1 - D_1D)D_2 \\ &= DD_1D_2 - DD_2D_1 - D_1D_2D + D_2D_1D + D_1D_2D - D_1DD_2 \\ &\quad - D_2DD_1 + DD_2D_1 + D_2DD_1 - D_2D_1D - DD_1D_2 + D_1DD_2 = 0. \end{aligned}$$

Now put the bracket on $(\mathfrak{g} \otimes R) \oplus \text{Der}(R)$ as $[D, a \otimes r] = a \otimes D(r)$. Skew-commutativity is immediate. Jacobi identity for $x = D$, $y = a \otimes r$ and $z = b \otimes s$ is

identical to the computation 2.6.1. In the case $x = D$, $y = D_1$ and $z = a \otimes r$, we get

$$\begin{aligned} & [D, [D_1, a \otimes r]] + [D_1, [a \otimes r, D]] + [a \otimes r, [D, D_1]] \\ &= [D, a \otimes D_1(r)] + [D_1, -a \otimes D(r)] + [a \otimes r, [D, D_1]] \\ &= a \otimes DD_1(r) - a \otimes D_1D(r) - a \otimes [D, D_1](r) = 0 \end{aligned}$$

□

3. KAC-MOODY ALGEBRAS

Recall the notion of the free Lie algebra on a vector space V of generators (or a set X , think of V as a vector space with basis X):

Definition 3.1. The *free Lie algebra* on V is characterized by the universal property

$$\begin{array}{ccc} V & \xrightarrow{f} & \mathfrak{g} \\ & \searrow i & \nearrow \exists! \tilde{f} \\ & F(V) & \end{array}$$

That is, for any linear map $f : V \rightarrow \mathfrak{g}$ with $\mathfrak{g}$ Lie algebra, there exists a unique $\tilde{f}$ homomorphism of Lie algebras $F(V) \rightarrow \mathfrak{g}$ such that $\tilde{f} \circ i = f$.

$$\text{Hom}_{\text{Lie}}(F(V), \mathfrak{g}) = \text{Hom}_{\text{Vec}}(V, \mathfrak{g})$$

naturally.

That is, F and the forgetful functor $G : \underline{\text{Lie}} \rightarrow \underline{\text{Vec}}$ are adjoint:

$$\text{Hom}_{\underline{\text{Lie}}}(F(V), \mathfrak{g}) \xrightarrow{\sim} \text{Hom}_{\underline{\text{Vec}}}(V, G(\mathfrak{g}))$$

A realisation of $F(V)$. Let

$$T(V) = \mathbb{C} \oplus V \oplus V^{\otimes 2} \oplus V^{\otimes 3} \oplus \dots$$

be the tensor algebra of V .

Then inside $T(V)$ consider $F(V)$ the span of iterated commutators of elements of V .

Proposition 3.2. *This realises the free Lie algebra.*

Proof. In online notes. □

In the finite dimensional simple case, we had

$$a_{ij} = \frac{2(\alpha_i, \alpha_j)}{(\alpha_i, \alpha_i)},$$

which we think also as $\alpha_i, \alpha_j \in \mathfrak{h}^*$, and $\alpha_i^\vee = \frac{2}{(\alpha_i, \alpha_i)} \nu^{-1}(\alpha_i) \in \mathfrak{h}$.

Clearly, $\alpha_{ii} = 2$ for all i . a_{ij} might not equal a_{ji} , but certainly $a_{ij} = 0 \iff a_{ji} = 0$. And $\forall i \neq j$, $a_{ij} \leq 0$.

One further property. Set

$$\varepsilon_i = \frac{2}{(\alpha_i, \alpha_i)}, \quad \text{and} \quad D = \begin{array}{c} \text{diagonal matrix} \\ \text{with entries } \varepsilon_i \end{array}$$

Then $A = DB$, where $B = ((\alpha_i, \alpha_j))$ is symmetric. If a matrix A is equal to (diag)(symm), we call it *symmetrizable*.

Definition 3.3. A *generalized Cartan matrix* is an integer matrix $A = (a_{ij})$ which is

- symmetrizable,
- $a_{ii} = 2$ for all i ,
- $a_{ij} = 0 \iff a_{ji} = 0$,
- $a_{ij} \leq 0$ for $i \neq j$.

Definition 3.4. A *realisation* of a generalized Cartan matrix is a complex vector space $\mathfrak{h}$, and two sets

$$\begin{aligned}\Pi^\vee &= \{\alpha_1^\vee, \alpha_2^\vee, \dots, \alpha_n^\vee\}, \quad \text{and,} \\ \Pi &= \{\alpha_1, \alpha_2, \dots, \alpha_n\}\end{aligned}$$

such that $\langle \alpha_i^\vee, \alpha_j \rangle = a_{ij}$, $1 \leq i, j \leq n$.

Exercise 3.5. $\dim(\mathfrak{h}) \geq 2n - \text{rank}(A)$.

Proof. For $A = \begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}$, a realisation is given by

$$\Pi^\vee = \{H_1, H_0\}, \quad \Pi = \{\alpha_0, \alpha_1\}$$

$$\mathfrak{h} = \mathbb{C}H \oplus \mathbb{C}d \oplus \mathbb{C}K,$$

$$\mathfrak{h}^* = \mathbb{C}\alpha_1 \oplus \mathbb{C}\delta \oplus \mathbb{C}\Lambda_0$$

(Canonical dual, $\langle \alpha_1, H \rangle = 2$, $\langle \delta, d \rangle = 1 = \langle \Lambda_0, K \rangle$, every other pairing 0.)

Then

$$\begin{cases} \alpha_1 = \alpha_1 \\ \alpha_0 = \delta - \alpha_1 \end{cases} \quad \begin{cases} \alpha_1^\vee = H \\ \alpha_0^\vee = K - H \end{cases}$$

So we obtain

$$\begin{aligned}\langle \alpha_0^\vee, \alpha_1 \rangle &= \langle K - H, \alpha_1 \rangle = 2 \\ \langle \alpha_1^\vee, \alpha_0 \rangle &= \langle H, \delta - \alpha_1 \rangle = -2 \\ \langle \alpha_0^\vee, \alpha_0 \rangle &= \langle K - H, \delta - \alpha_1 \rangle = +2\end{aligned}$$

□

Finally let's define Kac-Moody algebras.

Let A be a generalized Cartan matrix. Let

$$\tilde{\mathfrak{n}}_+ = F(e_1, \dots, e_n),$$

the free Lie algebra on n generators, and similarly

$$\tilde{\mathfrak{n}}_- = F(f_1, \dots, f_n).$$

Let $\mathfrak{h}$ be a realisation of A . Set $\tilde{\mathfrak{g}}(A) = \tilde{\mathfrak{n}}_- \oplus \mathfrak{h} \oplus \tilde{\mathfrak{n}}_+$.

Make $\tilde{\mathfrak{g}}(A)$ a Lie algebra by defining

- $[\mathfrak{h}, \mathfrak{h}] = 0$,
- $\forall H \in \mathfrak{h}, [H, e_i] = \langle \alpha_i, H \rangle e_i = \alpha_i(H)e_i$. And similarly, $[H, f_i] = -\alpha_i(H)f_i$.
- $[e_i, f_j] = \delta_{ij}\alpha_i^\vee$.

Then $\tilde{\mathfrak{g}}(A)$ is a Lie algebra (though not yet the Kac-Moody algebra). See Kac, [Kac90, Theorem 1.2].

Remark 3.6. In $\mathfrak{h}$ we have a lattice

$$\begin{aligned} Q^\vee &= \mathbb{Z}\alpha_1^\vee + \dots + \mathbb{Z}\alpha_n^\vee, \quad \text{and} \\ Q &= \mathbb{Z}\alpha_1 + \dots + \mathbb{Z}\alpha_n \text{ in } \mathfrak{h}^* \end{aligned}$$

(root and coroot lattices). $\tilde{\mathfrak{g}}(A)$ is naturally Q -graded, with

$$\tilde{\mathfrak{g}}(A)_\beta = \text{span}\{\text{commutators of } e_i \text{ with } \sum \alpha_i = \beta\}.$$

$$\tilde{\mathfrak{g}}(A) = \mathfrak{h}.$$

Theorem 3.7 (Gabber-Kac). Denote by $I \subset \tilde{\mathfrak{g}}(A)$ the maximal Q -graded ideal, such that $I \cap \mathfrak{h} = \{0\}$. Then I is generated by the Serre relations

$$\text{ad}(e_i)^{1-a_{ij}} e_j \quad \text{and} \quad \text{ad}(f_i)^{1-a_{ij}} f_j, \quad i \neq j.$$

Proof. [Kac90, Theorem 9.11]. □

(The existence of the ideal I does not need the theorem; the importance of the theorem is providing an expression for the generators.)

Definition 3.8. The Kac-Moody algebra $\mathfrak{g}(A)$ is $\tilde{\mathfrak{g}}(A)/I$.

4. AFFINE KAC-MOODY ALGEBRAS

Let $\mathfrak{g}$ be a finite-dimensional simple Lie algebra, with $(\cdot, \cdot) : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{C}$ invariant bilinear form,

$$([x, y], z) = (z, [y, z]) \quad \forall x, y, z \in \mathfrak{g}$$

(Eg. the Killing form $\kappa(x, y) = \text{Tr}_{\mathfrak{g}} \text{ad}(x) \text{ad}(y)$ is invariant.)

Typically we normalise $(\cdot, \cdot)$ so that $(\alpha, \alpha) = 2$ for the long roots of $\mathfrak{g}$.

Then $\hat{\mathfrak{g}} = \mathfrak{g}[t, t^{-1}] \oplus \mathbb{C}K$ (affine Lie algebra),

$$[at^m, bt^n] = [a, b]t^{m+n} + m\delta_{m,-n}(a, b)K, \quad [K, \hat{\mathfrak{g}}] = 0$$

and $\tilde{\mathfrak{g}} = \hat{\mathfrak{g}} \oplus \mathbb{C}d$, $[d, K] = 0$, $[d, at^m] = mat^m$, (affine Kac-Moody algebra or “extended affine Lie algebra”)

Theorem 4.1. $\tilde{\mathfrak{g}}$ is a Kac-Moody algebra.

Let $\mathfrak{g} = \mathfrak{h} \oplus \bigoplus_{\alpha \in \Delta} \mathfrak{g}_\alpha$, $(\mathfrak{g}_\alpha = \mathbb{C}E_\alpha)$

The simple roots and coroots. $\tilde{\mathfrak{h}} = \mathfrak{h} \oplus \mathbb{C}K \oplus \mathbb{C}d$. We identify $\tilde{\mathfrak{h}}^*$ with $\mathfrak{h}^* \oplus \mathbb{C}\Lambda_0 \oplus \mathbb{C}\delta$ where

$$\begin{aligned} \Lambda_0(\mathfrak{h}) &= \delta(\mathfrak{h}) = 0 \\ \Lambda_0(d) &= \delta(K) = 0 \\ \Lambda_0(K) &= \delta(d) = 1 \end{aligned}$$

The *real coroots* are

$$\hat{\Delta}^{V, re} = \{E_\alpha t^m | \alpha \in \Delta, m \in \mathbb{Z}\}$$

and there are also imaginary roots and coroots

$$\hat{\Delta}^{V, im} = \{Ht^m | H \in \mathfrak{h}, m \in \mathbb{Z} \setminus \{0\}\}$$

Roots:

$$\begin{aligned}\hat{\Delta}^{re} &= \{\alpha + m\delta \mid \alpha \in \Delta, m \in \mathbb{Z}\} \\ \hat{\Delta}^{im} &= \{m\delta \mid m \neq 0\}\end{aligned}$$

Xt^m :

$$\begin{aligned}[H, Xt^m] &= [H, x]t^m, & H \in \mathfrak{h} \\ [K, xt^m] &= 0 \\ [d, xt^m] &= mxt^m\end{aligned}$$

so it $x \in \mathfrak{g}_\alpha$, $xt^m \in \tilde{\mathfrak{g}}_{\alpha+m\delta}$.

The invariant bilinear form $(\cdot, \cdot)$ from $\mathfrak{g} \times \mathfrak{g}$ extends uniquely to $(\cdot, \cdot) : \tilde{\mathfrak{g}} \times \tilde{\mathfrak{g}} \rightarrow \mathbb{C}$.

$$(d, d) = (K, K) = 0, (d, K) = 1 \text{ and } (d, \mathfrak{h}) = (K, \mathfrak{h}) = 0.$$

So, in $\tilde{\mathfrak{h}}^*$:

$$\begin{aligned}(\Lambda_0, \Lambda_0) &= (\delta, \delta) = 0 \\ (\Lambda_0, \mathfrak{h}^*) &= (\delta, \mathfrak{h}^*) = 0 \\ (\Lambda_0, \delta) &= 1.\end{aligned}$$

Hence, $|\alpha + m\delta|^2 = |\alpha|^2$, $|m\delta|^2 = 0$.

Example 4.2. $\widetilde{\mathfrak{sl}}_2, \tilde{\mathfrak{h}}^* = \text{span}\{\alpha, \Lambda_0, \delta\}$ with Gram matrix ...

We can make a choice of positive roots,

$$\hat{\Delta}_+ = \{\alpha + m\delta \mid \alpha \in \Delta, m > 0\} \cup \{m\delta \mid m > 0\} \cup \Delta_+$$

Obviously, if $\alpha \in \Delta_+$ is simple, $\alpha \in \hat{\Delta}_+$ is simple.

Notation. Let $\theta \in \Delta_+$ be a the highest root. ($\nexists \alpha \in \Delta_+$ such that $\alpha - \theta \in \mathbb{Z}_+ \Delta_+$.) and $\alpha = \delta - \theta$.

Then $\alpha_0 \in \hat{\Delta}_+$ is simple and the set of simple roots is $\hat{\Pi} = \{\alpha_0, \underbrace{\alpha_1, \dots, \alpha_\ell}_{\text{the finite simple roots}}\}$.

where $\ell = \text{rank}(\mathfrak{g})$.

The coroot corresponding to α_0 is

$$\alpha_0^\vee = K - \theta^\vee, \quad \theta^\vee = \frac{2}{(\theta, \theta)} \nu^{-1}(\theta) \in \mathfrak{h}$$

$$\text{and} \quad E_{\alpha_0} = E_{-\theta}t.$$

5. WEYL GROUP

Upshot. The Weyl group is a semidirect product of pseudoreflections and translations.

In any Kac-Moody algebra, we have

$$\begin{aligned}\text{roots} \quad \Pi &= \{\alpha_1, \dots, \alpha_\ell\} \subset \mathfrak{h}^* \\ \text{coroots} \quad \Pi^\vee &= \{\alpha_1^\vee, \dots, \alpha_\ell^\vee\} \subset \mathfrak{h},\end{aligned}$$

and *reflections* $r_i \in \text{GL}(\mathfrak{h}^*)$, defined by

$$r_i(\lambda) = \lambda - \langle \lambda, \alpha_i^\vee \rangle \alpha_i.$$

One can check that

$$(r_i \lambda, r_i \mu) = (\lambda, \mu) \quad \forall \lambda, \mu \in \mathfrak{h}^*$$

The Weyl group W is $\langle r_i | i = 1, \dots, \ell \rangle \subset \text{GL}(\mathfrak{h}^*)$.

Example 5.1. For $\widetilde{\mathfrak{sl}}_2$, r_1 is easy,

$$\begin{aligned} r_1(\alpha) &= -\alpha & (\text{as in } \mathfrak{sl}_2) \\ r_1(\delta) &= \delta, & r_1(\Lambda_0) = \Lambda_0. \end{aligned}$$

To compute r_0 take an arbitrary element $m\alpha_1 + k\Lambda_0 + f\delta$ and do:

$$\begin{aligned} r_0(m\alpha_1 + k\Lambda_0 + f\delta) &= m\alpha_1 + k\Lambda_0 + f\delta - \langle \alpha_0^\vee, m\alpha_1 + k\Lambda_0 + f\delta \rangle \alpha_0 \\ \alpha_0 &= \delta - \alpha_1, & \alpha_0^\vee &= K - \alpha^\vee \end{aligned}$$

so we obtain

$$\begin{aligned} &= m\alpha_1 + k\Lambda_0 + f\delta - (k - 2m)(\delta - \alpha_1) \\ &= (k - m)\alpha_1 + k\Lambda_0 + (f - k + 2m)\delta. \end{aligned}$$

Relative to basis $\{\alpha_1, \Lambda_0, \delta\}$.

$$\begin{aligned} r_1 &= \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, r_0 = \begin{pmatrix} -1 & 1 & 0 \\ 0 & 1 & 0 \\ 2 & -1 & 1 \end{pmatrix} \begin{pmatrix} m \\ k \\ f \end{pmatrix} \\ t = r_1 r_0 &= \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 2 & -1 & 1 \end{pmatrix} \end{aligned}$$

Notice that δ is fixed by all r_i . Also $m\alpha + k\Lambda_0 + f\delta$, the *coefficient* of Λ_0 is fixed by all r_i .

Then

$$t(m\alpha_1 + k\Lambda_0 + f\delta) = (m - k)\alpha_1 + k\Lambda_0 + (f - k + 2m)\delta.$$

Think of t as a translation.

The number k in

$$\mathfrak{h}^* \ni \hat{\lambda} = \lambda + k\Lambda_0 + f\delta$$

is called the *level* of $\hat{\lambda}$.

$\hat{\mathfrak{h}}$ = union of (hyper)planes of constant level which are stable under W . The roots α are all of level 0.

[Picture] “ r_1 changes the sign of the finite path”. And $t = r_1 r_0$ is a sort of translation. Indeed, in general we can consider $t_{\alpha_i} = r_{\alpha_i} \circ r_0 \in W$,

$$t_\alpha(\beta + m\delta) = \beta + (m + (\beta, \alpha_i))\delta$$

One can describe the action of t_α on $\hat{\lambda}$ in general (e.g. see [Kac90, Chapter 6])

Proposition 5.2. *For the affine Kac-Moody algebra $\hat{\mathfrak{g}}$ with $\hat{W} = \langle r_0, r_1, \dots, r_\ell \rangle$ its Weyl group (and $W = \langle r_1, \dots, r_\ell \rangle \subset \hat{W}$ the Weyl group of $\mathfrak{g}$), then $\hat{W} \simeq W \times t_{Q^\vee}$ (where it should be semidirect product instead of $\times \dots$) where $Q^\vee$ is the coroot lattice of $\mathfrak{g}$.*

Remark 5.3. For general Kac-Moody algebras, the Weyl groups are much larger, hyperbolic reflection groups.

In the affine case, $\hat{W}$ fixes level k , and $|\hat{\lambda}|$. One gets, in the intersection, paraboloids [Picture of section of hyperboloid that is a parabola].

6. WEYL CHARACTER FORMULA

Highest weight representations of Kac-Moody algebras. Let $\lambda \in \mathfrak{h}^*$, where $\mathfrak{g}(A) = \mathfrak{n}_- \oplus \mathfrak{h} \oplus \mathfrak{n}_+$ is a Kac-Moody algebra. We define a *Verma module*

$$M(\Lambda) = U(\mathfrak{g}) \otimes_{U(\mathfrak{h} + \mathfrak{n}_+)} \mathbb{C}v_\Lambda$$

where $\mathfrak{h} + \mathfrak{n}_+$ acts on V_Λ by:

$$\begin{aligned} Xv_\Lambda &= 0, & \forall x \in \mathfrak{n}_+, \\ Hv_\Lambda &= \Lambda(H)v_\Lambda, & \forall H \in \mathfrak{h} \end{aligned}$$

So $\mathbb{C}v_\Lambda$ is a $U(\mathfrak{h} + \mathfrak{n}_+)$ -module,

$$\begin{array}{c} U(\mathfrak{h} + \mathfrak{n}_+) \\ \downarrow \\ U(\mathfrak{g}) \end{array}$$

By the PBW theorem, $M(\Lambda)$ has a linear $\mathbb{C}$ -basis.

Let $\{F_{\alpha,i} : i = 1, \dots, \dim \mathfrak{g}_{-\alpha}\}$ be a basis of $\mathfrak{g}_{-\alpha}$, $\forall \alpha \in \Delta_+$. Also choose a total order on Δ_+ . (Some sort of lexicographical order that takes longer to write than to say.)

$$F_{\alpha_1, i_1}, F_{\alpha_2, i_2}, \dots, F_{\alpha_s, i_s}, v_\Lambda$$

$\alpha_1 \leq \alpha_2 \leq \dots \leq \alpha_s$ and if $\alpha_p = \alpha_{p+1}$, $i_p \leq i_{p+1}$

We have $M(\Lambda)_\lambda = \{m | Hm = \lambda(H)m\}$ weight spaces.

$$M(\Lambda) = \bigoplus_{\lambda \in \mathfrak{h}^*} M(\Lambda)_\lambda$$

The vector v_Λ is in $M(\Lambda)_\Lambda$ by definition,

$$\begin{aligned} F_{\alpha,i} v_\Lambda &\in M(\Lambda)_{\Lambda - \alpha} \\ H(Fv_\Lambda) &= \underbrace{[H, F]v_\Lambda}_{= -\alpha(H)v_\Lambda} + \underbrace{FHv_\Lambda}_{= \Lambda(H)v_\Lambda} \end{aligned}$$

So $\chi_{M(\Lambda)} = \sum_{\lambda \in \mathfrak{h}^*} \dim M(\Lambda)_\lambda e^\lambda$ is computed by counting monomials y with fixed $\sum_i \alpha_i$.

$$(6.0.1) \quad \chi_{M(\Lambda)} = e^\Lambda \prod_{\alpha \in \Delta_+} \frac{1}{(1 - e^{-\alpha})^{\dim \mathfrak{g}_\alpha}}.$$

The product on Eq. 6.0.1 is called *Weyl denominator*.

Exercise 6.1. Convince yourself of this.

Example 6.2. $\mathfrak{g} = \mathfrak{sl}_2$, [Picture]

$$\begin{aligned} \chi_{M(\Lambda)} &= e^\Lambda + e^{\Lambda - \alpha} + e^{\Lambda - 2\alpha} + \dots \\ &= e^\Lambda (1 + e^{-\alpha} + e^{-2\alpha} + \dots) \\ &= e^\Lambda \frac{1}{1 - e^{-\alpha}}. \end{aligned}$$

For certain Λ , $M(\Lambda)$ is *reducible* (i.e. there exists a submodule $0 \neq N \subset M(\Lambda)$ (with proper contention).

Lemma 6.3. *For any submodule N ,*

$$N = \bigoplus_{\mu \in \mathfrak{h}^*} N \cap M(\Lambda)_\mu.$$

Corollary. The sum of all proper submodules of $M(\Lambda)$ is proper, in particular there is a maximal proper submodule.

Notation. $L(\Lambda) = M(\Lambda) / \left(\begin{smallmatrix} \text{max. proper} \\ \text{submodule} \end{smallmatrix} \right)$

Example 6.4. $\mathfrak{sl}_2$. $\Lambda = 3\omega$ (ω : fundamental weight, $\alpha = 2\omega$.) $L(3\omega) = \mathbb{C} \langle e^{3\omega}, e^{-\omega}, e^{-3\omega} \rangle$.
[Picture]

Definition 6.5. A $\mathfrak{g}$ -module is *integrable* if

- $V = \bigoplus_{\mu \in \mathfrak{h}^*} V_\mu$ (weight module).
- For all simple roots α_i ; e_i and f_i are locally nilpotent on V (i.e. for all $v \in V$ there exists N such that $e_i^N v = f_i^N v = 0$.)

Remark 6.6. • Vermas are not integrable.

- $\dim V < \infty \implies V$ integrable.
- $\mathfrak{g}$ itself (Kac-Moody) is integrable.

Dominant integrable weights. Let $\{\alpha_1^\vee, \dots, \alpha_\ell^\vee\} \subset \mathfrak{h}$ be the simple coroots.

Definition 6.7. The *dominant integral weights* are the weights that pair with the coroots to give integers:

$$P_+ = \{\lambda \in \mathfrak{h}^* : \langle \lambda, \alpha_i^\vee \rangle \in \mathbb{Z}_{\geq 0}, i = 1, \dots, \ell\}.$$

For $L(\Lambda)$ to be integrable, it is necessary that $\Lambda \in P_+$.

Indeed, suppose $L(\Lambda)$ is integrable. Then $f_i^N v_\Lambda = 0$ in $L(\Lambda)$, or rather

$$\underbrace{e_i f_i^{N+1} v_\Lambda}_{K f_i^N = 0} \in M(\Lambda),$$

and K can only be zero if $\langle \Lambda, \alpha_i^\vee \rangle \in \mathbb{Z}_{\geq 0}$. Applying for all i , we find $\Lambda \in P_+$ is necessary.

Proposition 6.8. $L(\Lambda)$ is integrable if and only if $\Lambda \in P_+$.

Proof. For the converse, use induction and Serre relations (we know the result for the highest weight, and want to prove for others). $\square$

Example 6.9. $\mathfrak{sl}_3$. [Picture]

Example 6.10. $\widehat{\mathfrak{sl}}_2$. [Picture, P_+ looks like diagonal lines.]

$$\alpha_0^\vee = K - H, \quad \alpha_1^\vee = H \in \mathfrak{sl}_2, \quad \langle \delta, \alpha_i^\vee \rangle = 0, \quad i = 0, 1$$

Remark 6.11. For affine Kac-Moody algebras, almost nothing about the structure of $M(\Lambda)$ depends on the coefficient of δ in Λ . So it's common to consider

$$M(\Lambda) = M_k(\lambda) = M(k\Lambda_0 + \lambda), \quad \lambda \in \mathfrak{h}^*$$

where k , the level of Λ , is super important.

Then

$$\underbrace{\hat{P}_+}_{\substack{\delta\text{-coef.} \\ =0}} = \bigcup_{k \in \mathbb{Z}_{\geq 0}} \{k\Lambda_0 + \lambda \mid \lambda \in P_+^k\} \{k\Lambda_0 + \lambda \mid \lambda \in P_+^k\}.$$

$$P_+^k = \{\lambda \in P_+ \mid \langle \lambda, \theta \rangle \leq k\} \subset P_+ \text{ for } \mathfrak{g}.$$

Consider $V = \bigoplus_{\mu \in \mathfrak{h}^*} V_\mu$ integrable, and $V_\lambda \neq 0$. Let $i \in \{1, \dots, \ell\}$. Consider $U = \bigoplus_{n \in \mathbb{Z}} V_{\lambda + n\alpha_i} \subset V$, and the action of e_i, f_i and $h_i = [e_i, f_i]$.

$\mathfrak{sl}_2 \curvearrowright U$, locally integrable. By structure of $\mathfrak{sl}_2$ -representations, U must be finite-dimensional with “symmetrical” weight space multiplicities, i.e.,

$$\{\lambda + n\alpha_i \mid n \in \mathbb{Z}\} \cap \{\text{weights of } V\} = \{\lambda + n\alpha_i \mid -p \leq n \leq q\}.$$

and

$$\langle \lambda - p\alpha_i, h_i \rangle = -\langle \lambda + q\alpha_i, h_i \rangle.$$

Consequently, the reflection $r_i(\lambda) := \lambda - \langle \lambda, \alpha_i^\vee \rangle \alpha_i$ has the same multiplicity as λ .

[Picture]

Now consider $M(\lambda)$ and $L(\Lambda)$ for $\Lambda \in P_+$. Actually, for general Λ , $M(\Lambda)$, while not necessarily irreducible, has an (Ω) -composition series* by irreducibles $L(\lambda)$.

$$M(\Lambda) = V_0 \supset V_1 \supset V_2 \supset \dots \supset V_n = 0,$$

such that V_i/V_{i+1} is $\simeq L(\lambda_i)$ (i.e. an irreducible highest weight module) for some $\lambda_i = \Lambda - \beta_i$, $\beta_i \in Q$. For Kac-Moody algebras we also consider the case that $(V_i/V_{i+1}) = 0$ for all $\mu \in \Omega + Q_+$, (which is the Ω -composition series).

[Picture.]

For an Ω -composition series, as above, we have

$$\text{ch}_{M(\Lambda)} = \sum_{\lambda \geq \Omega} \underbrace{[M(\Lambda) : L(\lambda)]}_{\substack{\# \text{ of times } L(\lambda) \\ \text{appears in the compos. series}}} \text{ch}_{L(\lambda)} \in \langle e^\mu : \mu \not\geq \Omega \rangle$$

Sending “ $\Omega \rightarrow -\infty$ ”, the identity

$$\text{ch}_{M(\Lambda)} = \sum_{\lambda \leq \Lambda} [M(\Lambda) : L(\lambda)] \text{ch}_{L(\lambda)}$$

makes sense.

Notation. $b_{\Lambda, \lambda} = [M(\Lambda) : L(\lambda)]$.

Remark 6.12. Recall the partial order on weights that $\lambda \leq \Lambda$ if $\Lambda - \lambda \in Q = \sum \mathbb{Z}_+ \alpha_i$. $b_{\Lambda, \lambda} = 1$ if $\lambda = \Lambda$ and $b_{\Lambda, \lambda} = 0$ if not ($\lambda \leq \Lambda$).

If we choose a total order on $\mathfrak{h}^*$, compatible with $\leq$. Then $\{b_{\Lambda, \lambda}\}$ is a lower triangular matrix with 1 on the diagonal. We can define $\{m_{\Lambda, \lambda}\}$ the *inverse matrix*. It's again lower triangular, 1 on the diagonal, and all $m_{\Lambda, \lambda}$ are integers (maybe negative now). And we have

$$(6.12.1) \quad \text{ch}_{L(\Lambda)} = \sum_{\lambda \leq \Lambda} m_{\Lambda, \lambda} \text{ch}_{M(\lambda)}.$$

Example 6.13. $\mathfrak{sl}_2$. $M(3) \underset{L(3)}{\supset} M(-5) \underset{L(-5)}{\supset} 0$. Since $M(-5)$ is already irreducible.

[Missing...]

We want to discover $m_{\Lambda, \lambda}$. In general massively difficult. For $\Lambda \in P_+$, $m_{\Lambda, \lambda}$ easy. Multiply Eq. 6.12.1 by R

$$R\text{ch}_{L(\Lambda)} = \sum_{\lambda \geq \Lambda} m_{\Lambda, \lambda} e^\lambda \cdot e^\rho R\text{ch}_{L(\Lambda)} = \sum_{\lambda \leq \Lambda} m_{\Lambda, \lambda} e^{\lambda + \rho}.$$

What's ρ ? It's $\rho \in \mathfrak{h}^*$ chosen so that $\langle \rho, \alpha_i^\vee \rangle = 1$, and it's called the *Weyl vector*.

Remark 6.14. For $\mathfrak{g}$ finite-dimensional, $\rho = \sum_{i=1}^\ell \omega_i$ necessarily. (And equals $\frac{1}{2} \sum_{\alpha \in \Delta_+} \alpha$.)

For $\mathfrak{g}$ finite dimensional and the affine $\hat{\mathfrak{g}}$, $\hat{\rho} = h^\vee \Lambda_0 + \rho$ works.

For $w \in W = \langle r_i | i = 1, \dots, \ell \rangle$, define $w(\text{ch}_V) = \sum \dim V_\mu e^{W(\mu)}$. We saw $w(\text{ch}_V) = \text{ch}_V$ if V integrable. In particular $w(\text{ch}_{L(\Lambda)}) = \text{ch}_{L(\Lambda)}$, $\Lambda \in P_+$.

Lemma 6.15. $m_{\Lambda, \lambda} = 0$ unless $\lambda + \rho = w(\Lambda + \rho)$ for some $w \in W$

Claim. $r_i(e^\rho R) = -e^\rho R$. So $w(e^\rho R) = \det(w)e^\rho R$ for all $w \in W$.

Proof.

$$R = \prod_{\alpha \in \Delta_+} (1 - e^{-\alpha})^{\text{mult}(\alpha)}.$$

Note that

- (1) $\text{mult}(r_i(\alpha)) = \text{mult}(\alpha)$ for all $\alpha \in \Delta$. (Since $\mathfrak{g}$ is integrable!)
- (2) $r_i(\Delta_+) = \{-\alpha_i\} \cup (\Delta_+ \setminus \{\alpha_i\})$.

Any $\alpha \in \Delta_+$ is of the form $\alpha = \sum_{i=1}^\ell k_i \alpha_i$. If $\alpha \neq \alpha_i$, some $k_{j_0} \neq 0$, $j_0 \neq i$ and

$$\begin{aligned} r_i(\alpha) &= \sum_j k_j \alpha_j \langle \alpha, \alpha_i^\vee \rangle \alpha_i \\ &= \sum k'_j \alpha_j \\ &= e^\rho (e^{-\alpha_i} - 1) \left(\prod_{\alpha \in \Delta_+ \setminus \alpha_i} \right) \\ &= -e^\rho R. \end{aligned}$$

for $k'_{j_0} = k_{j_0} > 0$. Can't have a mixture of signs, so $r_i(\alpha) \in \Delta_\perp$.

So

$$r_i(e^\rho R) = \prod_{\alpha \in \Delta_+ \setminus \alpha_i} (1 - e^{-\alpha})^{\text{mult}(\alpha)} \cdot (1 - e^{t\alpha_i} \cdot e^{\rho - \alpha_i})$$

$$r_i = \rho - \langle \alpha_i^\vee, \rho \rangle \alpha_i = \rho - \alpha_i$$

Hence

$$\sum_{\lambda \leq \Lambda} m_{\Lambda, \lambda} e^{\lambda + \rho}$$

is W -skew-invariant. (Lemma 6.15.)

Hence

$$\sum_{\lambda \leq \Lambda} m_{\Lambda, \lambda} e^{\lambda + \rho} = \sum_{w \in W} \det(w) \cdot e^{w(\Lambda + \rho)}$$

In conclusion, the *Weyl character formula*

$$\text{ch}_{L(\Lambda)} = \sum_{w \in W} \det(w) \cdot \frac{e^{w(\Lambda+\rho)-\rho}}{R}$$

□

Corollary (Weyl denominator formula).

$$e^\rho R = \sum_{w \in W} \det(w) e^{w(\rho)}.$$

Next time: Affine case, θ -functions. Modular forms (Poisson summation.)

7. CHARACTERS OF INTEGRABLE HIGHEST WEIGHT FOR AFFINE KAC-MOODY ALGEBRAS

Can be calculated using the Weyl character formula.

Recall: $\hat{\mathfrak{g}} = \mathfrak{g}[t, t^{-1}] \oplus \mathbb{C}K \oplus \mathbb{C}d$.

Dual Cartan $\hat{\mathfrak{h}}^* = \mathfrak{h}^* \oplus \mathbb{C}\Lambda_0 \oplus \mathbb{C}\delta$.

We consider weights $\Lambda = k\Lambda_0 + \lambda$, $\lambda \in \mathfrak{h}^*$, $k \in \mathbb{Z}_+$, $\lambda \in P_+^k$.

Simple roots: $\alpha_0 = \delta - \theta$, $\{\alpha_1, \dots, \alpha_\ell\} \subset \mathfrak{h}^*$.

The affine Weyl group is (the *semidirect* product!)

$$\widehat{W} = \langle r_0, \dots, r_\ell \rangle \cong W \times T$$

where $T = \{t_\alpha | \alpha \in Q\}$, where t_α is the translation

$$t_\alpha(\Lambda) = \Lambda + k\alpha - \left((\lambda, \alpha) + k \frac{|\alpha|^2}{2} \right) \delta$$

$$\Lambda = k\Lambda_0 + \lambda.$$

Let's compute $\chi_{L(\Lambda_0)}$ using the Weyl character formula

$$\chi_{L(\Lambda)} = \frac{\sum_{w \in \widehat{W}} \varepsilon(w) e^{w(\Lambda+\rho)-\rho}}{R}$$

Firstly $R = \prod_{\alpha \in \hat{\Delta}} (1 - e^{-\alpha})^{\text{mult}(\alpha)}$

Recall

$$\hat{\Delta}_+ = \{m\delta | m \in \mathbb{Z}_{\geq 1}\} \cup \{\alpha_1 + m\delta | m \in \mathbb{Z}_{\geq 0}\} \cup \{-\alpha_1 + m\delta | m \in \mathbb{Z}_{\geq 1}\}$$

So let's write $e^{m\delta} = q^m$ (i.e. $q = e^{-\delta}$, is a symbol) and $y = e^{\alpha_1}$.

So

$$R = \prod_{n=1}^{\infty} \underbrace{(1 - y^{-1}q^{n-1})}_{\alpha + (n-1)\delta} \underbrace{(1 - q^n)}_{n\delta} \underbrace{(1 - yq^n)}_{-\alpha + n\delta}$$

Now let's express numerator also in terms of q and y

$$\hat{\rho} = h^\vee \Lambda_0 + \rho$$

$h^\vee$ dual Coxeter number for $\mathfrak{g}$. For $\mathfrak{g} = \mathfrak{sl}_2$, $h^\vee = 2$. (For $\mathfrak{g} = \mathfrak{sl}_n$, $h^\vee = n$ and $\mathfrak{g} = E_8$, $h^\vee = 30$.)

For $\mathfrak{sl}_2$, $\rho = \frac{1}{2}\alpha_1 = \omega_1$.

$\Lambda = \Lambda_0$, $\Lambda + \rho = 3\Lambda_0 + \omega_1$.

Using the formula, we find

$$\begin{aligned} t_{m\alpha_1}(3\Lambda_0 + \omega_1) - (\Lambda_0 + \omega_1) = \\ \dots, \Lambda_0 - 3\alpha, -2\delta, \Lambda_0, \Lambda_0 + 3\alpha_1 - 4\delta, \dots \end{aligned}$$

[Picture]

$\hat{W} = T \cup T\sigma$, $\sigma = r_1$ finite reflection.

Notation. $w(\Lambda + \rho) := w \circ \Lambda$.

One finds

$$\sum_{w \in \hat{W}} \varepsilon(w) e^{w \circ \Lambda_0} = e^{\Lambda_0} \left(\underbrace{1 + y^3 q^4 + y^{-3} q^2 + \dots}_{+ \text{ signs because } w \in T} \underbrace{-y^{-1} - y^2 q^2 - \dots}_{- \text{ signs because } w \in T_\sigma} \right)$$

We find explicitly

$$\chi_{L(\Lambda_0)} = e^{\Lambda_0} \frac{\sum_{m \in \mathbb{Z}} y^{3m} q^{3m^2+m} - \sum_{m \in \mathbb{Z}} y^{3m-1} q^{3m^2-m}}{\prod_{n=1}^{\infty} (1 - y^{-1} q^{n-1})(1 - q^n)(1 - yq^n)}$$

Exercise 7.1. Put this formula in mathematica and confirm that [Picture]

$$\chi_{L(\Lambda_0)} = e^{\Lambda_0} (1 + q(y^{-1} + 1 + y) + q^2(y^{-1} + 2 + y) + (q^3(2y^{-1} + 3 + 2y) + \dots))$$

Appears that the central column here are *partitions*, i.e. $p(n) = \#$ of partitions of n . To see why recall the generating function $\sum_{n=0}^{\infty} q^n p(n) = \prod_{k=1}^{\infty} \frac{1}{1-q^k}$, and expand.

It would appear that

$$\chi_{L(\Lambda_0)} = \prod_{k=1}^{\infty} \frac{1}{1-q^k} \cdot \sum_{m \in \mathbb{Z}} y^m q^{m^2}$$

This identity is true, and we will see it has a vertex-algebra interpretation.

Remark 7.2. If we compute $L(0) = 1$ using the formula, we obtain

$$\prod_{n=1}^{\infty} (1 - yq^{n-1})(1 - q^n)(1 - y^{-1}q^n) = \sum (\text{exercise})$$

This identity is called the *Jacobi triple product identity*.

These functions

$$\sum_{n \in \mathbb{Z}} q^{n^2}, \sum_{n \in \mathbb{Z}} y^n q^{n^2}, \sum_{n \in \mathbb{Z}} y^{3n} q^{3n^2+n}, \text{ etc. } \dots$$

are all examples of θ -functions.

Remark 7.3. In the formula for $\chi_{L(\Lambda_0)}(y, q)$, we could put $y = 1$ to get

$$\begin{aligned} \chi_{L(\Lambda_0)}(q) &= 1 + 3q + 4q^2 + 7q^3 + \dots \\ &= \prod_{k=1}^{\infty} \frac{1}{1-q^k} \cdot \sum_{m \in \mathbb{Z}} q^{m^2} \end{aligned}$$

If one looks at $L(\Lambda_0 + \omega_1)$ (the other $\Lambda \in P_+^1$),

$$\chi_{L(\Lambda+\omega_1)} = \prod_{k=1}^{\infty} \frac{1}{1-q^k} \sum_{m \in \mathbb{Z}} q^{(m+1/2)^2 - 1/4}$$

8. THETA-FUNCTIONS

Let's consider

$$\theta(\tau) = \sum_{n \in \mathbb{Z}} q^{n^2/2}, \quad q = e^{2\pi i \tau}$$

This converges (absolutely on compact regions in) the domain $\text{Im}(\tau) > 0$.

Consider the Fourier transform

$$\hat{g}(y) = \int_{-\infty}^{\infty} g(x) e^{2\pi i x y} dx$$

Theorem 8.1 (Poisson summation).

$$\sum_{n \in \mathbb{Z}} g(n) = \sum_{n \in \mathbb{Z}} \hat{g}(n)$$

Let's take $g(x, t) = e^{-\pi t x^2}$. Then $\theta(it) = \sum_{n \in \mathbb{Z}} g(n, t)$.

In this case

$$\hat{g}(y) = \sqrt{t} e^{-\pi y^2/t}$$

(integral of Gaussian).

So we conclude that

$$\theta\left(-\frac{1}{\tau}\right) = \sqrt{\frac{\tau}{i}} \theta(\tau)$$

Note also that

$$\theta(\tau + 2) = \theta(\tau)$$

because $q^{\frac{1}{2}} = e^{\pi i \tau}$.

So $\theta(\tau)$ is an example of a modular form.

What is a modular form? Let

$$\text{SL}_2(\mathbb{Z}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid a, b, c, d \in \mathbb{Z}, \det = 1 \right\},$$

which acts on $\mathbb{H} = \{\tau \in \mathbb{C} \mid \text{Im}(\tau) > 0\}$ by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \cdot \tau = \frac{a\tau + b}{c\tau + d}.$$

Definition 8.2. Let $\Gamma = \text{SL}_2(\mathbb{Z})$ or some subgroup. A (weak) modular form (of weight k) is $f : \mathbb{H} \rightarrow \mathbb{C}$ holomorphic such that

$$(8.2.1) \quad f\left(\frac{a\tau + b}{c\tau + d}\right) = (c\tau + d)^k f(\tau).$$

If we demand that Γ contains $\begin{pmatrix} 1 & N \\ 0 & 1 \end{pmatrix}$ for some $N \geq 1$, (so $f(\tau + N) = f(\tau)$), and so $f(\tau) = \sum_{n=-\infty}^{\infty} a(n) q^{2\pi i n/2}$, for $f : \mathbb{H} \rightarrow \mathbb{C}$ satisfying Eq. 8.2.1 and such that $f(\tau)$ is “meromorphic at cusps”, i.e. $f(\tau) = \sum_{n \geq N_0} a(n) q^{n/N}$, etc.

Finally we would add a factor

$$f\left(\frac{a\tau + b}{c\tau + d}\right) = \varepsilon(A) (c\tau + d)^k f(\tau).$$

So in particular a weight-0 modular form is a modular function.

Denote $S = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$. So $S\tau = -\frac{1}{\tau}$, and denote $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, so $T\tau = \tau + 1$.

Then $\Gamma = \langle S, T^2 \rangle \subset \mathrm{SL}_2(\mathbb{Z})$.

We saw

$$\begin{aligned}\theta(T^2\tau) &= \theta(\tau) \\ \theta(S\tau) &= \sqrt{\frac{\tau}{i}}\theta(\tau) = i^{-1/2}(c\tau + d)^{1/2}\theta(\tau).\end{aligned}$$

So $\theta(\tau)$ is a modular form of weight $1/2$, for the group $\Gamma = \langle S, T^2 \rangle$ with multiplier system $\varepsilon : \Gamma \rightarrow \mathbb{C}^\times$ defined by

$$\begin{aligned}\varepsilon(S) &= i^{-1/2} \\ \varepsilon(T^2) &= 1\end{aligned}$$

Theorem 8.3. *The Dedekind eta function*

$$\eta(\tau) = q^{1/24} \prod_{n=1}^{\infty} (1 - q^n), \quad q = e^{2\pi i \tau}$$

is also a modular form of weight $1/2$ for $\mathrm{SL}_2(\mathbb{Z}) = \langle S, T \rangle$.

$$\begin{aligned}\eta\left(-\frac{1}{\tau}\right) &= \sqrt{\frac{\tau}{i}}\eta(\tau) \\ \eta(T\tau) &= e^{2\pi i/24}\eta(\tau).\end{aligned}$$

9. VERTEX ALGEBRAS

- (1) Recall $\hat{\mathfrak{a}} = \mathbb{C}K \oplus \bigoplus_{n \in \mathbb{Z}} \mathbb{C}h_n$ the *Heisenberg Lie algebra*, or *oscillator Lie algebra*,

$$[h_m, h_n] = m\delta_{m, -n}K, \quad [K, \hat{\mathfrak{a}}] = 0.$$

Remark 9.1. “It’s the simplest case of an affine Lie algebra: a 1-dimensional Lie algebra with a bilinear form”. It’s an example of the affine Lie algebra construction $\mathfrak{g} \rightsquigarrow \hat{\mathfrak{g}}$, where now $\mathfrak{a} = \mathbb{C}$, and

$$\begin{aligned}(\cdot, \cdot) : \mathfrak{a} \times \mathfrak{a} &\longrightarrow \mathbb{C} \\ (1, 1) &\longmapsto 1\end{aligned}$$

- (2) Witt Lie algebra

$$\begin{aligned}W &= \bigoplus_{n \in \mathbb{Z}} \mathbb{C}D_n, \\ [D_m, D_n] &= (m - n)D_{m+n}.\end{aligned}$$

The oscillator algebra $\hat{\mathfrak{a}}$ has a representation which we have seen already:

$$H = \mathbb{C}[x_1, x_2, \dots]$$

$$h_n \mapsto \begin{cases} n \frac{\partial}{\partial x_n} & \text{if } n > 0 \\ x_{-n} & \text{if } n < 0, \\ 0 & \text{if } n = 0 \end{cases}, \quad K \mapsto \mathrm{Id}$$

A picture of H [Picture].

Here I am introducing a grading of H :

$$H = \bigoplus_{n \in \mathbb{Z}_{\geq 0}} H_n, \quad H_n = \text{span}\{x_{m_1}, \dots, x_{m_s} \mid \sum m_s = n\}.$$

Remark 9.2. $\dim(H_n) = \# \{\text{integer partitions of } n\} = p(n)$ and $\sum_{n=0}^{\infty} \dim(H_n)q^n = \prod_{k=1}^{\infty} \frac{1}{1-q^k}$.

Remark 9.3 (Verma module style). H can be presented alternatively as

$$H = U(\hat{\mathfrak{a}}) \otimes_{U(\hat{\mathfrak{a}}_+)} \mathbb{C}1$$

where $\mathbb{C}1$ is a 1-dimensional representation of

$$\begin{aligned} \hat{\mathfrak{a}}_+ &= \bigoplus_{n \geq 0} \mathbb{C}h_n \oplus \mathbb{C}K. \\ h_n \cdot 1 &= 0 \quad \forall n \geq 0, \quad K \cdot 1 = 1 \end{aligned}$$

Exercise 9.4. H is irreducible.

Proof. We need to prove that there is no vector subspace $V \subset \mathbb{C}[x_1, x_2, \dots] = H$ such that $xV \subset V$ for all $x \in \hat{\mathfrak{a}}$. And then the proof is basically noticing that the orbit of the action of $\hat{\mathfrak{a}}$ on H is all of H . That is, if we had proper subspace of H , we can always find an element of $\hat{\mathfrak{a}}$ that takes some element in V to the one of the elements that is not in V . Indeed, by applying h_n for different values of positive n we can take any element of V to a constant. Then we apply h_n for different values of negative n to obtain any monomial. Then we add these monomials and obtain any polynomial in H . $\square$

Notation. Instead of 1 let's write $|0\rangle$.

Remark 9.5. We can easily generalise H to

$$H^\mu = U(\hat{\mathfrak{a}}) \otimes_{U(\hat{\mathfrak{a}}_+)} \mathbb{C}|\mu\rangle,$$

($\mu \in \mathbb{C}$), where $\hat{\mathfrak{a}}_+$ is the same, but now

$$h_n|\mu\rangle = \begin{cases} 0 & \text{if } n > 0 \\ \mu \cdot |\mu\rangle & \text{if } n = 0 \end{cases} \quad K|\mu\rangle = |\mu\rangle.$$

H^μ is again a (irreducible) $\hat{\mathfrak{a}}$ -module.

Generalise even more: Let $\mathfrak{h}$ be a finite-dimensional vector space, and $(\cdot, \cdot) : \mathfrak{h} \times \mathfrak{h} \rightarrow \mathbb{C}$ a symmetric bilinear form. $\hat{\mathfrak{h}} = \mathfrak{h}[t, t^{-1}] \oplus \mathbb{C}K$,

$$[at^m, bt^n] = m(a, b)\delta_{m, -n}K, \quad [K, \hat{\mathfrak{h}}] = 0.$$

Let $\mu \in \mathfrak{h}$. Define $H^\mu = U(\hat{\mathfrak{h}}) \otimes_{U(\hat{\mathfrak{h}}_+)} \mathbb{C}|\mu\rangle$, $\hat{\mathfrak{h}}_+ = \mathfrak{h}[t] \oplus \mathbb{C}K$, and

$$at^m \cdot |\mu\rangle = \begin{cases} (\mu, a)|\mu\rangle & \text{if } m = 0 \\ 0 & \text{if } m > 0. \end{cases}$$

Returning to $\hat{\mathfrak{a}} \curvearrowright H = H^0$. Introduce

$$L_m = \frac{1}{2} \sum_{k \in \mathbb{Z}} h_{m-k} h_k.$$

If $m \neq 0$, this sum is well-defined in $\text{End}(H)$.

Indeed, h_{m-k} and h_k commute ($m \neq 0$), and $\forall p(\underline{x}) \in H$, there exists N such that $p(\underline{x}) = p(x_1, \dots, x_{N-1})$, then $h_k \cdot p(x) = 0 \ \forall k \geq N$, and $h_{m-k} \cdot p(x) = 0 \ \forall m - k \geq N$.

So $(\sum h_{m-k} h_k) p(\underline{x})$ is a finite sum. Of course, the number of terms in the sum depends on $p(x)$ and can be arbitrarily large.

Exercise 9.6. If m, n and $m + n$ are not zero, then

$$[L_m, L_n] = (m - n)L_{m+n}$$

holds in $\text{End}(H)$.

(Trying to be a representation of W

$$D_n \mapsto L_n \in \text{End}(H).)$$

But $\sum_{k \in \mathbb{Z}} h_{-k} h_k$ is not well-defined. Indeed,

$$\begin{aligned} \left(\sum_{k \in \mathbb{Z}} h_{-k} h_k \right) 1 &= \sum_{k \geq 1} k \frac{\partial}{\partial x_k} (x_k 1) \\ &= (1 + 2 + 3 + 4 + \dots) 1 \end{aligned}$$

!

Normal ordering idea (“that physicist do”). Let’s cheat and redefine the product as

$$:h_{-k} h_k: = \begin{cases} h_{-k} h_k & \text{if } k \geq 0 \\ h_k h_{-k} & \text{if } k < 0. \end{cases}$$

Now $\sum_{k \in \mathbb{Z}} :h_{-k} h_k:$ is well-defined!

Notation. Let us consider series of the form

$$a(z) = \sum_{n \in \mathbb{Z}} a_{(n)} z^{-n-1}, \quad a_{(n)} \in \text{End}(V),$$

where V is a vector space.

Definition 9.7 (Most important in the course). We say $a(z)$ is a *quantum field* on V if, for every $v \in V$ there exists $N \in \mathbb{Z}$ such that $a_{(n)} v = 0 \ \forall n \geq N$.

Example 9.8. $V = H = \mathbb{C}[x_1, x_2, \dots]$,

$$a(z) = h(z) = \sum_{n \in \mathbb{Z}} h_n z^{-n-1}.$$

$$h(z) = \dots + 3z^{-4} \frac{\partial}{\partial x_3} + 2z^{-3} \frac{\partial}{\partial x_1} + x_1 + x_2 z + x_3 z^2 + \dots$$

Any fixed $v \in H$ is $v = p(x_1, \dots, x_{N-1})$ for some N . Then $h_N v = N \frac{\partial}{\partial x_N} v = 0$.

So $h(z) \curvearrowright H$ is a quantum field.

Definition 9.9. For a quantum field (or any series in fact) $a(z) = \sum_{n \in \mathbb{Z}} a_{(n)} z^{-n-1}$, define the *creation* and *annihilating operators*

$$a(z)_+ = \sum_{n \leq -1} a_{(n)} z^{-n-1}, \quad a(z)_- = \sum_{n \geq 0} a_{(n)} z^{-n-1}.$$

The quantum field condition solves the problem of the infinite series...

The normally order product of quantum fields $a(z) = \sum_{n \in \mathbb{Z}} a(n)z^{-n-1}$, and $b(z) = \sum_{n \in \mathbb{Z}} b(n)z^{-n-1}$ is.

$$:a(z)b(z): = a(z)_+b(z) + b(z)a(z)_-.$$

Exercise 9.10 (Most important of the course). Show that if $a(z)$ and $b(z)$ are quantum fields, then $:a(z)b(z):$ is well-defined, and is a quantum field.

Next idea: in our example of $h(z) \curvearrowright H$, the coefficients h_n were coming from a Lie algebra, so we had relations $[h_m, h_n] = (\dots)$. (Indeed, $[h_m, h_n] = m\delta_{m,-n}\text{Id}_H$ in this case.) We would like to interpret such relations at the level of $h(z)$.

$$\begin{aligned} [h(z), h(z)] &= h(z)h(z) - h(z)h(z) \\ &= \sum_p \sum_{m+n=p} h_m h_n z^{-(m+n)-2} - \sum_{m+n=p} h_n h_m z^{-(m+n)-2} \\ &= \sum_p z^{-p-2} \underbrace{\left(\sum_{m+n=p} [h_m, h_n] \right)}_{\text{infinite?}} \end{aligned}$$

which is a bad idea, since that sum can be infinite. Better idea: change a variable:

$$[h(z), h(w)] = \sum_{m,n \in \mathbb{Z}} [h_m, h_n] z^{-m-1} w^{-n-1}.$$

In this example,

$$\begin{aligned} [h(z), h(w)] &= \sum_{m,n \in \mathbb{Z}} m\delta_{m,-n} z^{-m-1} w^{-n-1} I_H \\ &= \sum_{m \in \mathbb{Z}} z^{-m-1} w^{m-1} I_H. \end{aligned}$$

Observe (geometric series):

$$\begin{aligned} \frac{1}{z-w} &= \sum_{k \geq 0} z^{-k-1} w^k \quad (\text{convergent for } |z| > |w|) \\ \frac{1}{z-w} &= - \sum_{k \geq 0} w^{-k-1} z^k \\ &= - \sum_{k < 0} z^{-k-1} w^k \quad (\text{convergent for } |z| < |w|) \end{aligned}$$

So, in a sense

$$\sum_{k \in \mathbb{Z}} z^{-k-1} w^k \stackrel{“}{=} \frac{1}{z-w} - \frac{1}{z-w}.$$

This motivates us to introduce the expression

Definition 9.11. The *formal delta function* is

$$\delta(z, w) = \sum_{k \in \mathbb{Z}} z^{-k-1} w^k \in \mathbb{C}[[z, z^{-1}, w, w^{-1}]].$$

Why delta function? It behaves like Dirac delta. Recall that $\delta(x)$ is the *distribution* on $\mathbb{R}$ defined by $\delta[f(x)] = f(0)$ for all test functions $f(x)$. Every function $k(x)$ gives a distribution D_k .

$$D_k[f] = \int_{-\infty}^{\infty} k(x)f(x)dx,$$

$f \in C_c^\infty(\mathbb{R})$.

If δ were of the form D_k (it's not) it would have to look like [Picture, positive part of y axis, all x axis.

One can show that (it's a theorem by Plameli)

$$\delta(x) = \frac{1}{2\pi i} \lim_{\varepsilon \rightarrow 0_+} \left(\frac{1}{x - i\varepsilon} - \frac{1}{x + i\varepsilon} \right)$$

as distributions (i.e. limit taken in “distributional sense”).

(So that's why we called the delta function that way.)

Denote by $i_{z,w}$ the “expansion in positive powers of w ”. E.g.

$$i_{z,w} \frac{1}{z-w} = \sum_{k \geq 0} z^{-k-1} w^k,$$

similarly,

$$i_{z,w} \frac{1}{(z-w)^2} = \sum_{k \geq 0} k z^{-k-1} w^{k-1},$$

$$i_{z,w}(w^{-2}) = w^{-2}.$$

Exercise 9.12. So, in fact,

$$[h(z), h(w)] = i_{z,w} \frac{1}{(z-w)^2} - i_{w,z} \frac{1}{(z-w)^2} = \partial_w \delta(z, w).$$

What exactly is $i_{z,w}$?

Notation.

$\mathbb{C}[z]$	ring of polynomials
$\mathbb{C}[z, z^{-1}]$	ring of Laurent polynomials
$\mathbb{C}[[z]]$	ring of power series
$\mathbb{C}[[z, z^{-1}]]$	vector space (not ring!) of formal distributions
$\mathbb{C}((z))$	field of Laurent series

where the last is $\sum_{n \geq N} f a_n z^n$ for some N .

Since $\mathbb{C}((z))$ is a field, $\mathbb{C}((z))((w))$ is also a field.

There are natural inclusions

$$\mathbb{C}[z, w] \rightarrow \mathbb{C}((z))((w)) \rightarrow \mathbb{C}[[z^{\pm 1}, w^{\pm 1}]].$$

Let $\mathbb{C}(z, w)$ denote the fraction field of the domain $\mathbb{C}[z, w]$. By the property of $\mathbb{C}(z, w)$, there exists an embedding

$$\begin{array}{ccc} \mathbb{C}[z, w] & \xrightarrow{\quad} & \mathbb{C}((z))((w)) \\ & \searrow & \nearrow i_{z,w} \\ & \mathbb{C}(z, w) & \end{array}$$

The following diagram does not commute: [diagram]

We computed

$$[h(z), h(w)] = \partial_w \delta(z, w)$$

We can similarly compute

$$\begin{aligned} h(z)h(w) &= :h(z)h(w): + i_{z,w} \frac{1}{(z-w)^2} \\ h(w)h(z) &= :h(z)h(w): + i_{w,z} \frac{1}{(z-w)^2}. \end{aligned}$$

Notation. We write $\partial_w = \frac{\partial}{\partial w}$ and $\partial_w^{(j)} = \frac{1}{j!} \partial_w^j$.

Lemma 9.13. (1) *If we multiply the delta function with $(z-w)$ we get zero, that is,*

$$(z-w)\delta(z, w) = 0.$$

(2)

$$i_{z,w} \frac{1}{(z-w)^{j+1}} - i_{w,z} \frac{1}{(z-w)^{j+1}} = \partial_w^{(j)} \delta(z, w).$$

(3) $\forall j \geq 0$,

$$(z-w)^{j+1} \partial_w^{(j)} \delta(z, w) = 0.$$

(4) *Whenever $m \leq j$,*

$$(z-w)^m \partial_w^{(j)} \delta(z, w) = \partial_w^{(j-m)} \delta(z, w).$$

Proof.

□

Remark 9.14. All this is completely parallel to the Dirac δ -distribution $\delta(x)$.

$$x\delta(x) = 0, \quad x\delta'(x) = \delta(x), \text{ etc., } x^2\delta'(x) = 0.$$

10. THE RESIDUE PAIRING

Let $f(z) \in \mathbb{C}[z, z^{-1}]$, which is a vector space with basis $\{z^n : n \in \mathbb{Z}\}$.

An element of the dual vector space is a formal linear combination

$$\sum_{n \in \mathbb{Z}} c_n \varphi_n, \quad \text{where } \varphi_n(z^k) = \begin{cases} 0 & \text{if } k \neq n \\ 1 & \text{if } k = n \end{cases}$$

(no restriction on the $c_n \in \mathbb{C}$).

Identify $\sum c_n \varphi_n$ with

$$c(z) = \sum c_n z^{-n-1} \in \mathbb{C}[z^{\pm 1}]$$

so that φ acts on $f(z) = \sum f_n z^n \in \mathbb{C}[z^{\pm 1}]$, as

$$\varphi(f) = \sum_n c_n f_n = \text{Res}_z c(z) f(z) dz,$$

where

Definition 10.1. If U is a vector space and $a(z) = \sum_n a_n z^n \in U[[z^{\pm 1}]]$,

$$\text{Res}_z a(z) dz = a_{-1}$$

Let's record a few properties of $\text{Res}_z(\cdot) dz := \text{Res}_z(\cdot)$.

Lemma 10.2. (1) $\text{Res}_z f(z) \delta(z, w) = f(w)$.

$$(2) \text{ Res}_z f(z)(\partial_z g(z)) = -\text{Res}_z(\partial_z f(z))fg(z).$$

Proof. (1) Straightforward computation.

(2) Product rule, and uses that derivatives have no residues. $\square$

So, in our example, $H, h(z)$ we see that

$$(z-w)^2[h(z), h(w)] = 0,$$

which is a nontrivial example: we really need to take the square, and also the bracket is not zero.

Definition 10.3. Two quantum fields $a(z)$ and $b(z)$ on a vector space V are *mutually local* if there exists N such that

$$(z-w)^N[a(z), b(w)] = 0.$$

Remark 10.4. See [?, Chapter 1] for physics motivation. More generally, if $D = \sum_{m,n} D_{m,n} z^m w^n \in U[[z^{\pm 1}, w^{\pm 1}]]$, we call D *local* if $(z-w)^N D = 0$ for some N .

Proposition 10.5. Let $a(z), b(z)$ be a local pair of quantum fields on V . Then

$$(10.5.1) \quad a(z)b(w) = :a(z)b(z): + \sum_{j=0}^{N-1} c_j(w) i_{z,w} \frac{1}{(z-w)^{j+1}}$$

and

$$(10.5.2) \quad b(w)a(z) = :a(z)b(z): + \sum_{j=0}^{N-1} c_j(w) i_{w,z} \frac{1}{(z-w)^{j+1}}$$

In particular,

$$(10.5.3) \quad [a(z), b(w)] = \sum_{j=0}^{N-1} c_j(w) \partial_w^{(j)} \delta(z, w).$$

Furthermore

$$(10.5.4) \quad c_j(w) = \text{Res}_z (z-w)^j [a(z), b(w)].$$

Proof. First we prove 10.5.3 using 10.5.4. $\square$

Exercise 10.6. Deduce 10.5.1 and 10.5.2 from 10.5.3.

Notation. If $a(w)$ and $b(w)$ are quantum fields on V , we denote

$$a(w)_{(j)} b(w) = \text{Res}_z (z-w)^j [a(z), b(w)]$$

for $j \in \mathbb{Z}_{\geq 0}$.

Example 10.7. On H , $[h(z), h(w)] = \partial_w \delta(z, w) I_H$. So $\text{Res}_z \partial_w \delta(z, w) I_H = 0$ and $\text{Res}_z (z-w) \partial_w \delta(z, w) I_H = \text{Res}_z \delta(z, w) I_H = I_H$.

Note that I_H is a (very simple) quantum field.

The following definition generalizes the j -product to any integer, possibly negative.

Definition 10.8. For $n \in \mathbb{Z}$, and $a(w), b(w)$ quantum fields, we define the n^{th} product $a(w)_{(n)} b(w)$ as

$$a(w)_{(n)} b(w) = \text{Res}_z [i_{z,w} (z-w)^n a(z) b(w) - i_{w,z} (z-w)^n b(w) a(z)].$$

Remark 10.9. (1) We recover the prior definition: if $n \geq 0$,

$$i_{z,w}(z-w)^n = i_{w,z}(z-w)^n = \sum_{r=0}^n \binom{n}{r} z^{n-r} (-w)^r,$$

a finite sum.

(2) (Exercise.) If $n = 1$, then

$$\begin{aligned} a(w)_{(-1)}b(w) &= a(w)_+b(w) + b(w)a(w)_- \\ &=: a(w)b(w): \end{aligned}$$

(3) (Exercise.) The more negative products are not something new: for $k \geq 0$,

$$a(w)_{(-k-1)}b(w) = (\partial_w^{(k)} a(w))b(w):$$

We have actually already used the following proposition:

Proposition 10.10. *If $a(w), b(w)$ are quantum fields, then $a(w)_{(n)}b(w)$ is also a quantum field for all $n \in \mathbb{Z}$.*

Definition 10.11. A *vertex algebra* consists of a vector space V , a set $\mathcal{F}$ of quantum fields on V , a nonzero vector $|0\rangle \in V$, and a linear map $T : V \rightarrow V$, such that

- (1) $T|0\rangle = 0$, and $[T, a(z)] = \partial_z a(z) \forall a(z) \in \mathcal{F}$.
- (2) V is spanned by $a_{(n_1)}^{i_1}, \dots, a_{(n_s)}^{i_s}|0\rangle$, where $a^{ij}(z) \in \mathcal{F}$ (and $s(z) = \sum a_{(n)} z^{-n-1}$ always).
- (3) All pairs $a(z), b(z) \in \mathcal{F}$ are mutually local. (We saw (ref?) that this is equivalent to $[a(z), b(w)] = \sum_{j=0}^{N-1} c_j(w) \partial_w^{(j)} \delta(z, w)$ and described the coefficients $c_j, \dots$)

We call $|0\rangle$ the *vacuum vector* and T the *translation operator*.

In fact, our Heisenberg example that we've been discussing so far is an example:

Example 10.12. $V = H = \mathbb{C}[x_1, x_2, \dots]$, $\mathcal{F} = \{h(z)\}$, $|0\rangle = 1$, $T = ?$, is a vertex algebra.

Answer 1. Recall $L(z) = \frac{1}{2} :h(z)h(z): = \frac{1}{2} h(z)_{(-1)}h(z)$, $L(z) = \sum_{m \in \mathbb{Z}} L_m z^{-m-z}$. In particular, $L_{-1} = \frac{1}{2} \sum_{k \in \mathbb{Z}} h_k h_{-1-k}$. Well, $T = L_{-1}$.

Answer 2. Given that T must satisfy $T1 = 0$, and $[T, h(z)] = \partial_z h(z)$, i.e.

$$[T, h_n] = -n h_{n-1},$$

and H is generated from 1 by $\{h_n | n \leq -1\}$, the action of T on H (if well-defined) is completely determined.

Exercise 10.13. Write a formula for $T(x_1^{m_1} x_2^{m_2} \dots x_s^{m_s})$.

11. A SECOND DEFINITION OF VERTEX ALGEBRA

Now we aim for a second definition of vertex algebras.

Let $(V, |0\rangle, T, \mathcal{F})$ be a vertex algebra. Define

$$\begin{aligned}\tilde{\mathcal{F}} &= \{\text{quantum fields } a(z) \text{ on } V \mid [T, a(z)] = \partial_z a(z)\} \\ \overline{\mathcal{F}} &= \bigcup_{k \geq 0} \mathcal{F}_k, \\ \mathcal{F}_0 &= \{I_V\}, \\ \mathcal{F}_k &= \{a(z)_{(n)}b(z) \mid a(z) \in \mathcal{F}, b(z) \in \mathcal{F}_{k-1}\} \\ \mathcal{F}' &= \{b(z) \in \tilde{\mathcal{F}} \mid a(z), b(z) \text{ is local}, \forall a(z) \in \mathcal{F}\}\end{aligned}$$

Where I_V is the identity of V considered as a quantum field. The idea is that this is like taking a subalgebra and taking the commutator over and over again.

Lemma 11.1. $a(z)_{(-1)}I_V = a(z)$.

Proof. Direct calculation. □

Lemma 11.2. $a(z)_{(-n-1)}I_V = \partial_z^{(n)}a(z)$.

Proof. Similar. □

Lemma 11.3 (Dong). *Suppose $a(z), b(z)$ and $c(z)$ are mutually local in pairs. Let $n \in \mathbb{Z}$. Then $a(z)_{(n)}b(z)$ and $c(z)$ is a local pair.*

Proof. Done in lecture. □

Therefore

$$\mathcal{F} \subset \overline{\mathcal{F}} \stackrel{\text{Dong}}{\subset} \mathcal{F}' \subset \tilde{\mathcal{F}}$$

Remark 11.4. To be sure, we should check $\overline{\mathcal{F}} \subset \tilde{\mathcal{F}}$.

$\mathcal{F}$ tilde is the translation invariant

Exercise 11.5. If $[T, a(z)] = \partial_z a(z)$, $[T, b(z)] = \partial_z b(z)$, then

$$[T, a(z)_{(n)}b(z)] = \partial_z(a(z)_{(n)}b(z)).$$

Theorem 11.6. (1) For any $a(z) \in \tilde{\mathcal{F}}$,

$$\begin{aligned}a(z)|0\rangle &= v + z \cdot Tv + \frac{z^2}{2}T^2v + \dots \\ &= e^{zT}v \quad (v \in V)\end{aligned}$$

Thus we define

$$\begin{aligned}s : \tilde{\mathcal{F}} &\longrightarrow V \\ a(z) &\longmapsto a(z)|0\rangle|_{z=0}\end{aligned}$$

(2)

$$s(z(z)_{(n)}b(z)) = a_{(n)}s(b(z)).$$

(3) Let $\mathcal{G} \subset \tilde{\mathcal{F}}$ be such that $s(\mathcal{G}) = V$, and suppose $a(z)$ is local with all $b(z) \in \mathcal{G}$. “If you commute with a large enough bunch of guys, you are close enough to being zero.” If $s(a(z)) = 0$, then $a(z) = 0$.

Proof. (1) We need to prove $a(z)|0\rangle$ does not have negative powers of z . So suppose it does, i.e. there is $n \geq 0$ such that $a_{(n)}|0\rangle z^{-n-1} \neq 0$.

Idea: use translation invariance to pair n and $-n$, and then the fact that $a(z)$ is a quantum field to get some vanishing.

So first we translate and find that:

$$Tv(z) = Ta((z)|0) = (Ta(z) - a(z)T)|0\rangle = \partial_z a(z)|0\rangle$$

so that

$$(11.6.1) \quad v_n = \frac{1}{n}Tv_{n-1}, \quad (n \neq 0).$$

So, $v_n \neq 0$ for some $n < 0 \implies v_{n-1}$ also not zero. But we also know by $a(z)$ being a quantum field that there exists N such that $a_{(n)}|0\rangle = 0$ for all $n \geq N$. So, we'd have a contradiction. So $v(z) = \sum_{n \geq 0} v_n z^n$.

Now Eq. 11.6.1 also implies

$$v_1 = Tv_0, v_0 = \frac{1}{2}Tv_1, \dots, v_n = \frac{1}{n!}T^n v_0.$$

So $a(z)|0\rangle = e^{zT}v_0$ ($v_0 \in V$).

(2) Consider

$$a(w)_{(n)}b(w)|0\rangle = \text{Res}_z(a(z)b(w)i_{z,w}(z-w)^n - \underbrace{b(w)a(z)i_{w,z}(z-w)^n}_{=0})$$

where that vanishing is because the $i_{w,z}$ expands the argument in positive powers, and $b(w)a(z)$ also has only positive powers since $a(z)|0\rangle$ also has only positive powers (and the residue picks the coefficient of the power -1). So we obtain

$$= \text{Res}_z a(z)b(w)i_{z,w}(z-w)^n.$$

But

$$\begin{aligned} s(a(w)_{(n)}b(w)) &= (\text{Res}_z s(z)b(w)i_{z,w}(z-w)^n|0\rangle)|_{w=0} \\ &= \text{Res}_z a(z)b(w)z^n|0\rangle|_{z=0} \\ &= \text{Res}_z z^n a(z)(s(b(w))) \\ &= \text{Res}_z z^n \sum_k a_{(k)}z^{-k-1}s(b) \\ &= a_{(n)}s(b). \end{aligned}$$

(3) By locality, there exists N such that

$$\begin{aligned} (z-w)^N a(z)b(w)|0\rangle &= (z-w)^N b(w)a(z)|0\rangle \\ &= (z-w)^N b(w)e^{2T}s(a) = 0 \end{aligned}$$

$$\begin{aligned} (z-w)^N a(z)b(w)|0\rangle|_{w=0} &= 0 \\ \implies z^N a(z)s(b) &= 0 \\ \implies a(z)v &= 0 \\ \implies a(z) &= 0 \end{aligned}$$

□

Corollary.

- (1) $S|_{\mathcal{F}'}$ is injective.
- (2) $S|_{\overline{\mathcal{F}}}$ is surjective.
- (3) $S|_{\mathcal{F}}$ is isomorphism $\overline{\mathcal{F}} \rightarrow V$.

So we define the inverse (linear) map $Y : V \rightarrow \overline{\mathcal{F}}$, i.e. for all $a \in V$, $Y(a, z)$ denotes $s^{-1}(a) \in \overline{\mathcal{F}}$.

- (4) $(V, |0\rangle, T, \overline{\mathcal{F}})$ is a vertex algebra, and

$$Y(a_{(n)}b, z) = Y(a, z)_{(n)}Y(b, z) \quad \forall a, b \in V.$$

So we have

$$\begin{array}{ccccccc} \mathcal{F} & \longrightarrow & \overline{\mathcal{F}} & \longrightarrow & \mathcal{F}' & \longrightarrow & \tilde{\mathcal{F}} \\ & & \downarrow & & \downarrow & & \downarrow s \\ & & V & \xrightarrow{\text{id}} & V & \xrightarrow{\text{id}} & V \end{array}$$

- Proof.* (1) Now we apply part 3 in the theorem to $a(z) \in \mathcal{F}'$, $\mathcal{G} = \overline{\mathcal{F}}$. If $s(a(z)) = 0$, then $a(z) = 0$. So $s : \mathcal{F}' \rightarrow V$ is injective (as in the diagram).
(2) By iterating the property in the theorem that $a_{(n)}s(b(z)) = s(a(z)_{(n)}b(z))$ so we can take out “one by one” in the following product to obtain

$$a_{(n_1)}^1 \dots a_{(n_s)}^s |0\rangle = s(a^1(z)_{(n_1)} \dots a^s(z)_{(n_s)}) I_V$$

This proves the double headed arrow in the diagram.

- (3) Now define $Y(-, z) : V \rightarrow \overline{\mathcal{F}}$.

$$s(Y(a, z)_{(n)}Y(b, z)) = a_{(n)}s(Y(b, z)) = a_{(n)}b.$$

- (4) Then all the axioms in our definition of vertex algebra are satisfied!

□

Remark 11.7. Let $(V, |0\rangle, T, \mathcal{F})$ be a vertex algebra. $\overline{\mathcal{F}}$ and Y as above. $\forall a, b \in V$,

$$(11.7.1) \quad [Y(a, z), Y(b, w)] = \sum_{j=0}^{N-1} c_j(w) \partial_w^{(j)} \delta(z, w).$$

$$c_j(w) = Y(a, w)_{(j)}Y(b, w) = Y(a_{(j)}b, w).$$

Which is like something we had written before. But now we know more. Let us extract the z^{-m-1}, w^{-n-1} coefficient of Eq. 11.7.1. We obtain

$$LHS = \sum_{m, n} [a_{(m)} z^{-m-1}, b_{(n)} w^{-n-1}] = [a_{(m)}, b_{(n)}]$$

$$RHS = \sum_j \left(\sum_k (a_{(j)}b)_{(k)} w^{-k-1} \right) \left(\sum_s \binom{s}{j} z^{-s-1} w^{s-j} \right)$$

The coefficient of $z^{-m-1} w^{-n-1}$ of RHS is, $s = m$,

$$\sum_j \binom{m}{j} w^{m-j} (a_{(j)}b)_{(k)} w^{-k-1} = \sum_j \binom{m}{j} (a_{(j)}b)_{(m+n-j)}$$

So we put this as a proposition:

Proposition 11.8 (Commutator formula). *In a vertex algebra $(V, |0\rangle, T, Y)$, we have*

$$[a_{(m)}, b_{(n)}] = \sum_{j \geq 0} \binom{m}{j} (a_{(j)}b)_{(m+n-j)} \quad \forall a, b \in V, \quad m, n \in \mathbb{Z}$$

Exercise 11.9. Prove that in a vertex algebra

(11.9.1)

$$[Y(a, z)Y(b, w)i_{z,w} - Y(b, w)Y(a, z)i_{w,z}](z - w)^n = \sum_{j \geq 0} Y(a_{(n+j)}b, w)\partial_w^{(j)}\delta(z, w).$$

Hint. Prove that the left hand side is local.

By extracting coefficients of Eq. 11.9.1, we obtain for all $a, b, c \in V$ and $m, n, k \in \mathbb{Z}$:

$$(11.9.2) \quad \begin{aligned} & \sum_{j \geq 0} \binom{m}{j} (a_{(n+j)}b)_{(m+k-j)}c \\ &= \sum_{j \geq 0} (-1)^j \binom{n}{j} (a_{(m+n-j)}b_{(k+j)}c - (-1)^n b_{(n+k-j)}a_{(m+j)}c) \end{aligned}$$

This is called *Borcherd's identity*, and is the key ingredient in our second definition of vertex algebra (see [Kac01, Proposition 4.8(b)]):

Definition 11.10. A *vertex algebra* is a vector space V , a nonzero vector $|0\rangle \in V$, and a set of bilinear products $V \times V \rightarrow V$, $a, b \mapsto a_{(n)}b$, $n \in \mathbb{Z}$ such that

- (1) $\forall a, b \in V \exists N$ such that $a_{(n)}b = 0 \forall n \geq N$.
- (2) $\forall a \in V, |0\rangle_{(n)}a = \delta_{n,-1}a$.
- (3) $\forall a \in V, a_{(-1)}|0\rangle = a$ and $a_{(\geq 0)}|0\rangle = 0$.
- (4) Equation 11.9.2 holds $\forall a, b, c \in V, m, n, k \in \mathbb{Z}$.

12. CALCULATING VERTEX ALGEBRAS

We begin with some notation.

Definition 12.1. Let V be a vertex algebra and $a, b \in V$. We package the elements $a_{(0)}b, a_{(1)}b, \dots$ into a series

$$[a_\lambda b] = \sum_{j \geq 0} a_{(j)}b \frac{\lambda^j}{j!}$$

called the λ -*bracket* or *operator product expansion (OPE)* of a and b .

Recall that elements in V correspond to fields $Y(a, z)$ and $Y(b, z)$. They are local, and the coefficients of their bracket are $c_j(w) = Y(a_{(j)}b, w)$. So we put that information in the λ -bracket.

We also denote by $:ab:$ the vector corresponding to the normally ordered product $:Y(a, z)Y(b, z):$. Again, by last time's theorem, $:ab: = a_{(-1)}b$.

Example 12.2. $V = H = \mathbb{C}[x_1, x_2, \dots]$, $\mathcal{F} = \{h(z)\}$ where $h(z) = \sum_{n \in \mathbb{Z}} z^{-n-1}$ (so in this example we are denoting $h_n \equiv h_{(n)}$), where, as before,

$$h_n = \begin{cases} n \frac{\partial}{\partial x_n} & n > 0 \\ x_n & n < 0 \end{cases}$$

Recall we completed $\mathcal{F}$ to $\overline{\mathcal{F}}$ and discovered

$$\begin{array}{ccc} V & \xrightarrow{\simeq} & \overline{\mathcal{F}} \\ a(z)|0\rangle|_{z=0} & \xleftarrow{s} & a(z) \\ a & \xrightarrow{Y} & Y(a, z) \end{array}$$

For $I_V \in \overline{\mathcal{F}}$, where

$$I_V = \sum I_{V(n)} z^{-n-1}$$

$$I_{V(n)} = \begin{cases} I_V & \text{if } n = -1 \\ 0 & \text{if } n \neq -1 \end{cases}$$

we get $I_V|0\rangle|_{z=0} = |0\rangle$. So,

$$Y(|0\rangle, z) = I_V$$

This shows that the vacuum $|0\rangle$ corresponds to the identity.

Next $h(z)$ in our example. We have

$$\begin{aligned} h(z)|0\rangle &= \sum_{n \in \mathbb{Z}} h_n 1 z^{-n-1} \\ &= \sum_{k \geq 1} x^k z^{k-1} + \sum_{k \geq 1} k \frac{\partial}{\partial x_k} 1 \\ \implies h(z)|0\rangle|_{z=0} &= x_1 \\ \implies S(h(z)) &= x_1 \\ \implies Y(x, z) &= h(z). \end{aligned}$$

So, x_1 corresponds to the sort of “generating field” $h(z)$.

What about x_2 ? Well, $h(z) = h(z)_{(-1)} I_V$. There was a lemma (maybe Remark 10.9?) that

$$a(z)_{(-2)} I_V = \partial_z a(z).$$

So $\partial_z h(z) \in \overline{\mathcal{F}}$,

$$\begin{aligned} a(\partial_z h(z)) &= \partial \sum_{k \geq 1} x_k z^{k-1} \Big|_{z=0} \\ &= \sum_{k \geq 1} (k-1) x_k z^{k-2} \Big|_{z=0} \\ &= x_2. \end{aligned}$$

So

$$Y(x_{n+1}, z) = \partial_z^{(n)} h(z).$$

Continuing:

$$Y(x_3, z) = \frac{1}{2} \partial_z^2 h(z),$$

and in general

$$Y(x_{n+1}, z) = \partial_z^{(n)} h(z).$$

The next question is: what about x_1^2 ? The trick here is that x_1 is identified with h , so

$$\begin{aligned} Y(x_1^2, z) &= Y(x_1 \cdot x_1, z) \\ &= Y(h_{(-1)} x_1, z) \\ &= Y(x_1, z)_{(-1)} Y(x, z) \\ &=: h(z)h(z): . \end{aligned}$$

Recall that $L(z) = \frac{1}{2} :h(z)h(z):$. We have claimed (Exercise 9.6) that $[L_m, L_n] = (m-n)L_{m+n}$ when $m, n, m+n \neq 0$ where

$$\begin{aligned} L(z) &= \sum_{m \in \mathbb{Z}} L_m z^{-m-2} \\ &= \sum_{n \in \mathbb{Z}} L_{(n)} z^{-n-1}. \end{aligned}$$

What about the λ -bracket? Let's denote $h = x_1$, (so $Y(h, z) = h(z)$, $L = \frac{1}{2}x_1^2$ and $Y(L, z) = L(z)$).

Well, long ago we computed that

$$\begin{aligned} [h(z), h(w)] &= \partial_w \delta(z, w) I_V \\ &= \sum_{j \geq 0} Y(h_{(j)} h, w) \partial_w^{(j)} \delta(z, w). \end{aligned}$$

Then

$$\begin{aligned} Y(h_{(0)} h, w) &= 0 \\ Y(h_{(1)} h, w) &= I_V \\ Y(h_{\geq 2} h, w) &= 0 \\ \implies h_{(1)} h &= |0\rangle, \text{ and} \\ h_{(j)} h &= 0, \quad j = 0 \text{ and } j \geq 2. \end{aligned}$$

So we see that

$$[h_\lambda h] = \lambda |0\rangle.$$

We often omit “ $|0\rangle$ ” from these computations, as if it were “1”.

So

$$[h_\lambda h] = \lambda.$$

$[h_\lambda |0\rangle] = 0 = [|0\rangle_\lambda h]$ because $[h(z), I_V] = 0$. In fact $[x_\lambda |0\rangle] = [|0\rangle_\lambda x] = 0$ for all $x \in V$.

What about $h_{(0)} h$? Well, $h_{(0)} = 0$ by definition of $h(z)$. So $h_{(0)} h = 0$. Next, $h_{(j)} = j \frac{\partial}{\partial x_j}$ for $j > 0$, so

$$h_{(1)} h = \frac{\partial}{\partial x_1} (x_1) = 1 = |0\rangle.$$

So $[h_\lambda h] = \lambda$ again.

What I really want is $[L_\lambda L] = ?$ Let's use Borcherd's formula 11.9.2.

First let's compute $(h_{(-1)} h)_{(0)} L$, i.e. we are putting $n = -1$, $m = 0$ and $k = 0$.

[missing computations]

Exercise 12.3. Check $(h_{(-1)} h)_{(1)} h = 2h$ and $(h_{(-1)} h)_{\geq 2} h = 0$. Thus $[L_\lambda h] = Th + \lambda h$.

If we wanted, we could continue to calculate $[L_\lambda L]$ by putting $c = \frac{1}{2}x_1^2$ above instead of $c = x_1$.

We find

$$[L_\lambda L] = \underbrace{TL}_{L_{(0)}L} + \underbrace{2\lambda L}_{L_{(1)}L} + \underbrace{\frac{\lambda^3}{12}|0\rangle}_{L_{(3)}L}.$$

This means

$$(12.3.1) \quad [L(z), L(w)] = (\partial_w L(w))\delta(z, w) + 2L(w)\partial_w \delta(z, w) + \frac{1}{12}\partial_w^3 \delta(z, w)I_V.$$

So we can extract coefficients:

$$LHS = \sum_{m,n} [L_m, L_n] w^{-m-2} z^{-n-2}$$

Coefficient of $w^{-m-2} z^{-n-2}$ on RHS is:

$$\begin{aligned} & \sum_k (-k-2)L_k w^{-k-3} \sum_a z^{-a-1} w^a \\ & + 2 \sum_k L_k w^{-k-2} \sum_a a z^{-a-1} w^{a-1} + \frac{1}{12} \sum_a a(a-1)(a-2) z^{-a-1} w^{a-3}. \end{aligned}$$

So actually we found

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{m^3-m}{12}\delta_{m,-n}I_V.$$

The moral is: one tries to define $L = \frac{1}{2}h^2$ naively, expecting $[L_m, L_n] = (m-n)L_{m+n}$. Need to “normally order/normalise” to make L well-defined, now $L = \frac{1}{2} :hh:$. The “cost” is that the expected relation $[L_m, L_n] = (m-n)L_{m+n}$ gets altered by an “anomaly” $\frac{m^3-m}{12}I_V$.

Right, so we showed how to compute this using Bochner’s formula, but there’s actually another way using λ -bracket:

Theorem 12.4. *Let V be a vertex algebra, $a, b, c \in V$. Then*

$$(12.4.1) \quad [Ta_\lambda b] = -\lambda[a_\lambda b]$$

$$(12.4.2) \quad [a_\lambda Tb] = (T + \lambda)[a_\lambda b]$$

$$(12.4.3) \quad T(:ab:) = :Ta)b: + :a(Tb):$$

$$(12.4.4) \quad [b_\lambda a] = -[a_{-\lambda-T}b]$$

$$(12.4.5) \quad [a_\lambda :bc:] = :[a_\lambda b]c: + :b[a_\lambda c]: + \int_0^\lambda [[a_\lambda b]_\mu c] d\mu$$

$$(12.4.6) \quad [a_\lambda [b_\mu c]] - [b_\mu [a_\lambda c]] = [[a_\lambda b]_{\lambda+\mu} c].$$

Proof. For the first one use $\partial_z \delta(z, w) = -\partial_w \delta(z, w)$.

For the last one notice that

$$-[a_{-\lambda-T}b] = \sum_j \frac{1}{j!} (-\lambda - T)^j (a_{(j)}b)$$

$$\text{if } [a_\lambda b] = \sum_j \frac{\lambda^j}{j!} b.$$

□

Remark 12.5. By noting that $[a_\lambda b] = \text{Res}_z e^{\lambda z} Y(a, z)b$, these identities can be proved pretty efficiently.

That is,

Exercise 12.6. Use the fifth equation to prove what we proved with Bochner's formula.

And another one:

Exercise 12.7. Let $V = H = \mathbb{C}[x_1, x_2, \dots]$ again. Define $B = \frac{1}{2} :hh: + \beta Th$ with $\beta \in \mathbb{C}$. Confirm that

$$[B_\lambda B] = TB + 2\lambda B = \frac{1}{12}(1 - 12\beta^2)|0\rangle.$$

Definition 12.8. The *Virasoro Lie algebra* is

$$\text{Vir} = \bigoplus_{n \in \mathbb{Z}} \mathbb{C}L_n \oplus \mathbb{C}C$$

with Lie bracket

$$[L_m, L_n] = (m - n)L_{m+n} + \frac{m^3 - m}{12}\delta_{m, -n}C, \quad [C, \text{Vir}] = 0.$$

By the discussion above (all we've said so far!), there is a representation ρ of Vir in H given by

$$\rho(\underbrace{L_n}_{\substack{\text{as an abstract} \\ \text{object in Vir}}}) = \underbrace{L_n}_{\substack{\text{as a complicated} \\ \text{operator}}}, \quad \rho(C) = I_H.$$

Also H carries a representation ρ_β of Vir,

$$\rho_\beta(L_n) = B_n, \quad \rho_\beta(C) = (1 - 12\beta^2)I_V$$

If (M, ρ) is a representation of Vir, in which $\rho(C) = c \cdot I_M$ for some scalar c , we say M has a *central charge* c .

Looks like we have a second example of a vertex algebra. Consider $V = H$, $|0\rangle = 1$, $T = T$ from before, but this time put $F = \{B(z)\}$. Then all the axioms of the first definition of vertex algebra are satisfied, except the second: $B_{(n_1)}B_{(n_2)} \dots B_{(n_s)}|0\rangle$ might not span all of $V = H$. So take $V = \text{span}\{\text{these monomials}\} \subset H$. One can check that $T(V) \subset V$, and $B(z)$ is a quantum field on V . So $(V, 1, T, \{B(z)\})$ is a vertex algebra, called the *Virasoro vertex algebra*.

13. THE CHARGED FREE FERMIONS: A VERTEX SUPERALGEBRA

Also known as:

- The Clifford (vertex) algebra.
- The Dirac sea.
- The bc system.

First we consider a vector superspace with basis φ and φ^*

$$\mathfrak{a} = \mathbb{C}\varphi + \mathbb{C}\varphi^*$$

(where φ and φ^* are odd, so this is a O,2-dimensional superspace).

(A vector superspace is a $\mathbb{Z}/2$ -graded vector space, i.e. a vector space split in two pieces, $V = V_0 \oplus V_1$, which we call even and odd.)

We consider a bilinear form

$$\begin{aligned} \langle \cdot, \cdot \rangle : \mathfrak{a} \times \mathfrak{a} &\longrightarrow \mathbb{C} \\ \langle \varphi^*, \varphi \rangle &= 1 \\ \langle \varphi, \varphi^* \rangle &= 1 \text{ (why? see below)} \end{aligned}$$

Now consider

$$\tilde{\mathfrak{a}} = \underbrace{t^{1/2} \mathfrak{a}[t, t^{-1}]}_{\text{odd}} \oplus \underbrace{\mathbb{C}K}_{\text{even}}.$$

with Lie bracket

$$\begin{aligned} [at^m, b^n] &= \delta_{m, -n} \langle a, b \rangle K, \\ [K, \tilde{\mathfrak{a}}] &= 0. \end{aligned}$$

(Notice the small difference from the Heisenberg case: there is no m before the δ !)

To see why $\langle \varphi, \varphi^* \rangle = 1$, notice that in a Lie superalgebra we always want $[x, y] = (-1)(-1)^{p(x)p(y)}[y, x]$ where p denotes the parity. Since both at^m and bt^n are odd, we have $[at^m, bt^n] = [bt^n, at^m]$. Then apply the definition of bracket.

Since a, b odd here, we say $\langle a, b \rangle = \langle b, a \rangle$ means $\langle \cdot, \cdot \rangle$ is skew-super symmetric.

Definition 13.1. A bilinear form

$$(\cdot, \cdot) : U \times U \rightarrow \mathbb{C}$$

is *supersymmetric* if $(b, a) = (-1)^{p(a)p(b)}(a, b)$, and *skew-supersymmetric* if $(b, a) = -(-1)^{p(a)p(b)}(a, b)$.

Exercise 13.2. Invent the definition of Lie superalgebra, and confirm that if $U = U_0 \oplus U_1$ is a vector superspace, $\text{End}(U)$ is a Lie superalgebra with $[X, Y] := XY - (-1)^{p(X)p(Y)}YX$.

Let's build a Fock-type representation of $\tilde{\mathfrak{a}}$.

Construction 1.

$$\tilde{\mathfrak{a}}_+ = \bigoplus_{n>0} t^n \mathfrak{a} \oplus \mathbb{C}K, \quad (n \in \frac{1}{2} + \mathbb{Z}, \text{ recall})$$

$\tilde{\mathfrak{a}}_+ \subset \tilde{\mathfrak{a}}$ is a superalgebra.

Consider $U(\tilde{\mathfrak{a}})$ and $U(\tilde{\mathfrak{a}}) \subset U(\tilde{\mathfrak{a}})$.

Recall that the PWB theorem says that $U(\mathfrak{g})$ “looks like” polynomials on $\mathfrak{g}$.

When $\mathfrak{g}$ is pure even, this says: take $\{a_1, a_1, \dots\}$ a basis of $\mathfrak{g}$, then

$$\{a_{i_1}^{m_1}, a_{i_2}^{m_2} \dots a_{i_s}^{m_s} : i_1 \leq i_2 \leq \dots \leq i_s, m_j \geq 1\}.$$

For $\mathfrak{g} = \mathfrak{g}_0 \oplus \mathfrak{g}_1$ super Lie algebra, the statement is similar, only now, we take each a^i homogeneous ($\in \mathfrak{g}_0$ or $\mathfrak{g}_1$):

$$\{a_{i_1}^{m_1} a_{i_2}^{m_2} \dots a_{i_s}^{m_s}, i_1 \leq i_2 \leq \dots \leq i_s, m_j \geq 1, \text{ if } a_{i_j} \text{ odd, } m_j = 1\}.$$

The point is that if we put an odd guy twice it becomes one.

Why? Because in $U(\mathfrak{g})$, $X \in \mathfrak{g}_1$ should satisfy $XX + XX = [X, X] = 0$. So $X^2 = 0$. So we only allow odd basis vectors to appear 0 or 1 times each.

Define $F = U(\tilde{\mathfrak{a}}) \otimes_{U(\tilde{\mathfrak{a}}_+)} \mathbb{C}|0\rangle$, where $at^n \cdot |0\rangle$ for all $n > \frac{1}{2}$, and $K|0\rangle = |0\rangle$.

[Picture of F]

We can introduce a bi-grading of F :

$$F = \bigoplus_{\substack{c \in \mathbb{Z} \\ \Delta \in \frac{1}{2}\mathbb{Z}_+}} F_{\Delta}^c$$

where, for a monomial

$$\varphi_{-n}^* \varphi_{-n_2}^* \dots \varphi_{-n_s}^* \varphi_{-m}, \varphi_{-m_2} \dots \varphi_{-m_t} |0\rangle$$

we define its *energy* by $\Delta = \sum_i n_i + \sum_j m_j$ and its *charge* by $c = s - t$.

The point is that F is an infinite-dimensional vector space, but we get a reasonable mental picture of what it looks like.

What is $\dim F_\Delta^c$? Let's write...

So we have a vector superspace F , and an action on F of $\tilde{\mathfrak{a}}$. Let's build some quantum fields!

There are going to be two: $\varphi(z)$ and $\varphi^*(z)$.

$$\varphi(z) = \sum_{n \in \frac{1}{2} + \mathbb{Z}} \varphi_n z^{-n - \frac{1}{2}}$$

(Notice the exponent on z is an integer!)

$$\varphi^*(z) = \sum_{n \in \frac{1}{2} + \mathbb{Z}} \varphi_n^* z^{-n - \frac{1}{2}}.$$

Now let's compute $[\varphi^*(z), \varphi(w)]$.

$$\begin{aligned} [\varphi^*(z), \varphi(w)] &= \sum_{m, n \in \frac{1}{2} + \mathbb{Z}} [\varphi_m^*, \varphi_n] z^{-m - \frac{1}{2}} w^{-n - \frac{1}{2}} \\ &= \sum_{n \in \frac{1}{2} + \mathbb{Z}} K z^{n - \frac{1}{2}} w^{-n - \frac{1}{2}} \\ &= \sum_{\substack{r \in \mathbb{Z} \\ r = n - \frac{1}{2}}} z^r w^{-r - 1} \\ &= \delta(z, w). \end{aligned}$$

So $F, |0\rangle, \mathcal{F} = \{\varphi(z), \varphi^*(z)\}$, $T : F \rightarrow F$, is a vertex superalgebra. Here $T : F \rightarrow F$ has to satisfy $T|0\rangle = 0$ and $[T, \varphi(z)] = \partial_z \varphi(z)$,

$$\begin{aligned} \sum [T, \varphi_n] z^{-n - \frac{1}{2}} &= \sum \left(-n - \frac{1}{2}\right) \varphi_n z^{-n - \frac{3}{2}} \\ &= \sum \left(-k + \frac{1}{2}\right) \varphi_{k-1} z^{-k - \frac{1}{2}} \end{aligned}$$

after putting $n = k - 1$. So

$$[T, \varphi_n] = -\left(n - \frac{1}{2}\right) \varphi_{n-1}.$$

Inductively, this determines how T acts on F .

In λ -bracket notation, this just says

$$[\varphi_\lambda^* \varphi] = |0\rangle.$$

Skew-symmetry says $[b_\lambda a] = -(-1)^{p(a)p(b)}[a_{-\lambda-T}b]$.

For example,

$$\begin{aligned} [h_\lambda h] &= \lambda |0\rangle, \\ [h_\lambda h] &= -[h_{-\lambda-T}h] = -(-\lambda - T)|0\rangle \\ &= T\lambda |0\rangle. \end{aligned}$$

In this case $[\varphi_\lambda \varphi^*] = +|0\rangle$ too.

Notice $\sum \varphi_n z^{-n - \frac{1}{2}} = \sum \varphi_{(m)} z^{-m - 1}$.

So

$$\begin{aligned}\varphi_{(m)} &= \varphi_{m-\frac{1}{2}} \text{ and} \\ \varphi_{(m)}^* &= \varphi_{m-\frac{1}{2}}^*.\end{aligned}$$

So

$$\begin{aligned}:\varphi^*\varphi: &= \varphi_{(-1)}^*\varphi_{(-1)}|0\rangle \\ &= \varphi_{-\frac{1}{2}}^*\varphi_{-\frac{1}{2}}|0\rangle.\end{aligned}$$

I.e.

$$\varphi = \varphi_{(-1)}|0\rangle = \varphi_{-\frac{1}{2}}|0\rangle$$

Now we can put the former picture but with the associated operators using the second definition of vertex algebra: [Picture]

Let's compute some brackets: [lots of computations]

So,

$$[\alpha_\lambda \alpha] = \lambda|0\rangle.$$

Just like $[h_\lambda h] = \lambda|0\rangle$. This relation is the basis of the “boson-fermion correspondence”, i.e. consider the subspace $U \subset F$ defined as

$$U = \text{span}\{\alpha_{n_1}\alpha_{n_2}, \dots, \alpha_{n_s}|0\rangle : n_i \in \mathbb{Z}\}.$$

We claim that since $[T, \alpha(z)] = \partial_z \alpha(z)$, $[T, \alpha_n] = -n\alpha_{n-1}$.

So $T(U) \subset U$. Put $\mathcal{F}_1 = \{\alpha(z)\}$. Then $(U, |0\rangle, T|_U, \{\alpha(z)\})$ is a vertex algebra (an honest one, not super).

Proposition 13.3. *U is isomorphic to the Heisenberg vertex algebra. (In particular, $U \simeq \mathbb{C}[x_1, x_2, \dots]$ as a vector space.)*

Proof. Next time. □

Since $[\alpha_0, \varphi_n] = -\varphi_n$, $[\alpha_0, \varphi_m^*] = \varphi_m^*$, and $\alpha_0|0\rangle = 0$, in fact the charge of a monomial is exactly the eigenvalue of α_0 acting on it.

$$\begin{aligned}\alpha_0 \cdot \varphi_{-m}^* \varphi_{-m_2}^* \varphi_{-n}|0\rangle &= [\alpha_0, \varphi_{-m}^*, \varphi_{-m_2}^*, \varphi_{-n}]|0\rangle \\ &= \underbrace{(+1 + 1 - 1)}_{\text{charge}} \varphi_{-m_1}^* \varphi_{-m_2} \varphi_{-n}|0\rangle.\end{aligned}$$

From $[\alpha_\lambda \alpha] = \lambda|0\rangle$, we get

$$[\alpha_m, \alpha_n] = m\delta_{m, -n}I_F \text{ (as for } h)$$

So $[\alpha_0, \alpha_n] = 0$ for all $n \in \mathbb{Z}$. Thus $\alpha_0|_U = 0$ and thus $U \subset F^{(0)}$.

Proposition 13.4. $U = F^{(0)}$.

Proof. Next time. □

Let's take two copies of F , i.e. $F_2 = F \otimes F$ with fields $\varphi^1(z), \varphi^2(z), \varphi^{1*}(z), \varphi^{2*}(z)$

$$[\varphi_\lambda^{i*} \varphi^j] = \delta_{ij}.$$

Basis of F_2 is

$$\varphi_{-n_1}^1 \cdots \varphi_{-n_2}^1 \varphi_{-m_1}^2 \cdots \varphi_{-m_t}^2 \varphi_{-p_1}^{*-1} \cdots \varphi_{-p_u}^{*-1} \varphi_{-q_1}^{*2} \cdots \varphi_{-q_v}^{*2} |0\rangle.$$

Let $\alpha^{ij} = : \varphi^{j*} \varphi^i :$.

$$[\alpha_\lambda^{ij} \alpha^{k\ell}] = ?$$

(certainly $[\alpha_\lambda^{ii} \alpha^{ii}] = \lambda |0\rangle$.) In general

$$[\varphi_\lambda^k \alpha^{ij}] = \text{computation} = \delta_{ki} \varphi^j.$$

Similarly

$$[\varphi_\lambda^{k*} \alpha^{ij}] = -\delta_{kj} \varphi^{i*}.$$

$$[\alpha_\lambda^{ij} \varphi^k] = -\delta_{ik} \varphi^j$$

$$[\alpha_\lambda^{ij} \varphi^{k*}] = +\delta_{jk} \varphi^{i*}.$$

We would like to consider

$$F_n = F^{\otimes n} = F \otimes \dots \otimes F$$

where on the i -th factor we have φ_i, φ_i^* .

Then we have relations among the generators:

$$[\varphi_\lambda^i \varphi_j^*] = [\varphi_j^* \lambda \varphi_i] = \delta_{ij} |0\rangle$$

Today let's define

$$\alpha_{ij} = : \varphi_i \varphi_j^* :$$

We notice that

- the φ_i behave like e_i ,
- the α_{ij} behave like E_{ij} ,
- the φ_i^* behave like e_i^* .

[Computations of some λ -brackets]

Recall, the E_{ij} span a Lie algebra: $\mathfrak{gl}_n$. It has a representation on $\mathbb{C}^n = \langle e_1, \dots, e_n \rangle$ by $E_{ij} e_k = \delta_{ik} e_j$, and on $(\mathbb{C}^n)^*$ we have

[computations]

$$E_{ij} e_k^* = -\delta_{ki} e_j^*.$$

Let $A \in \mathfrak{gl}_n$. Let's write

$$\alpha^A = \sum_{i,j} a_{ij} \alpha_{ij} \in F_n$$

We'd like to compute $[\alpha_\lambda^A \alpha^B]$.

Some computations done in lecture are the proof of

Theorem 13.5. For $\alpha_{ij} = : \varphi_i \varphi_j^* : \in F_n$, and α^A as above,

$$[\alpha_\lambda^A \alpha^B] = \alpha^{[A,B]} + \lambda \text{Tr}(AB) |0\rangle.$$

Which says that α^A behaves like matrices, but with that correction term.

Nex, let $\mathfrak{g} \subset \mathfrak{gl}_n$ be a Lie subalgebra. For $A, B \in \mathfrak{g}$, $[A, B] \in \mathfrak{g}$ also, and so the set

$$\mathcal{F} = \{ \alpha^A : A \in \mathfrak{g} \cup \{0\} \}$$

is "closed under λ -brackets".

More precisely, for

$$F_n \supset V = \text{span} \{ \alpha_{(n_1)}^{A_1} \dots \alpha_{(n_s)}^{A_s} : n_i \in \mathbb{Z}, A_i \in \mathfrak{g} \},$$

with $|0\rangle = |0\rangle$, $T = T$ and $\mathcal{F} = \{ \alpha^A(z) : A \in \mathfrak{g} \} \cup \{I\}$ is a vertex algebra.

14. UNIVERSAL AFFINE VERTEX ALGEBRA

Recall the construction of the affine vertex algebra, Definition 2.3.

Let (M, ρ) be a representation of $\hat{\mathfrak{g}}$.

For $a \in \mathfrak{g}$, define

$$(14.0.1) \quad a^M(z) = \sum_{n \in \mathbb{Z}} \rho(at^n) z^{-n-1} \in \text{End}(M)[[z^{\pm 1}]].$$

Definition 14.1. A *smooth $\hat{\mathfrak{g}}$ -module* is a $\hat{\mathfrak{g}}$ -module (M, ρ) such that for each $m \in M$ there is $N \in \mathbb{Z}$ such that $\rho(at^n)m = 0$ for all $a \in \mathfrak{g}$ and $n \geq N$.

Notice that on a smooth module M , the fields $a^M(z)$ are quantum fields.

We may calculate:

$$\begin{aligned} [a^m(z), b^m(w)] &= \sum_{m,n} [\rho(at^m), \rho(bt^n)] z^{-m-1} w^{-n-1} \\ &= \sum_{m,n} z^{-m-1} w^{-n-1} ([a, b]t^{m+n} + m\delta_{m,-n}(a, b)\rho(K)) \\ &= \sum_{m,n} ([a, b]t^{m+n}w^{-(m+n)-1}) z^{-m-1} w^m + \sum_{m,n} z^{-m-1} w^{-n-1} \delta_{m,-n}(a, b)\rho(K) \\ &= [a, b]^m(w)\delta(z, w) + (a, b)\partial_w\delta(z, w)\rho(K). \end{aligned}$$

Now suppose the K acts by some constant, i.e. $\rho(K) = kI_M$. Then our quantum fields $a^M(z)$ are mutually local, with exponent 2, i.e. $(z - w)^2[a^M(z), b^M(w)] = 0$.

As a special case, we may take M to be a sort of “Fock space”:

$$\hat{\mathfrak{g}}_+ = \mathfrak{g}[t] \oplus \mathbb{C}K \subset \hat{\mathfrak{g}}.$$

Let $\hat{\mathfrak{g}}_+$ act on $\mathbb{C}|0\rangle$ by

$$\begin{aligned} at^m|0\rangle &= 0 \quad \forall m \geq 0, \\ K \cdot |0\rangle &= k|0\rangle \quad (k \in \mathbb{C} \text{ called } level). \end{aligned}$$

Consider

$$V^k(\mathfrak{g}) = U(\hat{\mathfrak{g}}) \otimes_{U(\hat{\mathfrak{g}}_+)} \mathbb{C}|0\rangle.$$

Remark 14.2. $H = \mathbb{C}[x_1, x_2, \dots]$ is an example of this construction with $\mathfrak{g}$ one-dimensional and $(h, h) = 1$.

Exercise 14.3. $V^k(\mathfrak{g})$ is a smooth $\hat{\mathfrak{g}}$ -module, in which $K \mapsto k\text{Id}$.

Proof. To prove $V^k(\mathfrak{g})$ is smooth we need to show that for every formal power series of endomorphisms of $V^k(\mathfrak{g})$ of the form Equation 14.0.1, the coefficient operators vanish at every v for sufficiently large n . But to define these power series we need to find a copy of $\hat{\mathfrak{g}}$ inside $V^k(\mathfrak{g})$. $\square$

The set $\mathcal{F} = \{a(z) = \sum_{n \in \mathbb{Z}} (at)^n z^{-n-1} : a \in \mathfrak{g}\}$ are mutually local quantum fields.

Define $T : V^k(\mathfrak{g}) \curvearrowright V^k(\mathfrak{g})$ by the relation $T|0\rangle = 0$ and $[T, at^m] = -mat^{m-1}$. $V^k(\mathfrak{g})$ is a vertex algebra called the *universal affine vertex algebra* of level k associated with $\mathfrak{g}$.

[Picture of $V^k(\mathfrak{g})$ for $\mathfrak{g} = \mathfrak{sl}_2$.]

We have a bigrading, vertical grading Δ (energy) is

$$\Delta(a_{(-n_1)}^1 a_{(-n_2)}^2 \dots a_{(-n_s)}^s |0\rangle) = n_1 + n_2 + \dots + n_s.$$

Notice that for $a \in \mathfrak{g}$, we have

$$\begin{aligned} \mathcal{F} \ni a(z) &\xrightarrow{s} a(z)|0\rangle|_{z=0} \\ &= \sum_m at^m|0\rangle z^{-m-1} \Big|_{z=0} \\ &= (at^{-1})|0\rangle + (at^{-2})z|0\rangle + \dots \\ &= at^{-1}|0\rangle. \end{aligned}$$

Thus we can think of $\{at^{-1}|0\rangle : a \in \mathfrak{g}\} \subset V^k(\mathfrak{g})$ as a “copy” of $\mathfrak{g}$ inside $V^k(\mathfrak{g})$. Let's **abuse notation** and write a for $at^{-1}|0\rangle$.

Then $at^{-2}|0\rangle = Ta$, also $(at^{-1})(bt^{-1})|0\rangle = :ab:$, etc. $e = et^{-1}|0\rangle = e_{(-1)}|0\rangle$, $\Delta(e) = 1$.

In fact $\Delta(a) = 1$ for all $a \in \mathfrak{g}$, $\Delta(Ta) = 2$, $\Delta(:ab:) = 2$, etc.

In this example, i.e. $\mathfrak{g} = \mathfrak{sl}_2$, $\mathfrak{sl}_2$ is itself a $\mathbb{Z}$ -graded Lie algebra,

$$\begin{aligned} w(E) &= 2 & [H, E] &= 2E \\ W(H) &= 0 & [H, H] &= 0 \\ w(F) &= -2 & [H, F] &= -2F. \end{aligned}$$

(That is, $\mathfrak{g} = \bigoplus_{n \in \mathbb{Z}} \mathfrak{g}_n$, $[\mathfrak{g}_m, \mathfrak{g}_n] = \mathfrak{g}_{m+n}$.) w = eigenvalue of $\text{ad}(H)$. This induces a $\mathbb{Z}$ -grading on $V^k(\mathfrak{sl}_2)$, compatible, i.e.

$$w(a_{(-n_1)}^1 \dots a_{(-n_s)}^s)|0\rangle = 2\#(E) - 2\#(F).$$

As for the character,

$$\begin{aligned} \chi(q, u) &= \sum_{\Delta, w} \dim V^k(\mathfrak{g})_{\Delta}^w q^{\Delta} u^w \\ &= 1 + q(u^2 + 1 + u^{-2}) + q^2(u^4 + 2u^2 + 2 + 2u^{-2} + u^{-4} + \dots \end{aligned}$$

Then by the generating function argument we have discussed before,

$$\chi(q, u) = \prod_{n=1}^{\infty} \frac{1}{(1 - u^2 q^n)(1 - q^n)(1 - u^{-2} q^n)}.$$

Can discard u to get vertical grading

$$\chi(q) = \prod_{n=1}^{\infty} \frac{1}{(1 - q^n)^3}.$$

[Missing: we changed a little the definition of $V^k(M)$; now it also depends on a bilinear form B .]

Here's another example:

Example 14.4. $\mathfrak{g} = \mathfrak{gl}_n$, with $B(X, Y) = \text{Tr}(XY)$, and $k = 1$.

Remark 14.5. In general $V^1(\mathfrak{g}, kB) = V^k(\mathfrak{g}, B)$. If $\mathfrak{g}$ is finite-dimensional simple, we typically take $B(a, b) = \frac{1}{2\hbar} \kappa(a, b)$ where κ is the Killing form. (This B has the property that $B(\theta, \theta) = 2$ for **long** roots $\theta \in \Delta \subset \mathfrak{h}^*$.) (In fact, in a simple f.d. there's all invariant forms are proportional.) Then we write this vertex algebra as $V^k(\mathfrak{g})$ without specifying B .

So perhaps that's why last time we didn't put B .

Back to the Fermion algebra F_n , and now denoting J instead of α , we built fields $\{J^A : A \in \mathfrak{gl}_n\}$ with λ -bracket $[J_\lambda^A J^B] = J^{[A,B]} + \lambda \text{Tr}(AB)|0\rangle$. This suggests a relation with $V^1(\mathfrak{gl}_n, B)$, where $B(X, Y) = \text{Tr}(X, Y)$.

Consider $\mathcal{F} = \{J^A(z) : A \in \mathfrak{gl}_n\} \subset \mathbb{F}_{F_n} \cong F_n$. Let $V = \text{span}\{J_{(n_1)}^{A_1} \dots J_{(n_s)}^{A_s}|0\rangle : A_i \in \mathfrak{gl}_n, n_i \in \mathbb{Z}\} \subset F_n$.

Then $(V, \mathcal{F}, T|_V, |0\rangle)$ is a vertex algebra. $V \subset F_n$. Notice that $V \neq F^n$ since V consists only of even fields, indeed, $V = \langle : \varphi_i \varphi_j^* : \rangle$.

Here $[T, \varphi_{(n)}] = -n\varphi_{(n-1)}$.

$$T(\varphi_{(n_1)}^{i_1} \dots \varphi_{(n_s)}^{i_s}|0\rangle) = - \sum_{j=1}^s n_j \varphi_{(n_1)}^{i_1} \dots v o_{(n_j-1)}^{i_j} \dots \varphi_{(n_s)}^{i_s}|0\rangle.$$

Does $V = F_n^{(0)}$ then?

Let's examine $n = 1$ first. $\mathfrak{g} = \mathbb{C}1$, $B(1, 1) = 1$, $k = 1$. We just remarked that $V^1(\mathfrak{g}, B) = H$ is Heisenberg. And $F_1 = F = \langle \varphi, \varphi^* \rangle$ which we have drawn previously. We also computed the character as an infinite product.

We have $F^{(0)} \ni J = : \varphi \varphi^* :$ with λ -bracket relation $[J_\lambda J] = \lambda|0\rangle$. Recall $J(z) = \sum_{n \in \mathbb{Z}} J_{(n)} z^{-n-1} = \sum_{n \in \mathbb{Z}} J_n z^{-n-1}$. The λ -bracket relation implies

$$\begin{aligned} [J(z), J(w)] &= \partial_w \delta(z, w) I \\ \implies [J_m, J_n] &= m \delta_{m, -n} I. \end{aligned}$$

This implies that $F^{(0)}$ is a representation of the oscillator Lie algebra $\hat{\mathfrak{a}} = \bigoplus_{m \in \mathbb{Z}} \mathbb{C} h_m \oplus \mathbb{C} K$, in which $h_m \mapsto J_m$, $k \mapsto I$ and $h_m|0\rangle$ for all $m \geq 0$.

Proposition 14.6. *As representations of $\mathfrak{a}$, $F^{(0)} \simeq H = \mathbb{C}[x_1, x_2, \dots]$.*

Proof. Using $H \simeq U(\hat{\mathfrak{a}}) \otimes_{U(\hat{\mathfrak{a}}_+)} \mathbb{C}1$, by its universal property, there exists a morphism of $\hat{\mathfrak{a}}$ -representations $f : H \rightarrow F^{(0)}$, such that $f(1) = |0\rangle$.

By Exercise 9.4 we know H is irreducible, so $f : H \rightarrow F^{(0)}$ is injective. Is it an isomorphism? Consider $\tilde{F} = F^{(0)}/f(H)$. We want to prove $\tilde{F} = \{0\}$. The trick is to consider $L = \frac{1}{2} : JJ :$.

We have seen that

$$F^{(m)} = \{v \in F : J_0 v = mv\}$$

(since $[J_\lambda \varphi] = -\varphi$ and $[J_\lambda \varphi^*] = +\varphi^*$, $[J_0, \varphi_n] = -\varphi_n$).

Write $L(z) = \sum_{n \in \mathbb{Z}} L_n z^{-n-2}$, then $L_0 = \frac{1}{2} J_0^2 + \sum_{m>0} J_{-m} J_m$. This implies (for example) that $L_0|0\rangle = 0$.

One may compute

$$\begin{aligned} [L_\lambda \varphi] &= T\varphi + \frac{1}{2} \lambda \varphi \\ [L_\lambda \varphi^*] &= T\varphi^* + \frac{1}{2} \lambda \varphi^*. \end{aligned}$$

If you expand these in terms of coefficients, you find $[L_0, \varphi_n] = -n\varphi_n$, $[L_0, \varphi_n^*] = -n\varphi_n^*$.

From these relations we conclude that $\Delta(\text{monomial}) = L_0$ -eigenvalue (monomial), i.e., the (charge, energy)-grading is the eigenspace grading by (J_0, L_0) . Suppose $\tilde{F} \neq 0$. Then $\exists \bar{v} \in \tilde{F}$ for which $J_m \bar{v} = 0$ for all $m > 0$. Indeed, by Lemma

(?) the $\mathbb{Z}_+$ -grading of $F^{(0)}$ is inherited by $\tilde{F}$. Since $|0\rangle \in f(H)$, $\tilde{F} = \bigoplus_{\Delta \geq N} \tilde{F}_\Delta$ for some $N > 0$. Let $\bar{v} \in \tilde{F}_N$, $\bar{v} \neq 0$. For $m > 0$, $J_m \bar{v} \in \tilde{F}_{N-m} = 0$.

Key point: $J_0 \bar{v}$, because $\tilde{F}$ is quotient of $F^{(0)}$. Hence $L_0 \bar{v} = 0$. But $L_0 = \Delta$, and we know the only element of $F^{(0)}$ with $\Delta = 0$ is $|0\rangle$. This contradiction implies $\tilde{F} = 0$, hence $F^{(0)} = f(H) \simeq H$. $\square$

In the process we saw that Δ coincides with L_0 , where $L = \frac{1}{2} : JJ :$, and $J = f(h)$. So consider $L = \frac{1}{2} : hh_i :$.

We have $[L_\lambda h] = Th + \lambda h$ so $[L_0, h_n] = -nh_n$ in particular, and

$$L_0(h_{-n_1}, h_{-n_2}, \dots, h_{-n_s} | 0\rangle = L_0(x_{n_1} x_{n_2} \dots x_{n_s} | 0\rangle) = \left(\sum_j n_j x_{n_1} \dots x_{n_s} | 0\rangle \right)$$

Using this we obtain

$$\chi_{F^{(0)}}(q) = \sum_{\Delta} \dim(F_{\Delta}^{(0)}) q^{\Delta} = \prod_{m=1}^{\infty} \frac{1}{1 - q^m}.$$

In $F^{(m)}$, let us denote

$$|m\rangle = \begin{cases} \varphi_{-1/2} \varphi_{-3/2} \varphi_{-5/2} \dots \varphi_{-\frac{2(-m)-1}{2}} | \rangle & \text{if } m < 0 \\ \varphi_{-1/2}^* \varphi_{3/2}^* \dots \varphi_{-\frac{2m-1}{2}}^* & \text{if } m > 0 \end{cases}$$

We observe that $\Delta(|m\rangle) = \frac{m^2}{2}$ and $F_{m^2/2}^{(m)} = \mathbb{C}|m\rangle$.

Proposition 14.7. *As an $\hat{\mathfrak{a}}$ -module, $F^{(m)} \simeq H^m$, and is irreducible.*

Proof. Same as above. $\square$

This gives a formula

$$\chi_{F^{(m)}}(q) = \sum_{\Delta} \dim F_{\Delta}^{(m)} q^{\Delta} = q^{m/2} \prod_{k=1}^{\infty} \frac{1}{1 - q^k}.$$

As a corollary, we obtain again Jacobi's triple product formula. (Again!)

$$\prod_{m=1}^{\infty} (1 + y q^{m-1/2}) (1 + y^{-1} q^{m-1/2}) = \prod_{k=1}^{\infty} \frac{1}{1 - q^k} \left(\sum_{n \in \mathbb{Z}} y^n q^{n^2/2} \right).$$

15. ANOTHER PRESENTATION THE CHARGED FREE FERMIONS

The idea is to consider $X = \bigoplus_{i \in \mathbb{Z}} \mathbb{C} e_i$, a countably infinite dimensional space, and define

$$(15.0.1) \quad \Lambda^{\infty/2} = \left\{ \begin{array}{l} \text{span of symbols of the form} \\ e_{i_0} \wedge e_{i_1} \wedge e_{i_2} \wedge e_{i_3} \wedge \dots \\ \text{where } \exists N \text{ s.t. } \forall n \geq N, i_{n+1} = i_n + 1 \end{array} \right\} / \left\{ \begin{array}{l} \text{relation that} \\ e_i \wedge e_j = -e_j \wedge e_i \\ \text{"wherever it occurs"} \end{array} \right\}.$$

That is, the indices can dance around as they like but at some point they will become consecutive. Clearly a basis of $\Lambda^{\infty/2}$ is given by those "semi-infinite monomials" for which $i_0 < i_1 < i_2 < \dots$

For any semi infinite monomial $\underline{e} = e_{i_0} \wedge e_{i_1} \wedge \dots$ there exists a number m such that $i_j = -m + j$ for all $j \gg 0$. This number is called the *charge* of $\underline{e}$.

We have a decomposition in vector spaces

$$\Lambda^{\infty/2} = \bigoplus_{m \in \mathbb{Z}} \Lambda^{\infty/2, (m)}$$

called *charge decomposition*.

Let's introduce some operators in $\Lambda^{\infty/2}$:

$$\begin{aligned} \varphi_{(n)} : \Lambda^{\infty/2} &\longrightarrow \Lambda^{\infty/2} \\ \varphi_{(n)}(\underline{e}) &= e_n \wedge \underline{e} \end{aligned}$$

i.e. we just put e_n at the beginning of the monomial.

Next

$$\begin{aligned} \varphi_{(n)}^* : \Lambda^{\infty/2} &\longrightarrow \Lambda^{\infty/2} \\ \varphi_{(n)}^* &= \sum_{k \geq 0} (-1)^k \delta_{n, i_k} e_{i_0} \wedge e_{i_1} \wedge \dots \wedge \underbrace{e_{i_k}}_{\text{remove this}} \wedge \dots \end{aligned}$$

so in analogy, we remove this term. The $(-1)^k$ accounts for moving around the unwanted term to the beginning of the monomial.

Exercise 15.1. Something like

$$\varphi_{(m)} \varphi_{(n)}^* + \varphi_{(n)}^* \varphi_{(m)} = \delta_{m, +n} I_{\Lambda^{\infty/2}}$$

Let's introduce some quantum fields now:

$$\varphi(w) = \sum_{n \in \mathbb{Z}} \varphi_{(n)} w^{-n-1}$$

which vanishes for very large n because we eventually get to the “consecutive” region, and

$$\varphi^*(w) = \sum_{n \in \mathbb{Z}} \varphi_{(n)}^* w^n$$

which vanishes for very negative n since the star operators are defined to give zero if the e_n is not found in the monomial.

Notice that the convention $\varphi^*(w) = \sum \varphi_{(n)}^* w^n$, instead of $\sum \varphi_{(n)}^* w^{-n}$ is so that $\varphi^*(w)$ is a quantum field (indeed, by our convention on how we write quantum fields, see Definition 9.7). The convention $\varphi^*(w) = \sum \varphi_{(n)}^* w^n$ instead of, say, $\sum \varphi_{(n)}^* w^{n+1}$ is so that the equation in Exercise 15.1 turns into a nice relation

$$[\varphi(z), \varphi^*(z)] = \delta(z, w) I,$$

(i.e. $[\varphi_\lambda \varphi^*] = |0\rangle$, just like in the charged fermions F^{ch} !).

Exercise 15.2. These fields will match with our previous φ, φ^* as

$$\begin{aligned} \sum \varphi_{(n)} z^{-n-1} &= \sum \varphi_n z^{-n-1/2} : \varphi_n = \varphi_{n-1/2} \\ \sum \varphi_{(n)}^* z^n &= \sum \varphi_n^* z^{-n-1/2} : \varphi_n^* = \varphi_i \end{aligned}$$

The space $\Lambda^{\infty/2}$ has a super-structure with parity $\Lambda^{\infty/2, (m)}$ = the parity of m . Setting $|0\rangle = e_0 \wedge e_1 \wedge e_2 \dots$, $\mathcal{F} = \{\varphi(z), \varphi^*(z)\}$, T what it needs to be, we get a vertex superalgebra. In fact,

$$F^{\text{ch}} = U(\hat{\mathfrak{a}}) \otimes_{U(\hat{\mathfrak{a}}_+)} \mathbb{C}|0\rangle \xrightarrow{\cong} \Lambda^{\infty/2}$$

is an isomorphism of vertex algebras. Indeed, if $n_j, m_k \geq 1$,

$$\varphi_{(n-1)}\varphi_{(-n_2)}\cdots\varphi_{(-n_s)}\varphi_{(m_1)}^*\cdots\varphi_{(m_t)}^*|0\rangle = \pm e_{-n_1} \wedge e_{-n_2} \wedge \cdots \wedge e_{-n_s} \wedge \underbrace{(e_0 \wedge e_1 \wedge \cdots)}_{\substack{\text{but with} \\ e_{m_1} \dots e_{m_t} \\ \text{missing}}}$$

But these two pictures are bases of F^{ch} and $\Lambda^{\infty/2}$, so we have our linear isomorphism $F^{\text{ch}} \rightarrow \Lambda^{\infty/2}$. Just need to check carefully φ and φ^* are compatible with it.

Pauli exclusion principle. In an ensemble state of fermions, two cannot occupy the same state, (so they are modelled by odd variables like “ e_n ” such that $e_n \wedge e_n = 0$.)

Electron waves should obey “quantized” wave-equation, which is something like

$$\begin{aligned} \partial_t^2 \psi &= (-\hbar^2 \nabla + m) \psi \\ \partial_t \psi &= \sqrt{-\hbar \nabla + m} \psi. \end{aligned}$$

A solution to this is, instead of

$$\sqrt{\partial_{xx} + m},$$

try to define

$$\sqrt{(\partial_{xx} + m)I_4}$$

where I_4 is the identity 4×4 matrix. Then the “square root” of $-\hbar^2 \nabla + m$ becomes a matrix valued differential operator, denoted sometimes as $\not{\partial} + \gamma$, where γ is an explicit 4×4 matrix. In $\partial_t \psi = \dots$, ψ has become a vector-valued function on space, with a splitting in spin up and spin down parts of the electron wavefunction, and also an “unphysical” part. The latter can be interpreted as the positrons.

Problem: the Dirac equation has positive and negative energy solutions.

Let’s examine the neutral part of $\Lambda^{\infty/2(0)}$. Consider the element

$$\underline{e} = e_{-4} \wedge e_{-2} \wedge e_{-1} \wedge e_0 \wedge e_1 \wedge e_4 \wedge e_5 \wedge \underbrace{e_7 \wedge e_8 \wedge e_9 \wedge \cdots}_{\text{consecutive part}}$$

The successive differences are

$$2 \quad 1 \quad 1 \quad 1 \quad 3 \quad 1 \quad 2 \quad 1 \quad 1 \quad \dots$$

Since the sequence will stabilize at 1, we might as well subtract 1 and obtain the finite sequence

$$(1 \quad 0 \quad 0 \quad 0 \quad 2 \quad 0 \quad 1)$$

In fact, you could reconstruct $\underline{e}$ from this list, and knowing that $\text{charge}(\underline{e}) = 0$.

Form the partial sums

$$1000201 \rightarrow 4333311$$

these numbers are non-increasing, so define uniquely a partition of some integer (in this case 18), and if we compare with

$$\underline{e} = \pm \varphi_{-4} \varphi_{-2} \varphi_{-1} \varphi_2^* \varphi_3^* \varphi_6^* |0\rangle$$

which has energy

$$\Delta = (4 - 1/2) + (2 - 1/2) + (1 - 1/2) + (2 + 1/2) + (3 + 1/2) + (6 + 1/2) = 18.$$

We denote by $\underline{e}_\lambda$ the semi-infinite monomial of charge 0 given by this procedure.

So we have a basis $\{\underline{e}_\lambda | \lambda \text{ integer partitions}\}$ of $F^{(0)} \simeq H = \mathbb{C}[x_1, x_2, x_3, \dots]$, which also has the basis $\{\underline{x}_\lambda = x_1^{\lambda_1} \dots x_5^{\lambda_5} | \lambda \text{ integer partitions}\}$ where $\lambda = (1^{\lambda_1}, 2^{\lambda_2}, \dots, s^{\lambda_s})$, that is, 1 appears λ_1 times and so on. We do **not** have $\underline{e}_\lambda = \underline{x}_\lambda!$

16. SCHUR POLYNOMIALS

The Schur polynomials are given by the exponential generating function of $\sum z^k x_k$. More precisely,

Definition 16.1. We set $S_k(\underline{x})$ by

$$\sum_{k \geq 0} z^k S_k(\underline{x}) = \exp \left(\sum_{k \geq 1} z^k x_k \right).$$

For a partition $\lambda = (1^{\lambda_1} 2^{\lambda_2} \dots k^{\lambda_k})$, define

$$S_\lambda(\underline{x}) = \det \begin{pmatrix} S_{\lambda_1} & S_{\lambda_1+1} & \cdots & S_{\lambda_1+k-1} \\ S_{\lambda_2-1} & S_{\lambda_2} & \cdots & \\ \vdots & & & \vdots \\ S_{\lambda_k-k+1} & & & S_{\lambda_k} \end{pmatrix}$$

Recall we defined $X = \bigoplus_{i \in \mathbb{Z}} \mathbb{C}e_i$ and $\Lambda^{\infty/2} = \langle e_{i_0} \wedge \dots \rangle$. Let $g : X \rightarrow X$ be an invertible endomorphism. We would like to define an endomorphism

$$R(g) : \Lambda^{\infty/2} \longrightarrow \Lambda^{\infty/2}$$

$$R(g)(e_{i_0} \wedge e_{i_1} \wedge \dots) = (ge_{i_0}) \wedge (ge_{i_1}) \wedge \dots$$

For some $g, \varepsilon g, g(e_n) = e_{-n}$, $R(g)$ does not make sense (or does not send $\Lambda^{\infty/2} \rightarrow \Lambda^{\infty/2}$).

As a note, if X were finite-dimensional, say $X = \langle e_1, \dots, e_n \rangle$, then $R(g) : \Lambda^{k+1} X \rightarrow \Lambda^{k+1} X$ has matrix entry

$$(e_{i_0} \wedge e_{i_1} \wedge \dots \wedge e_{i_k}) \rightarrow (e_{j_0} \wedge e_{j_1} \wedge \dots \wedge e_{j_k}),$$

(with strictly increasing indices on both sides, up to a sign maybe) given by the determinant of the $(k+1) \times (k+1)$ matrix given by selecting the rows $i_0, i_1, \dots, i_k$, and the columns $j_0, j_1, \dots, j_k$ of the matrix of g .

For our X and $\Lambda^{\infty/2}$, $R(g)$ makes sense if g is such that

$$g(e_i) - e_i \in \text{span}\{e_{>i}\} \quad \forall i.$$

Proposition 16.2. $\underline{e}_\lambda = S_\lambda(\underline{x})$ where

Proof. Write $\underline{e}_\lambda = P(x)$. We wish to show $P(x) = S_\lambda(x)$. Introduce new variables $y_1, y_2, \dots$ and consider

$$F(y) = \exp \left(\sum_{k \geq 0} y_k \frac{\partial}{\partial x_k} \right) P(x) \Big|_{x=0}$$

First we observe that

$$F(y) = P(x_1 + y_1, x_2 + y_2, \dots) \Big|_{x=0} = P(x + y) \Big|_{x=0} = P(y)$$

since in general exponentiating a differential operator gives shifting by the coefficient, i.e. $e^{a \frac{\partial}{\partial t}} f(t) = f(t + a)$.

Considering

$$P(x) \in \mathbb{C}[x_1, x_2, \dots] = H \simeq F^{(0)},$$

observe that for $k > 0$,

$$\frac{\partial}{\partial x_k} = h_k = (: \varphi \varphi^* :)_k = \sum_{i \in \mathbb{Z}} \varphi_{k+1} \varphi_i^*$$

i.e. it's a sum of “remove e_i ” and “insert e_{k+i} ”.

Denote

$$\begin{aligned} \Lambda_k : X &\longrightarrow X \\ \Lambda_k(e_n) &= e_{n+k} \quad \forall n \in \mathbb{Z} \end{aligned}$$

$R(\Lambda_k) = \frac{\partial}{\partial x_k}$, since $\Lambda^{\infty/2(0)} \xrightarrow{\simeq} H$.

Therefore

$$F(y) = \text{Rexp} \left(\sum_{k>0} y_k \Lambda_k \right) \underline{e}_\lambda \Big|_{\text{coef. in } |0\rangle}.$$

Notice that $\text{Rexp}(\Lambda_k)$ is of the form $g(e_i) - e_i$.

Finally, notice that $\Lambda_k = \Lambda_1^k$, so

$$\begin{aligned} &\text{Rexp} \left(\sum_{k>0} y_k \Lambda_k \right) \underline{e}_\lambda \Big|_{\text{coef. in } |0\rangle} \\ &= \text{Rexp} \left(\sum y_k \Lambda_1^k \right) \underline{e}_\lambda \\ &= R((S_k \Lambda_k) \underline{e}_\lambda). \end{aligned}$$

□

17. THE TATE EXTENSION AND THE JAPANESE COCYCLE

Recall $X = \bigoplus_{n \in \mathbb{Z}} \mathbb{C} e_n$ and $\Lambda^{\infty/2}$. (You can be floppy and think that $\Lambda^{\infty/2}$ is the exterior power of X , though that's not completely right.) Define

$$\mathfrak{gl}_\infty = \left\{ (a_{ij}) : \begin{array}{l} i, j \in \mathbb{Z}, a_{ij} \in \mathbb{C} \\ \text{all but finitely many of the } a_{ij} \text{ vanish} \end{array} \right\},$$

add and multiply as usual.

$\text{GL}_\infty = \{I + (a_{ij}) : a_{ij} \in \mathfrak{gl}_\infty\}$, with $I = (I_{ij})$ and $I_{ij} = \delta_{ij}$. A typical element of $\mathfrak{gl}_\infty$ is E_{ij} with $E_{ij}(e_k) = \delta_{jk} e_i$.

Last time we used shift operators $\Lambda_k : e_n \mapsto e_{n+k}$ for all n . These are **not** in $\mathfrak{gl}_\infty$, but **are** in

$$\widetilde{\mathfrak{gl}_\infty} = \left\{ (a_{ij}) : \begin{array}{l} \exists N \text{ s.t. } a_{ij} = 0 \\ \text{whenever } |i-j| > N \end{array} \right\}.$$

Then $\Lambda_k = \sum E_{i+k, k} \in \widetilde{\mathfrak{gl}_\infty}$.

Remark 17.1. $\widetilde{\mathfrak{gl}_\infty}$ is an associative Lie algebra (hence a Lie algebra with commutator). This is because the condition of finiteness in the definition excludes the possibility of infinite sums.

We have a representation r of $\mathfrak{gl}_\infty$ on $\Lambda^{\infty/2}$ by

$$r : E_{ij} \mapsto \varphi_i \varphi_j^*$$

But this doesn't extend to $\widetilde{\mathfrak{gl}_\infty}$. For instance, $r(\Lambda_0) = \sum_{j \in \mathbb{Z}} \varphi_{-j} \varphi_j^*$ diverges.

A “solution” is to use normal order, so $r(\Lambda_0) = \sum_{j \in \mathbb{Z}} : \varphi_{-j} \varphi_j^* :$, is now well-defined, but r is no longer a representation.

Today: more conceptual point of view. Consider the vector space $\mathbb{C}((t)) = X$. (Which is *uncountably* infinite-dimensional.) Give X a linear topology with a base of open neighbourhoods of 0 being $t^N \mathbb{C}[[t]]$ for $N \in \mathbb{Z}$. The idea is that $t, t^2, t^3, \dots$ “tends to 0” in this topology.

Definition 17.2. A *Tate vector space* is a linearly topologised vector space X and a set $\mathcal{L}$ of linear subspaces $L \subset X$ (called *lattices*) such that

- (1) (Separated.) Any neighbourhood of 0 contains a lattice.
- (2) (Exhaustive.) Every $x \in X$ is contained in some lattice.
- (3) (Commensurable.) For all $L_1, L_2 \in \mathcal{L}$, $L_1 \cap L_2$ has finite codimension in L_1 (and in L_2). (This says that any two lattices cannot be “infinitely far apart”.)
- (4) (Complete.) For all $L_1, L_2 \in \mathcal{L}$, all vector subspaces $L_1 \cap L_2 \subset S \subset L_1 + L_2$ then $S \in \mathcal{L}$.
- (5) (Another completeness.) By the universal property of limit, we know there exists a map $X \rightarrow \lim_{\substack{L \in \mathcal{L} \\ L \rightarrow 0}} X/L$. We ask this is an isomorphism.

Example 17.3. (1) Laurent series is an example of a Tate vector space. Let $X = \mathbb{C}((t))$ with

$$\mathcal{L} = \{L \subset X : \exists N \text{ s.t. } t^N \mathbb{C}[[t]] \subset L \subset t^{-N} \mathbb{C}[[t]]\}.$$

We can also say that $\mathcal{L}$ is the unique structure such that $\mathcal{L} \ni \mathbb{C}[[t]]$.

- (2) If $\mathcal{L} = \{0\}$, then $\mathcal{L} = \{L \subset X \mid \dim(L) < \infty\}$.
- (3) If $\mathcal{L} \ni X$, then $\mathcal{L} = \{L \subset X \mid \dim(X/L) < \infty\}$.

Since we introduced a topology on X , we can consider the continuous endomorphisms $\text{End}_{\text{cont}}(X)$. Let’s denote throughout this section

$$\text{End}(X) = \{f \in \text{End}_{\text{cont}}(X) : \exists U, V \in \mathcal{L}, f(U) \subset V\}.$$

Relative to the “basis” $\{t^n \mid n \in \mathbb{Z}\}$, the matrix of $f \in \text{End}(X)$ looks like

$$\begin{pmatrix} * & 0 \\ * & * \end{pmatrix} \begin{pmatrix} 0 \\ * \end{pmatrix}$$

since the vectors in $V \subset t^m \mathbb{C}[[t]]$ have zeroes before a certain index n .

$$\begin{aligned} \text{End}_c(X) &= \{f \in \text{End}(X) \mid \exists V \in \mathcal{L} \text{ s.t. } f(X) \subset V\} = \left\{ \begin{pmatrix} 0 & 0 \\ * & * \end{pmatrix} \right\} \\ \text{End}_d(X) &= \{f \in \text{End}(X) \mid \exists U \in \mathcal{L} \text{ s.t. } f(U) = 0\} = \left\{ \begin{pmatrix} * & 0 \\ * & 0 \end{pmatrix} \right\} \\ \text{End}_f(X) &= \text{End}_c(X) \cap \text{End}_d(X) = \left\{ \begin{pmatrix} 0 & 0 \\ * & 0 \end{pmatrix} \right\}. \end{aligned}$$

Remark 17.4. For $f \in \text{End}_f(X)$, the trace $\text{Tr}(f) \in \mathbb{C}$ is well-defined because the intersection of the diagonal with the lower-left part of any matrix in $\text{End}_f(X)$ is finite. For End_c and End_d trace is not well-defined.

We introduce the map

$$\begin{aligned} p : \text{End}_c(X) \oplus \text{End}_d(X) &\longrightarrow \text{End}(X) \rightarrow 0 \\ (f, g) &\longmapsto f + g \end{aligned}$$

which is clearly surjective since we can put

$$\begin{pmatrix} A & 0 \\ B & C \end{pmatrix} = \begin{pmatrix} A & 0 \\ B & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & D \end{pmatrix}.$$

We can complete this to a short exact sequence (of vector spaces) by putting

$$\begin{aligned} i : \text{End}_f(X) &\longrightarrow \text{End}_c \oplus \text{End}_d \\ h &\longmapsto (h, -h). \end{aligned}$$

Considering the trace $\text{Tr} : \text{End}_f(X) \rightarrow \mathbb{C}$, we can form the pushout L in the category of **vector spaces**

$$\begin{array}{ccccccc} 0 & \longrightarrow & \text{End}_f & \longrightarrow & \text{End}_c(X) \oplus \text{End}_d(X) & \longrightarrow & \text{End}(X) \longrightarrow 0 \\ & & \text{Tr} \downarrow & & \downarrow & & \\ & & \mathbb{C} & \longrightarrow & L & & \end{array}$$

Concretely,

$$L = \frac{(\text{End}_c(X) \oplus \text{End}_d(X)) \oplus \mathbb{C}}{\langle (h, -h) = (0, 0, \text{Tr}(h)) | \forall h \in \text{End}_f(X) \rangle}.$$

Notice that the map i is **not** a Lie algebra map: for this it would have to be a Lie algebra map in both entries, which is true in the first entry but not on the second. Meanwhile, p is a morphism of Lie algebras. We could correct i to map $h \mapsto (h, h)$ and then change p to be $\beta - \gamma$, but then p would stop being a Lie algebra morphism. So this is not a short exact sequence of Lie algebras; indeed it's not sensible to think of short exact sequences of Lie algebras since Lie algebras do not form an Abelian category.

But it is a short exact sequence of vector spaces and we can form its exact sequence pushout by the following exercise. (Also see Stacks Project tag 010I.)

Exercise 17.5. Let

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

be a short exact sequence of vector spaces and $t : A \rightarrow K$ a linear map of vector spaces. Let X be the pushout

$$\begin{array}{ccc} A & \xrightarrow{i} & B \\ t \downarrow & \lrcorner & \downarrow \\ K & \longrightarrow & X. \end{array}$$

Explicitly,

$$X = \frac{K \oplus B}{\langle (t(a), 0) - (0, i(a)) | a \in A \rangle}.$$

Show that there exists a short exact sequence

$$0 \longrightarrow K \longrightarrow X \longrightarrow C \longrightarrow 0$$

such that

$$\begin{array}{ccccccc}
 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C \longrightarrow 0 \\
 & & \downarrow t & & \downarrow \ulcorner & & \downarrow \text{id} \\
 0 & \longrightarrow & K & \longrightarrow & X & \longrightarrow & C \longrightarrow 0
 \end{array}$$

commutes.

Proof. To define a map $X \rightarrow C$ we first notice that all we have to do is define the map on $K \oplus B$ at points not coming from A , i.e. whose entries are not of the form $(i(a), t(a))$ for $a \in A$. Indeed, those are cancelled by the pushout definition as a quotient. Further, we already have a map $\varphi : B \rightarrow C$ defined on elements not of the form $i(a)$ for $a \in A$ by exactness of the given short exact sequence. Then we must define $(k, b) \mapsto \varphi(b)$ for the diagram to commute.

The resulting exact sequence is exact once we know that $K \rightarrow B$ in the pushout is given by $k \mapsto (k, 0) \bmod \sim$. $\square$

Then we obtain

$$\begin{array}{ccccccc}
 0 & \longrightarrow & \text{End}_f & \longrightarrow & \text{End}_c(X) \oplus \text{End}_d(X) & \longrightarrow & \text{End}(X) \longrightarrow 0 \\
 & & \downarrow \text{Tr} & & \downarrow \ulcorner & & \downarrow = \\
 0 & \longrightarrow & \mathbb{C} & \longrightarrow & L & \longrightarrow & \text{End}(X) \longrightarrow 0
 \end{array}$$

Notice that the exact sequence

$$0 \longrightarrow \text{End}_f \xrightarrow{i} \text{End}_d \oplus \text{End}_f \xrightarrow{p} \text{End} \longrightarrow 0$$

is a bit more than just a short exact sequence of vector spaces. Notice that $\text{End}_d, \text{End}_c \subset \text{End}$ are ideals. Indeed, if $f \in \text{End}_c$ and $g \in \text{End}$, then $f \circ g \in \text{End}_c$ obviously, $f(X) \subset V$ so $g \circ f(X) \subset g(V)$, continuity of g (plus axioms of Tate vector space) implies that there exists $\tilde{V} \in \mathcal{L}$ such that $g(V) \subset \tilde{V}$.

Exercise 17.6. Spell this out.

So $g \circ f \in \text{End}_c$.

Exercise 17.7. $\text{End}_d \subset \text{End}$ is an ideal too.

So $\text{End}_c, \text{End}_d$ are End -modules (as associative algebras and as Lie algebras), and

$$0 \longrightarrow \text{End}_f \longrightarrow \text{End}_d \oplus \text{End}_c \longrightarrow \text{End} \longrightarrow 0$$

is a short exact sequence of End -modules.

Let $\mathfrak{g}$ be a Lie algebra, E a $\mathfrak{g}$ -module with $E \xrightarrow{p} \mathfrak{g}$ surjective $\mathfrak{g}$ -module morphism. Consider

$$0 \longrightarrow \mathfrak{a} \xrightarrow{i} E \xrightarrow{p} \mathfrak{g} \longrightarrow 0$$

i.e. $\mathfrak{a} = \text{Ker}(p)$.

Then one obtains a symmetric bilinear form defined by

$$(a, b) = p(a)b + p(b)a.$$

Exercise 17.8. This form is (1) symmetric and (2) $\mathfrak{g}$ -invariant.

Exercise 17.9. Check that the data above, satisfying $(\cdot, \cdot) = 0$ is the same as a central extension of $\mathfrak{g}$ by $\mathfrak{a}$, i.e. Lie bracket $[\cdot, \cdot] : E \times E \rightarrow E$ compatible with $[\cdot, \cdot]$ on $\mathfrak{g}$ such that $\mathfrak{a} \subset E$ is central. **Hint.** Set $[a, b]^E = p(a)b$.

As a particular case of this, if we have

- $\mathfrak{g}$ a Lie algebra,
- $\mathfrak{a}_1, \mathfrak{a}_2 \subset \mathfrak{g}$ two ideals such that $\mathfrak{g} = \mathfrak{a}_1 + \mathfrak{a}_2$,
- setting $\mathfrak{a}_0 = \mathfrak{a}_1 \cap \mathfrak{a}_2$, a linear map $\mathfrak{a}_0 \xrightarrow{T} K$ such that $T([x_1, x_2]) = 0$ for all $x_1 \in \mathfrak{a}_1$ and $x_2 \in \mathfrak{a}_2$.

Then we can form

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathfrak{a}_0 & \xrightarrow{i} & \mathfrak{a}_1 \oplus \mathfrak{a}_2 & \xrightarrow{p} & \mathfrak{g} \longrightarrow 0 \\ & & \downarrow T & & \downarrow & & \downarrow \text{id} \\ 0 & \longrightarrow & K & \longrightarrow & X & \longrightarrow & \mathfrak{g} \longrightarrow 0 \end{array}$$

with pushout at level of vector spaces and $i : \alpha \mapsto (\alpha, -\alpha)$ and $p : (\beta, \gamma) \mapsto \beta + \gamma$.

Exercise 17.10. Then X will become a Lie algebra with K central and $\mathfrak{a}_1 \oplus \mathfrak{a}_2 \rightarrow X$ a map of Lie algebras.

The main point of this construction is to use $T([x_1, x_2])$ to confirm that

$$T(p(a)b + p(b)a) = 0 \quad \forall a, b \in \mathfrak{a}_1 \oplus \mathfrak{a}_2.$$

Indeed, write $a = (a_1, a_2)$ and $b = (b_1, b_2)$. Then

$$\begin{aligned} & T([a_1, a_2, b_1], [a_1 + a_2, b_2]) + ([b_1 + b_2, b_1], [b_1 + b_2, a_2]) \\ &= T([a_1, b_1] + [a_2, b_1] + [b_1, a_1] + [b_2, a_1], [a_1, b_2] + [a_2, b_2] + [b_1, a_2] + [b_2, a_1]) \\ &= T(i(\underbrace{[a_2, b_1]}_{\in [\mathfrak{a}_1, \mathfrak{a}_2]} + \underbrace{[b_2, a_1]}_{\in [\mathfrak{a}_2, \mathfrak{a}_1]})) \\ &= 0 \quad \text{by hypothesis.} \end{aligned}$$

We apply this, in particular, to

$$\mathfrak{g} = \text{End}(X) \quad \mathfrak{a}_1 = \text{End}_d(X) \quad \mathfrak{a}_2 = \text{End}_c(X) \quad \mathfrak{a}_0 = \text{End}_f(X)$$

$$X \text{ our Tate vector space} \quad T \text{ trace.}$$

For the construction above to work we still have to check that

$$\text{Tr}([A_d, A_c]) = 0$$

whenever $A_d \in \text{End}_d$ and $A_c \in \text{End}_c$. You would think this is obvious, but it isn't. (I.e., $\text{Tr}(AB - BA) = 0$ for $A, B \in \text{End}(\text{finite-dimensional vector space})$.)

Let us sketch the proof. Let $f \in \text{End}_c$ and $f(X) \subset V$ ($V \in \mathcal{L}$), and $g \in \text{End}_d$ and $g(U) = 0$ ($U \in \mathcal{L}$). Then

$$X \xrightarrow{f} \underbrace{f(X)}_{\subset V} \xrightarrow{g} \underbrace{gf(X)}_{\subset g(V)}.$$

And

$$X \xrightarrow{g} g(X) \rightarrow \underbrace{fg(X)}_{\subset V}.$$

So $\text{Im}[f, g] \subset V + g(V)$ So $[f, g](U \cap f^{-1}(U)) = 0$.

$$\underbrace{f^{-1}g^{-1}(0)}_{\subset f^{-1}(U)} \xrightarrow{f} \underbrace{g^{-1}(0)}_{\subset U} \xrightarrow{g} 0$$

and

$$\underbrace{g^{-1}f^{-1}(U)}_{\subset U} \xrightarrow{g} f^{-1}(0) \xrightarrow{f} 0.$$

Now the idea is to argue that

$$\text{Tr}[f, g] = \text{Tr}_Q[f, g],$$

where

$$Q = (V + g(V)) / (U \cap f^{-1}(U))$$

is a finite-dimensional vector space. But on finite dimensional vector spaces the trace of commutator vanishes.

You might imagine that $\text{Tr}[f, g] = 0$ whenever $[f, g] \in \text{End}_f(X)$, but this is false.

Example 17.11. Let $A \subset \text{End}(X)$ be a commutative subalgebra. (For instance, if $X = \mathbb{C}((t))$, then $f(t) \in \mathbb{C}((t))$ we have $\mu_f \in \text{End}(X)$ and $\mu_f(g) = fg$.) Then I claim, if $f, g \in A$, $f = f_c + f_d$, $g = g_c + g_d$, that $[f_c, g_c] \in \text{End}_f$. Obviously $[f_c, g_c] \in \text{End}_c$, but also

$$\begin{aligned} [f_c, g_c] &= [f - f_d, g - g_d] \\ &= \underbrace{[f, g]}_{=0} - \underbrace{[f, g_d] + [f_d, g] - [f_d, g_d]}_{\substack{\in \text{End}_d \text{ because} \\ \text{ideal} \\ \text{End}_d \subset \text{End}}} \end{aligned}$$

I claim that $\text{Tr}[f_c, g_c]$ might not vanish.

In fact let $X = \mathbb{C}((t))$, $f = \mu_{t^{-2}}$, $g = \mu_{t^2}$ (or with any $N \geq 0$ in place of 2).

To fix f_c, g_c , lwt's make the following choice:

$$\begin{aligned} \pi : X &\longrightarrow X \\ \pi(t^n) &= \delta_{n \geq 0} t^n. \end{aligned}$$

Then set $f_c = \pi \circ f$, etc. Let's compute $[f_c, g_c]$. [Picture]. We obtain $\text{Tr}[f_c, g_c] = 2$.

The next proposition is "Tate's definition of the residue". Tate was trying to generalize the Residue Theorem, i.e. that $\sum_{p \in C} \text{Res}_p \omega = 0$ for points on a curve C and a differential form ω .

Proposition 17.12. For any choice of splittings, $f = f_c + f_d$, etc, and for all $f, g \in \mathbb{C}((t))$,

$$\text{Tr}[f_c, g_c] = \text{Res}_t f \cdot dg.$$

18. REPRESENTING THE ENDOMORPHISMS ALGEBRA ON THE CHARGED FERMIONS

We want to represent $\text{End}(X)$ on $\Lambda^{\infty/2}$. We start presenting the naive idea. Let $x \in X$ and $\varphi \in X^*$. We already have

$$\rho(x) = x \wedge (-) \in \text{End}(\Lambda^{\infty/2})$$

and

$$\rho(\varphi) = \sum_{j=0}^{\infty} (-1)^j \varphi(x_{i_j}) x_{i_0} \wedge x_{i_1} \wedge \dots \wedge \widehat{x_{i_j}} \wedge \dots$$

We already saw that $\rho(\hat{x}_i)\rho(\hat{\varphi}_j) + \rho(\hat{\varphi}_j)\rho(\hat{x}_i) = \delta_{ij}I$ (here I'm identifying $\hat{x}_i \equiv t^i \in \mathbb{C}((t)) = X$, $\varphi_i \in X^*$ is $\hat{\varphi}_i(\sum c_j t^j) = c_j$). See Exercise 15.1.

If $f \in \text{End}_d$, we can think of f as

$$f = \sum_{i=0}^a x_i \otimes \varphi_i,$$

where $\varphi_i \rightarrow 0$ as $i \rightarrow \infty$.

Here $x_i \rightarrow 0$ means $\forall N \geq 0 \exists n_0$ such that $x_n \in t^N \mathbb{C}[[t]]$ for all $n \geq n_0$. And $\varphi_i \rightarrow 0$ means $\forall N \geq 0 \exists n_0$ such that $\varphi_n(t^{-N} \mathbb{C}[[t]]) = 0$ for all $n \geq n_0$.

One can confirm that for $f \in \text{End}_d$,

$$\rho_d(f) := \sum_i \rho(x_i) \rho(\varphi_i)$$

acts a finite sum, when applied to any fixed vector of $\Lambda^{\infty/2}$. So

$$\rho_d : \text{End}_d \rightarrow \text{End}(\Lambda^{\infty/2} X)$$

is well defined.

For $\text{End}_c \ni f$, we can write

$$f = \sum_i x_i \otimes \varphi_i,$$

where $x_i \rightarrow 0$. Now $\rho_d(f)$ makes no sense, but

$$\rho_c(f) := \sum_{i=0}^{\infty} \rho(\varphi_i) \rho(x_i)$$

does.

Exercise 18.1. Confirm that $\rho_d : \text{End}_d \rightarrow \text{End}(\Lambda^{\infty/2})$ and $\rho_c : \text{End}_c \rightarrow \text{End}(\Lambda^{\infty/2} X)$ are morphisms of Lie algebras.

(Neither of these morphisms work at the level of associative algebras.)

We notice that, for $f \in \text{End}_f(X)$,

$$\rho_d(f) - \rho_c(f) = \text{Tr}(f).$$

This means

$$\text{End}_d \oplus \text{End}_c \xrightarrow{(\rho_d, \rho_c)} \text{End}(\Lambda^{\infty/2} X)$$

descends to a morphism

$$\begin{array}{ccc} \text{End}_d \oplus \text{End}_c & \xrightarrow{(\rho_d, \rho_c)} & \text{End}(\Lambda^{\infty/2} X) \\ \downarrow & \nearrow \rho & \\ \mathfrak{gl}^b(X). & & \end{array}$$

Theorem 18.2. Let X be a Tate vector space. There exists a natural representation of $\mathfrak{gl}^b(X)$ on $\Lambda^{\infty/2}(X)$.

Next time we'll apply this to the case X itself is already a Lie algebra!

19. THE 26-DIMENSIONALITY OF THE UNIVERSE

Last time: X a Tate vector space (for us $X = \mathbb{C}((t))$). We say that $\text{End}(X) = \text{End}_c(X) + \text{End}_d(X)$ comes with a canonical central extension (as a Lie algebra

$$0 \longrightarrow \mathbb{C} \longrightarrow \mathfrak{gl}(X)^b \longrightarrow \text{End}(X) \longrightarrow 0$$

with

$$\mathfrak{gl}(X)^b = \frac{\text{End}_c \oplus \text{End}_d \oplus \mathbb{C}}{\langle (h, -h, 0) = (0, 0, \text{Tr}(h)) | h \in \text{End}_f(X) \rangle},$$

and there is a canonical representation of $\mathfrak{gl}(X)^b$ on the semi-infinite wedge space $\Lambda^{\infty/2}(X) = \langle i_{i_0} \wedge e_{i_1} \wedge \dots \rangle$ defined by

$$(19.0.1) \quad \begin{aligned} \rho_d(\underbrace{\sum_i x_i \varphi_i}_{\in \text{End}_d?}) &= \sum_i \rho(x_i) \rho(\varphi_i) \\ \rho_c(\underbrace{\sum_i x_i \varphi_i}_{\in \text{End}_c}) &= - \sum_i \rho(\varphi_i) \rho(x_i), \end{aligned}$$

where

$$\rho(x) = x \wedge (-), \quad \rho(\varphi) = \sum_{j=0}^{\infty} (-1)^j \varphi(e_{i_j}) e_{i_0} \wedge e_{i_1} \wedge \dots \wedge \widehat{e_{i_j}} \wedge \dots$$

Now suppose X itself were a Lie algebra.

Example 19.1. $X = \mathbb{C}((t))$ as above, but we identify $t^m \rightsquigarrow L_m$ in $\text{Vir} @ c = 0$. So $[t^m, t^n] = (m - n)t^{m+n}$. To avoid confusion, let's write t^m as L_m in fact. In such situation, we have a linear map $\text{ad} : X \rightarrow \text{End}(X)$.

We can pull back $\mathfrak{gl}(X)^b$

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathbb{C} & \longrightarrow & \mathfrak{gl}(X)^b & \xrightarrow{\pi} & \text{End}(X) \longrightarrow 0 \\ & & & & \uparrow & & \uparrow \text{ad} \\ & & & & \hat{X} & \longrightarrow & X. \end{array}$$

Explicitly,

$$\hat{X} = \{(x, A) \in X \oplus \mathfrak{gl}(X)^b | \text{ad}(X) = \pi(A) \text{ in } \text{End}(X)\}.$$

Exercise 19.2. Similarly to Exercise 17.5, show $\hat{X}$ fits into a sequence

$$0 \longrightarrow \mathbb{C} \longrightarrow \hat{X} \longrightarrow X \longrightarrow 0.$$

So we get a (canonical!) central extension of X :

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathbb{C} & \longrightarrow & \mathfrak{gl}(X)^b & \xrightarrow{\pi} & \text{End}(X) \longrightarrow 0 \\ & & \downarrow \text{id} & & \uparrow & & \uparrow \text{ad} \\ 0 & \longrightarrow & \mathbb{C} & \longrightarrow & \hat{X} & \longrightarrow & X \longrightarrow 0 \end{array}$$

For $X = \text{Der}(\mathbb{C}[t^{\pm 1}]) = \text{centerless Vir}$, we will find the canonical central extension is $\text{Vir} @ c = -26$. (i.e. charge is -26 .)

What might be amazing here is that any infinite-dimensional Lie algebra has a central extension — just by being infinite-dimensional.

We'll exploit

$$\Lambda^{\infty/2}(X) = \langle L_{m_0} \wedge L_{m_1} \wedge L_{m_2} \wedge \dots \rangle.$$

We have quantum fields

$$\begin{aligned} \varphi(w) &= \sum_n \varphi_n w^{-n-1}, & \varphi &= L_n \wedge (-), \\ \varphi^*(w) &= \sum_n \varphi_n^* w^n, & \varphi_n^*(\underline{L}) &= \sum (-1)^j \varphi(L_{i_j}) L_{i_0} \wedge L_{i_1} \wedge \dots \end{aligned}$$

Now for

$$\begin{aligned} L_m \in X, & \quad L_m : L_n \mapsto (m-n)L_{m+n}, \\ \text{ad}(L_m) &= \underbrace{\sum_{n \in \mathbb{Z}} (m-n)L_{m+n}L_n^*}_{\in \text{End}_c + \text{End}_d}. \end{aligned}$$

To represent $\text{ad}(L_m)$ on $\Lambda^{\infty/2}$ we have to split $\text{ad}(L_m)$ into pieces in End_d and End_c and send each part to their corresponding pieces according to Equation 19.0.1.

Let's work at the level of fields.

$$\begin{aligned} L(w) &= \sum_m \rho(\text{ad}(L_m)) w^{-m-2} \\ &= \sum_{m,n} (m-n) \rho(L_{m+n}L_n^*) w^{-m-2} \\ &= \sum_{m,n} (m-n) : \varphi_{m+n} \varphi_n^* : w^{-m-2} \\ &= \sum_{m,n} (m-n) : (\varphi_{m+n} w^{-(m+n)-?}) (\varphi_n^* w^{+n+?}) : \end{aligned}$$

where

$$\rho(L_{m+n}L_n^*) = \begin{cases} \rho(L_{m+n})\rho(L_n^*) & \text{when } n \ll 0 \\ -\rho(L_n^*)\rho(L_{m+n}) & \text{when } n \gg 0. \end{cases}$$

Notice that

$$\begin{aligned} \partial_w \varphi(w) &= \partial_w \sum \varphi_{m+n} w^{-m+n} \\ &= - \sum (m+n) \varphi_{m+n} w^{-(m+n)-1} \\ &= \sum n \varphi_n^* w^{n-1} \end{aligned}$$

since $m-n = (m+n) - 2n$, so it seems we should consider

$$- : (\partial \varphi) \varphi^* : - 2 : \varphi (\partial \varphi^*) :$$

So, we have a vertex superalgebra $V = \Lambda^{\infty/2}$, with quantum fields φ and φ^* , OPE relation $[\varphi_\lambda \varphi^*] = 1$.

We are defining a field

$$L = - : (\partial \varphi) \varphi^* : - 2 : \varphi (\partial \varphi^*) :,$$

(which is a quantum field, $L(w) = \sum_m L_m w^{-m-2}$), and we want to know/check

$$[L_m, L_n] = (m-n)L_{m+n} + \delta_{m,-n} \frac{m^3 - m}{12} CI_{\Lambda^{\infty/2}}.$$

So, what's c ? Let's do it.

- (Method 1.) Brute force.
- (Method 2.) Use mathematica (Thielman's package).
- (Method 3.) Recall $\alpha = : \varphi \varphi^* :$, satisfies $[\alpha_\lambda \alpha] = \lambda$ and $L^0 = \frac{1}{2} : \alpha \alpha :$ is Virasoro @ $c = 1$.

$$[L_\lambda^0 L^0] - TL^0 + 2\lambda L^0 + \frac{\lambda^3}{12}.$$

$$\begin{aligned} L^0 &= \frac{1}{2} : \alpha \alpha : \\ &= \frac{1}{2} : (: \varphi \varphi^* :) (: \varphi \varphi^* :) : \\ &= \dots \\ &= \frac{1}{2} (: (T\varphi) \varphi^* : + : \varphi (T\varphi^*) :). \end{aligned}$$

How to see this? Recall Borchers identity (Equation 11.9.2):

$$\begin{aligned} &\sum_{j \geq 0} \binom{m}{j} (a_{(n+j)} b)_{(m+k-j)} c \\ &= \sum_{j \geq 0} (-1)^j \binom{n}{j} (a_{(a+m-j)} b_{(k+j)})^c - (-1)^n b_{(n+k-j)} a_{(m+j)}^c. \end{aligned}$$

So put

$$\begin{aligned} a &= \varphi & m &= 0 \\ b &= \varphi^* & n &= -1 \\ c &= : \varphi \varphi^* : & k &= -1. \end{aligned}$$

Then

$$LHS = (a_{(-1)} b)_{(-1)} c = : (: \varphi \varphi^* :) (: \varphi \varphi^* :) :$$

and

$$RHS = \sum_{j \geq 0} (\varphi_{(-i-j)} \varphi_{(-1+j)}^* \varphi_{(-1)} \varphi^* + \varphi_{(-2-j)}^* \varphi_{(j)} \varphi_{(-1)} \varphi^*).$$

[Picture]

$$RHS = \varphi_{(-1)} \varphi_{(-1)}^* \alpha + \varphi_{(-2)} \varphi_{(0)}^* \alpha + \varphi_{(-2)}^* \varphi_{(0)} \alpha.$$

Note:

$$\varphi_{(0)} \alpha = -\alpha_{(0)} \varphi = +\varphi.$$

In general

$$b_{(0)} a = - \sum_{j \geq 0} \frac{1}{j!} T^j (a_{(j)} b) = -a_{(0)} b + T(\text{stuff}).$$

Remark 19.3. Recall (ref?) that if V is any vertex algebra, V/TV , $[\bar{a}, \bar{b}] = a_{(0)} b$, is a Lie algebra.

So,

$$RHS = \varphi_{(-1)}\varphi_{(-1)}^*\alpha - \varphi_{(-2)}\varphi^* + \varphi_{(-2)}^*\varphi.$$

Now, since in general

$$b_{(n)}a = -(-1)^{p(a)p(b)} \sum_{j \geq 0} \frac{(-1)^j}{j!} T^j(a_{(n+j)}b),$$

then

$$\varphi_{(-2)}^*\varphi = \sum_{j \geq 0} \frac{(-1)^j}{j!} T^j(\varphi_{(-2+j)}\varphi^*).$$

But this led to mistaken calculations. I was trying to do the following: we know that $\alpha = : \varphi \varphi^* :$ satisfies $[\alpha_\lambda \alpha] = \lambda$, and $L^0 = \frac{1}{2} : \alpha \alpha : \text{ is } \text{Vir}[L_\lambda^0 L^0] = TL^0 + 2\lambda L^0 + \frac{\lambda^3}{12}$ ($c = 1$). Also, if $B = L^0 + kT_\alpha$ then

$$[B_\lambda B] = TB + 2\lambda B + \frac{\lambda^3}{12}c_k, \quad c_k = 1 - 12k^2.$$

Writing B in terms of $v\varphi, \varphi^*$, one gets something like

$$B = b : (T\varphi)\varphi^* : + (1-b) : \varphi(T\varphi^*) : .$$

The correct answer is: if $L = - : (T\varphi)\varphi^* : - 2 : \varphi(T\varphi^*) :$, we may verify that

$$[L_\lambda L] = TL + 2\lambda L - \frac{26}{12}\lambda^3.$$

In summary, today we did the following. For φ, φ^* odd, $[\varphi_\lambda \varphi^*] = 1$, we define the operator

$$L = - : (T\varphi)\varphi^* : = 2 : \varphi(T\varphi^*) : .$$

Then compute and find out that

$$[L_\lambda L] = TL + 2\lambda L + \frac{c}{12}\lambda^3,$$

where $c = -26$.

That is, after computing the central extension of the Virasoro algebra, we turn to the vertex algebra language to find that the central charge is -26 .

Even more explicitly: by the Tate vector space construction we have the central extension

$$0 \longrightarrow \mathbb{C}C \longrightarrow \mathfrak{gl}(X)^\flat \longrightarrow \mathfrak{gl}(X) \longrightarrow 0$$

where $W = \mathbb{C}((t))$ is the Witt algebra, with bracket $[L_m, L_n] = (m-n)L_{m+n}$.

And then we do the pullback in the following way:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathbb{C}K & \longrightarrow & \mathfrak{gl}(W)^\flat & \xrightarrow{\pi} & \text{End}(X) \longrightarrow 0 \\ & & \downarrow \text{id} & & \uparrow & & \uparrow \text{ad} \\ 0 & \longrightarrow & \mathbb{C}K & \longrightarrow & W \oplus \mathbb{C}K & \longrightarrow & W \longrightarrow 0 \end{array}$$

Then we get a representation in which K goes to the identity, namely

$$\mathfrak{gl}(X)^\flat \curvearrowright \Lambda^{\infty/2}, \quad K \mapsto \text{Id}.$$

So, the vertex algebra language allows us to compute the central charge of the new algebra we obtained, $W \oplus \mathbb{C}K$, and realise it's -26 . So the bracket in the new algebra is

$$[L_m, L_n] = (m - n)L_{m+n} + \frac{m^3 - m}{12}(-26)K.$$

20. LATTICES AND THE NEIGHBOR CONSTRUCTION

Definition 20.1. A *lattice* (for us) is a discrete subgroup $L \subset \mathbb{R}^n$, $L \simeq \mathbb{Z}^n$, and such that $(\alpha, \beta) \in \mathbb{Z}$ for all $\alpha, \beta \in L$, where $(\cdot, \cdot) : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ is the standard bilinear form $((u_1, \dots, u_n), (v_1, \dots, v_n)) = \sum u_i v_i$.

In particular, L contains a basis of the ambient space $\mathbb{R}^n$.

Example 20.2. (1) $\mathbb{Z}^n \subset \mathbb{R}^n$ is a lattice.
 (2) A_2 , the root lattice of $\mathfrak{sl}_3$, is a lattice. Recall this looks like a triangular tiling with $(\alpha_1, \alpha_1) = 2$, $(\alpha_2, \alpha_2) = 2$ and $(\alpha_1, \alpha_2) = -1$, which says that the angle between α_1 and α_2 is 120 degrees.

Definition 20.3. A lattice is called *even* if $(\alpha, \alpha) \in 2\mathbb{Z}$ for all $\alpha \in L$.

(That is, the product of every element with *itself* is even, not that $(\alpha, \beta) \in 2\mathbb{Z}$ for all $\alpha, \beta \in L$.)

So A_2 is even. In fact, whenever the basis elements are even, say $(\alpha, \alpha), (\beta, \beta) \in 2\mathbb{Z}$, then the lattice is even since $(\alpha \pm \beta, \alpha \pm \beta) = (\alpha, \alpha) \pm 2(\alpha, \beta) + (\beta, \beta) \in 2\mathbb{Z}$ too.

Definition 20.4. The *dual* of L is

$$L^\vee = \{x \in \mathbb{R}^n \mid (x, \alpha) \in \mathbb{Z} \forall \alpha \in L\}.$$

Clearly $L \subset L^\vee$. But $L^\vee$ might be strictly larger.

Exercise 20.5. $L^\vee/L$ is a finite group.

Notice that for $\gamma, \delta \in L^\vee$ it might happen that $(\gamma, \delta) \notin \mathbb{Z}$.

For $L = A_2$ we find that

$$L^\vee = A_2 \cup (\omega + A_2) \cup (2\omega + A_2).$$

So $L^\vee/L \simeq \mathbb{Z}/3$ as groups.

For $L = \mathbb{Z}^n$, we have $L^\vee = \mathbb{Z}^n = L$, that is, $\mathbb{Z}^n$ is self dual. On the other hand $\mathbb{Z}^n$ is not even.

Are there any other even self-dual lattices (other than $\{0\}$)?

The simplest nontrivial example is

$$E_8 = \{(x_1, x_2, \dots, x_8) \in \mathbb{Z}^8 \mid \sum_{i=1}^8 x_i \equiv 0 \pmod{2}\}.$$

Lemma 20.6. E_8 is even and self-dual.

Theorem 20.7. (1) If $L \subset \mathbb{R}^n$ is even and self dual, then $8 \mid n$
 (2) If, also, $n = 8$, then $L \simeq E_8$.

Remark 20.8. Let L be an even lattice. On the group $D = L^\vee/L$ we define

$$\begin{aligned} q : D &\longrightarrow \mathbb{Q}/\mathbb{Z} \\ q(a) &= (\alpha, \alpha)/2 \pmod{\mathbb{Z}}. \end{aligned}$$

Then q is a well-defined quadratic form.

Indeed, for $\beta \in L$,

$$q(\alpha + \beta) = \frac{(\alpha + \beta, \alpha + \beta)}{2} = \underbrace{\frac{(\alpha, \alpha)}{2}}_{=q(\alpha)} + \underbrace{\frac{2(\alpha, \beta)}{2}}_{\in \mathbb{Z}} + \underbrace{\frac{(\beta, \beta)}{2}}_{\in \mathbb{Z}}.$$

We call (D, q) the *discriminant form* of L .

Now we explain the neighbour construction/orbifold. Let L be a self-dual even lattice. Let $\phi : L \rightarrow \mathbb{Z}/2$ a homomorphism (i.e. if $\{e_1, \dots, e_n\}$ is a basis of L , set $\phi(e_i) = \varepsilon_i \in \{0, 1\}$ and $\phi(\sum m_i e_i) = \sum m_i \varepsilon_i \pmod 2$).

Assume ϕ is nontrivial and surjective, so that its kernel $L_0 = \text{Ker } \phi \subset L$ has index 2. That is, $L/L_0 \simeq \mathbb{Z}/2$.

Now we have $L_0 \subset L = L^\vee \subset L_0^\vee$.
index 2
index 2
exercise

Remark 20.9. $D = L_0^\vee/L_0 \simeq \mathbb{Z}/2 \times \mathbb{Z}/2$.

Why? If $\omega \in L_0^\vee$, then $2\omega \dots$ Exercise.

What about (D, q) ? I claim there are two possibilities (up to $\simeq$):

$$(20.9.1) \quad \begin{pmatrix} 0 & 0 \\ 0 & 1/2 \end{pmatrix} \quad \begin{pmatrix} 0 & 1/4 \\ 0 & 3/4 \end{pmatrix}$$

Exercise.

Suppose (by choice of ϕ) we are in the first case. Then $L = L_0 \cup (\alpha + L_0)$, $q(\beta) = 0$, i.e. $(\beta, \beta) \equiv 0 \pmod 2$,

$$\begin{pmatrix} L_0 & \beta + L_0 \\ \alpha + L_0 & \gamma + L_0 \end{pmatrix}.$$

Define $L^{\text{orb}(\phi)} = L_0 \cup (\beta + L_0)$.

Lemma 20.10. $L^{\text{orb}(\phi)}$ is an even lattice.

Proof. Exercise. Idea: $(L_0, L_0) \subset \mathbb{Z}$ and $(L_0, \beta + L_0) \subset \mathbb{Z}$ by definition. □

For $\beta, \beta' \in \beta + L_0$,

$$\begin{aligned} (\beta', \beta) &= (\beta + \alpha, \beta), \quad \alpha \in L_0 \\ &= \underbrace{(\beta, \beta)}_{\in \mathbb{Z}} + \underbrace{(\alpha, \beta)}_{\in \mathbb{Z}} \in \mathbb{Z} \end{aligned}$$

Also, $(L^{\text{orb}(\phi)})^\vee = L^{\text{orb}(\phi)}$. Indeed

$$L_0 \subset L^{\text{orb}(\phi)} \subset (L^{\text{orb}(\phi)})^\vee \subset L_0^\vee.$$

index 2
by above
index 2

Theorem 20.11 (Niemeier). *If we consider the set Γ_n of all even self-dual lattices of rank n ($8|n$) as a graph, with edge whenever there exists an operation $L \rightarrow L^{\text{orb}(\phi)}$, then Γ_n is connected for all n .*

Remark 20.12. There is a reverse orbifold construction.

Let L be even self dual, $L = L_0 \cup L_1$, and

$$\begin{pmatrix} L_0 & L_+ \\ L_1 & L_- \end{pmatrix}$$

$L_0^\vee = L_0 \cup L_1 \cup L_+ \cup L_-$, say L_+ is $q(\alpha) = 0$ for all $\alpha \in L_+$ and $q(\beta) = 1/2$ for all $\beta \in L_-$ since we are fixed in the first case of Equation 20.9.1.

Define

$$\begin{aligned} \psi : L^{\text{orb}(\phi)} &\longrightarrow \mathbb{Z}/2 \\ \psi(\alpha) &= \begin{cases} 0 & \text{for } \alpha \in L_0 \\ 1 & \text{for } \alpha \in L_+. \end{cases} \end{aligned}$$

By definition, $\text{Ker}(\psi) = (L^{\text{orb}(\phi)})_0 = L_0 \subset L^{\text{orb}(\phi)}$. So $(L^{\text{orb}(\phi)})_0^\vee = L_0^\vee$ and the discriminant form $(L^{\text{orb}(\phi)})_0^\vee / L_0^{\text{orb}(\phi)}$ brecoveres the same picture.

$$(L^{\text{orb}(\phi)})^{\text{orb}(\phi)} = L.$$

$$\begin{array}{ccccc} L & \underbrace{0} & 0 & M \\ & L_0^{\text{orb}(\phi)} = L_0 & & \\ & 0 & 1/2 & \end{array}$$

Definition 20.13. Let $L \subset \mathbb{R}^n$ be an even lattice. The *root system* of L is

$$\Delta(L) = \{\alpha \in L \mid (\alpha, \alpha) = 2\}.$$

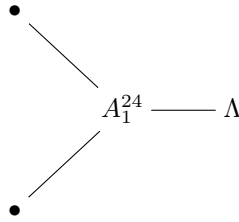
Proposition 20.14. The root system of an even lattice is a root systems.

Example 20.15. The root system of E_8 (the lattice) is E_8 (the root system).

If $\Delta(L_1) \not\cong \Delta(L_2)$ then $L_1 \not\cong L_2$.

In fact, in $\mathbb{R}^{16}$ there exists two even self-dual lattices, with root systems $E_8 \oplus E_8$ and D_{16} .

For rank 24, there exist (coincidentally) 24 even self-dual lattices called *Neimeier lattices*. They are constructed as a graph going from one another by applying the orbifold construction on different homomorphisms ϕ . Eventually the graph looks like this:



where Λ is the *Leech lattice*, which has *empty* root system, that is, Λ contains 0, and no vectors of norm 2, and 196560 (?) vectors of squared norm 4.

There is something called the Siegel mass formula. Consider the orthogonal group of a lattice, namely

$$\begin{aligned} \text{Aut}(L) &= \{g \in \text{GL}_n(\mathbb{R}) \mid (g\alpha, g\beta) = (\alpha, \beta) \forall \alpha, \beta \in L \text{ and } g(L) \subset L\} \\ &= \{g \in O_n(\mathbb{R}) \mid g(L) \subset L\}, \end{aligned}$$

which is a finite group.

Then

$$\sum_{L \in \Gamma_n} \frac{1}{\#\text{Aut}(L)} = \frac{|B_{n/2}|}{n} \prod_{1 \leq j \leq n/2} \frac{|B_{2j}|}{4j}$$

where B_k is a Bernoulli number. (See Wikipedia page for Neimeier lattice.)

21. VERTEX ALGEBRA MODULES

Let V be a vertex algebra, so we have, for every $v \in V$ a quantum field $Y(a, z)$ on V , also $|0\rangle \in V$ and $T : V \rightarrow V$, such that various properties hold, in particular

$$Y(a_{(n)}b, z) = Y(a, z)_{(n)}Y(b, z), \quad Y(|0\rangle, z) = \text{Id}, \quad [T, Y(a, z)] = \partial_z Y(a, z).$$

Definition 21.1. A V -module is a vector space M and, for all $a \in V$, a quantum field $Y^M(a, z)$ on M (i.e. $Y^M(a, z) = \sum_{n \in \mathbb{Z}} a_{(n)}^M z^{-n-1}$ such that $\forall x \in M, \exists N$ such that $a_{(n)}x = 0$ for all $n \geq N$) such that

- (1) $Y^M(a_{(n)}b, z) = Y^M(a, z)_{(n)}Y^M(b, z)$.
- (2) $Y^M(|0\rangle, z) = I_M$.
- (3) Ambiguity! We might either omit (3), or... demand M come with an endomorphism $T_M : M \rightarrow M$ such that $[T_M, Y^M(a, z)] = \partial_z Y^M(a, z)$.

But we will not use 3, i.e. for us a V -module just satisfies 1 and 2.

Let $\mathfrak{g}$ be a finite-dimensional Lie algebra, $(\cdot, \cdot) : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathbb{C}$ an invariant symmetric bilinear form, and $\hat{\mathfrak{g}}$ the corresponding affine Lie algebra, i.e. $\hat{\mathfrak{g}} = \mathfrak{g}[t^{\pm 1}] \oplus \mathbb{C}K$, $[at^m, bt^n] = [a, b]t^{m+n} + m\delta_{m, -n}(a, b)K$, K central. Recall the affine vertex algebra $V^k(\mathfrak{g}) = U(\hat{\mathfrak{g}}) \otimes_{U(\hat{\mathfrak{g}}_+)} \mathbb{C}|0\rangle$, $K|0\rangle = k|0\rangle$, $\mathfrak{g}[t]|0\rangle = 0$, which is a $\hat{\mathfrak{g}}$ -representation with quantum fields for all $a \in \mathfrak{g}$,

$$Y(a_{-1}|0\rangle, z) = a(z) = \sum_{n \in \mathbb{Z}} at^m \cdot z^{-m-1},$$

where we denote $a_m = at^m$.

Recall that the Lie bracket definition for $\hat{\mathfrak{g}}$ implies that

$$[a(z), b(w)] = [a, b](w)\delta(z, w) + k(a, b)\partial_w\delta(z, w)I,$$

i.e. in $\lambda.[a_\lambda b] = [a, b] + \lambda(a, b)k|0\rangle$.

Recall also that a $\hat{\mathfrak{g}}$ -module M is called smooth if $\forall x \in M$ and $a \in \mathfrak{g}$ there is N such that $at^n x = 0$ for all $n \geq N$.

It M is a smooth $\hat{\mathfrak{g}}$ -module, then

$$Y^M(a, z) = \sum_{n \in \mathbb{Z}} (at^n)|_M z^{-n-1}$$

is a quantum field.

If M is a smooth $\hat{\mathfrak{g}}$ -module of level k (i.e. $K|_M = kI_M$), then

$$[Y^M(a, z), Y^M(b, w)] = Y^M([a, b], w)\delta(z, w) + (a, b)k\partial_w\delta(z, w)I_M.$$

The following proposition says that there is an equivalence of categories between smooth level k $\hat{\mathfrak{g}}$ -modules and $V^k(\mathfrak{g})$ -modules:

Proposition 21.2. *If M is a smooth level k $\hat{\mathfrak{g}}$ -module then there exists a structure of $V^k(\mathfrak{g})$ -module on M such that $Y^M(a_{-1}|0\rangle, z) = Y^M(a, z)$ for all $a \in \mathfrak{g}$.*

As a special case, $\mathfrak{g} = \mathfrak{h}$ a finite-dimensional vector space, $(\cdot, \cdot) : \mathfrak{h} \times \mathfrak{h} \rightarrow \mathbb{C}$ some nondegenerate symmetric bilinear form, $k = 1$. Then we get $V^k(\mathfrak{g}) = H(\mathfrak{h})$ Heisenberg. Let $\mu \in \mathfrak{h}$, then we have the Fock module $H^\mu = U(\hat{\mathfrak{h}}) \otimes_{U(\hat{\mathfrak{h}}_+)} \mathbb{C}|\mu\rangle$,

$$ht^m|\mu\rangle = \begin{cases} 0 & \text{if } m \geq 1 \\ (\mu, h)|\mu\rangle & \text{if } m = 0. \end{cases}$$

As a vector space, $H^\mu \cong \mathbb{C}[h_{(n_1)}^{i_1}, h_{(n_2)}^{i_2} |_{n=1,2,\dots}^{i=1,\dots,\dim \mathfrak{h}}]|\mu\rangle$.

H^μ , $\mu = 0$ is the Heisenberg vertex algebra itself, the H^μ for various μ are all modules over $H = H^0$.

We saw an example in $F^{\text{ch}} = \bigoplus_{c \in \mathbb{Z}} F^c$. [Picture of upside-down parabola. The center column is $F^{(0)} \cong H \cong H(\mathbb{C}) \cong \mathbb{C}[h_{-1}, h_{-2}, \dots] \cong H^0$. In fact we can show that the H^i are V -modules over H^0 .]

Remark 21.3. The fact that $F = \bigoplus_{c \in \mathbb{Z}} F^c$ is a vertex superalgebra implies that F^c with $a \in F^0 \curvearrowright F^c$ by restriction of $Y(a, z)$, is a F^0 -module.

Can we go the other way and build a vertex (super)algebra from an infinite sum of $H(\mathfrak{h})$ -modules? Yes.

Remark 21.4 (Technical). In $H(\mathbb{C})$, we defined $L = \frac{1}{2} : hh :$, which is a Virasoro vector.

$$[L_\lambda L] = TL + 2\lambda L + \frac{\lambda^3}{12}|0\rangle.$$

($L(z) = \sum_{n \in \mathbb{Z}} L_n z^{-n-2}$.) We observed that $L_{-1} = \frac{1}{2} \sum_{k \in \mathbb{Z}} h_{-1-k} h_k$ coincides with T .

Exercise 21.5. Check it.

More generally, if $\{a^1, \dots, a^n\}$ is a basis of $\mathfrak{h}$, and $\{b^1, \dots, b^n\}$ the dual basis (i.e., $(a^i, b^j) = \delta_{i,j}$). Define

$$L = \frac{1}{2} \sum_{i=1}^{\dim \mathfrak{h}} : a^i b^i :.$$

Exercise 21.6. $[L_\lambda L] = TL + 2\lambda L + \frac{\lambda^3}{12} cI$, where $c = \dim(\mathfrak{h})$. We call this a *Virasoro or conformal vector* of central charge. $\dim(\mathfrak{h})$.

Exercise 21.7. $L_{-1} = T$ once again.

(Consider $\mathfrak{h} \otimes \mathfrak{h}$, the element $\sum_i a_i \otimes b_i$ is well-defined (i.e. depends on $\mathfrak{h}$ and $(\cdot, \cdot)$ but not the choice of basis $\{a^i\}$.)

So if we chose a different basis $\{a^i\}$ we'd get the same $L \in H(\mathfrak{h})$.

Definition 21.8. If V contains a Virasoro vector L , such that $T = L_{-1}$, we call V (more precisely, the pair (V, L)) a *conformal vertex algebra* of ventral charge c .

Now we can reinterpret Exercise 21.6 as saying that the Heisenberg (?) algebra is conformal.

Remark 21.9. If (V, L) is a conformal vertex algebra, and M a V -module, then we may define $T_M := L_{-1}^M$, i.e. (where $Y^M(L, Zz) = \sum L_n^m z^{-n-2}$) then we will have

$$[T_M, Y^M(a, z)] = \partial_z Y^M(a, z).$$

(So for conformal vertex algebras, axiom 3 is superfluous).

Observe that in the Fock space H^μ , we have

$$\begin{aligned}
 L_{-1}|\mu\rangle &= \frac{1}{2} \sum_i \left(\sum_{k \in \mathbb{Z}} :a_{-1-k}^i b_k^i: \right) \left(\sum_{k \in \mathbb{Z}} a_{-1-k}^i b_k^i \right), \quad \text{since they commute} \\
 &= \sum_i (a_{-1} b_0^i + a_0^i b_{-1}) |\mu\rangle \\
 &= \frac{1}{2} \sum_i (a_{-1}^i |\mu\rangle (b^i, \mu) + b_{-1}^i |\mu\rangle (a^i, \mu)) \\
 &= \frac{1}{2} (\mu_{-1} |\mu\rangle + \mu_{-1} |\mu\rangle) \quad \text{lin. alg. fact} \\
 &= \mu_{-1} |\mu\rangle \quad \sum_i (\mu, b^i) a^i = \mu \\
 &=: \underbrace{\mu}_{\substack{\text{element of generator of} \\ \text{Heis. alg. Fock module}}} \quad \mu \quad :.
 \end{aligned}$$

It's also fun to do:

$$\begin{aligned}
 L_0 |\mu\rangle &= \frac{1}{2} \sum_i \left(\sum_k :a_{-k}^i b_k^i: \right) |\mu\rangle \\
 &= \frac{1}{2} \sum_i a_0^i b_0^i |\mu\rangle \\
 &= \frac{1}{2} \sum_i (b^i, \mu) (a^i, \mu) |\mu\rangle \quad \text{lin. alg. fact} \\
 &= \frac{1}{2} (\mu, \mu) |\mu\rangle \quad \sum_i (x, a^i) (y, b^i) = (x, y) \\
 &= \underbrace{\frac{1}{2} (\mu, \mu)}_{\text{a number}} |\mu\rangle.
 \end{aligned}$$

Let us try to build a vertex (super)algebra $V_S = \bigoplus_{\mu \in S} H^\mu$, where S is some subset $S \subset \mathfrak{h}$ (assume $0 \in S$) such that $T = L_{-1}$, where $L = \frac{1}{2} :a^i b^i: \in H^0$. In particular

$$(21.9.1) \quad T|\mu\rangle = : \mu |\mu\rangle : \quad \text{for all } \mu \in S.$$

Let $\Gamma^\mu(z)$ denote $Y(|\mu\rangle, z)$ now. Equation 21.9.1 implies

$$\partial_z \Gamma^\mu(z) \underbrace{=}_{\substack{\text{holds for} \\ \text{vertex algebras}}} [T, \Gamma^\mu(z)] \underbrace{=}_{\text{now}} : \mu(z) \Gamma^\mu(z) :.$$

So

$$(21.9.2) \quad \partial_z \Gamma(z) = : \mu(z) \Gamma(z) :$$

(Note that this is formally similar to $\partial_z f(z) = \mu(z) f(z)$, whose solution is $f(z) = c e^{\int \mu(z) dz}$. So informally we should have $\Gamma(z) = c : e^{\int \mu(z) dz} :.$)

So we kind of have

$$\mu(z) = \sum_{n \in \mathbb{Z}} \mu_n z^{-n-1}, \quad \int \mu(z) dz = \mu_0 \cdot \underbrace{\log z}_{\substack{\text{not very} \\ \text{well defined}}} + \sum_{n \neq 0} \frac{z^{-n}}{-n} \mu_n,$$

$$\exp(\mu_0 \log(z)) = z^{\mu_0}.$$

Maybe this means

$$(21.9.3) \quad \Gamma(z) = c \exp \left(\sum_{j < 0} \frac{z^{-j}}{-j} \mu_j \right) \exp \left(\sum_{j > 0} \frac{z^{-j}}{-j} \mu_j \right) z^{\mu_0}.$$

This crazy argument turns out to get us to the right definition.

Now we describe the precise argument. Firstly $h_0|\mu\rangle = (h, \mu)|\mu\rangle$ for all $h \in \mathfrak{h}$, $h_{\geq 1}|\mu\rangle = 0$. So

$$\underbrace{[h_\lambda|\mu\rangle]}_{\sum_j h_j|\mu\rangle \cdot \frac{\lambda_j^j}{j!}} = (h, \mu)|\mu\rangle.$$

This implies that $[h_m, \Gamma^\mu(w)] = (h, \mu)w^m\Gamma^\mu(w)$. In particular $[h_0, \Gamma^\mu(w)] = (h, \mu)\Gamma^\mu(w)$. It follows that

$$\Gamma^\mu(w) : H^\beta \rightarrow H^{\beta+\mu}((w)).$$

(where H^β is where elements of H act by pairing with β , i.e. $\forall x \in H^\beta$, $h_0x = (h, \beta)x$ for all $h \in \mathfrak{h}$.)

Recall that all Fock spaces are spaces of polynomials, they are only different by the constant by which h_0 acts. So they are all isomorphic as Lie algebras over the oscillator Lie algebra $\mathfrak{a}' = \bigoplus_{n \neq 0} \mathbb{C}h_n \oplus \mathbb{C}K$. This means we can introduce

$$u : H^\beta \rightarrow H^{\beta+\mu}$$

such that

- (1) $u(|\beta\rangle) = |\beta + \mu\rangle$,
- (2) $u(h_n x) = h_n u(x)$ for all $n \neq 0$ (extend linearly).

Consider on H^β

$$(21.9.4) \quad A = u^{-1} \exp \left(- \sum_{j < 0} \frac{z^{-j}}{-j} \mu_j \right) \Gamma^\mu(z) \exp \left(- \sum_{j > 0} \frac{z^{-j}}{-j} \mu_j \right)$$

$H^\beta \rightarrow H^\beta((z))$. (Which already looks similar to Equation 21.9.3!)

Exercise 21.10. $[h_n, A] = 0$ for all $n \neq 0$.

So $A|_{H^\beta} = p_{\mu, \beta}(z)I_{H^\beta}$.

Proposition 21.11.

$$\Gamma^\mu(z)|_{H^\beta} = C_{\mu, \beta} \exp \left(\sum_{j < 0} \frac{z^{-j}}{-j} \mu_j \right) \exp \left(\sum_{j > 0} \frac{z^{-j}}{-j} \mu_j \right) z^{(\mu, \beta)} u.$$

Proof. Consider $\Gamma^\mu(z)|\beta\rangle = \sum_{n \in \mathbb{Z}} \Gamma_{(n)}^\mu|\mu\rangle z^{-n-1}$. Take two elements a, b with conformal weights given by $L_0 a = \Delta(a)a$ and $L_0 b = \Delta(b)b$. Then $L_0(a_{(n)}b) = \Delta(a) + \Delta(b) - n - 1$ (exercise: deduce this from the commutator formula).

We computed earlier that

$$\Delta(|\beta\rangle) = \frac{(\beta, \beta)}{2}, \quad \Delta(|\beta + \mu\rangle) = \frac{(\beta + \mu, \beta + \mu)}{2},$$

and

$$\Delta(|\mu\rangle) = \frac{(\mu, \mu)}{2}$$

This implies

$$\Gamma_{(-(\beta,\mu)-1)}^\mu |\beta\rangle = P_{\mu,\beta}(z) |\beta + \mu\rangle.$$

It follows that

$$P_{\mu,\beta}(z) = cz^{(\mu,\beta)}.$$

□

Here's a summary of this section so far (new lecture). $\mathfrak{h}$ complex vector space with nondegenerate symmetric bilinear form $(\cdot, \cdot)$. $V = \bigoplus_{\mu \in S} H^\mu$ vertex (super)algebra? extending H^0 -module structure $H^0 \hookrightarrow H^\mu$. We saw (putting $\Gamma^\mu(z) = T(|\mu\rangle, z)$)

$$\Gamma^\mu(z) : H^\beta \rightarrow H^{\beta+\mu}[[z^{\pm 1}]],$$

so S should be closed under addition (we hadn't said this so far!) and

$$(21.11.1) \quad \Gamma^\mu(z)|_{H^\beta} = C_{\mu,\beta} \exp\left(\sum_{n<0} \frac{z^{-n}}{-n} \mu_n\right) \exp\left(\sum_{n>0} \frac{z^{-n}}{-n} \mu_n\right) z^{(\mu,\beta)} u$$

(see Equation 21.9.4). Here u is the unique linear map $u : H^\beta \rightarrow H^{\beta+\mu}$ such that $u(|\beta\rangle) = |\beta + \mu\rangle$, and u commutes with the action of h_n for all $n \in \mathbb{Z} \setminus \{0\}$. (Intuitively, it takes a monomial and takes it to "the same" monomial.)

Let's assume the constant $C_{\mu,\beta}$ are all nonzero. Also want $z^{(\mu,\beta)}$ to be an integer power of z , so $(\alpha, \beta) \in \mathbb{Z}$ for all $\alpha, \beta \in S$, i.e. $S \subset \mathfrak{h}$ should be an integral lattice. Call S as Q from now on.

In particular

$$\begin{aligned} H^\beta &= \mathbb{C}[t^{-1}\mathfrak{h}[t^{-1}]] \otimes |\beta\rangle, \\ H^{\mu+\beta} &= \mathbb{C}[t^{-1}\mathfrak{h}[t^{-1}]] \otimes |\mu + \beta\rangle, \end{aligned}$$

and

$$u(\underbrace{h_{-n_1}^{i_1} h_{-n_2}^{i_2} \dots h_{-n_s}^{i_s}}_{\substack{h_0 \text{ acts by} \\ \text{the scalar } (h,\beta) \in \mathbb{C}}} \otimes |\beta\rangle = \underbrace{h_{-n_1}^{i_1} h_{-n_2}^{i_2} \dots h_{-n_s}^{i_s}}_{\substack{h_0 \text{ acts by} \\ \text{the scalar } (h,\mu+\beta) \in \mathbb{C}}} \otimes |\mu + \beta\rangle.$$

(We can picture this as a lattice with a vector space sitting at every point, and a map from every vector space with tag β to the vector space with tag $\beta + \mu$. The vector spaces are infinite-dimensional and are drawn as cones with, e.g., $|\beta\rangle$ and $|\beta + \mu\rangle$ in the vertex of the cone.)

Notice that in the vector space H^β , $h_0 = (h, \beta)$ constant, we can write $z^{(\mu,\beta)} I_{H^\beta}$ as z^{μ_0} , therefore, let's write a symbol e^μ for the map(s) u , more precisely

$$e^\mu : V \rightarrow V,$$

whose restriction to H^β is $C_{\mu,\beta}$ times the unique $u : H^\beta \rightarrow H^{\beta+\mu}$ as above.

Let's also start writing $|\beta\rangle$ as e^β . So now elements of V are linear combinations of vectors of the form $h_{-n_1}^{i_1} \dots h_{-n_s}^{i_s} \otimes e^\beta$, for $\beta \in Q$.

$$\begin{aligned} e^\mu(h_{-n_1}^{i_1} \dots h_{-n_s}^{i_s} \otimes e^\beta) &= C_{\mu,\beta} h_{-n_1}^{i_1} \dots h_{-n_s}^{i_s} \otimes e^{\beta+\mu} \\ z^{\mu_0}(h_{-n_1}^{i_1} \dots h_{-n_s}^{i_s} \otimes e^\beta) &= z^{(\mu,\beta)} h_{-n_1}^{i_1} \dots h_{-n_s}^{i_s} \otimes e^\beta. \end{aligned}$$

Now

$$(21.11.2) \quad \Gamma^\mu(z) = e^\mu \exp\left(\sum_{n<0} \frac{z^{-n}}{-n} \mu_n\right) \exp\left(\sum_{n>0} \frac{z^{-n}}{-n} \mu_n\right) z^{\mu_0}.$$

Note this expression works in all of V , and not in a fixed H^β , as Equation 21.11.1 does!

Exercise 21.12. Each $\Gamma^\mu(z)$ is a quantum field.

Are the $\Gamma^\mu(z)$ mutually local? We claim that

$$(21.12.1) \quad \Gamma^\mu(z)\Gamma^\beta(w) = i_{z,w}(z-w)^{(\mu,\beta)} : \Gamma^\mu(z)\Gamma^\beta(w) :$$

We can sketch a proof of this:

$$LHS = e^\mu e^\beta z^? w^? \exp\left(\sum_{n<0} \frac{z^{-n}}{-n} \mu_n\right) \underbrace{\exp\left(\sum_{n>0} \frac{z^{-n}}{-n} \mu_n\right) \exp\left(\sum_{n<0} \frac{w^{-n}}{-n} \beta_n\right) \exp\left(\sum_{n>0} \frac{w^{-n}}{-n} \beta_n\right)}_{\substack{\text{to pass to RHS we have to swap these terms} \\ \text{(at least morally)}}}$$

Consider $h \in \mathfrak{h}$, $(h, h) = 1$ for simplicity, $\mu = \beta = h$. Now we claim

$$\begin{aligned} \left[\sum_{m>0} \frac{z^{-m}}{-m} h_m, \sum_{n<0} \frac{w^{-n}}{-n} h_n \right] &= - \sum_{n>0} \frac{1}{n} \left(\frac{w}{z} \right)^n (\mu, \beta) \\ &= \log\left(1 - \frac{w}{z}\right)(\mu, \beta) \end{aligned}$$

Where the logarithm here is defined by $\log(1-x) = -x + \frac{1}{2}x^2 + \frac{1}{3}x^3 + \dots$. $(\mu, \beta)\log(1 - \frac{w}{z})$ goes into the exponential and becomes $i_{z,w}(z-w)^{(\mu,\beta)}$. (Check the details of this argument!)

Consider this argument fixing conformal weights:

$$\Delta(a_{(n)}b) = \Delta(a) + \Delta(b) - n - 1.$$

If $a = |\mu\rangle$, $b = |\beta\rangle$, then

$$\begin{aligned} \Delta(|\beta\rangle) &= (\beta, \beta)/2, & \Delta(|\mu\rangle) &= (\mu, \mu)/2 \\ \Delta(|\mu + \beta\rangle) &= (\mu + \beta, \mu + \beta)/2. \end{aligned}$$

Knowing that $|\mu\rangle_{(n)}|\beta\rangle \in H^{\mu+\beta}$ and

$$\begin{aligned} \Delta(|\mu\rangle_{(n)}|\beta\rangle) &= \frac{(\mu, \mu)}{2} + \frac{(\beta, \beta)}{2} - n - 1 \\ &= \frac{(\mu + \beta, \mu + \beta)}{2}, \end{aligned}$$

IF

$$(\mu, \beta) = -n - 1 \iff n = -(\mu, \beta) - 1.$$

Therefore,

$$\begin{aligned} \Gamma^\mu_{(-(\mu,\beta)-1)}|\beta\rangle &= C \cdot |\mu + \beta\rangle. \\ :(\partial_z^{(\mu,\beta)}\Gamma^\mu(z)\Gamma^\beta(z))&:= CT^{\mu+\beta}(z) \end{aligned}$$

if $(\mu, \beta) \geq 0$.

One finds

$$\begin{aligned} :\Gamma^\mu(z)\Gamma^\beta(w): &= e^\mu e^\beta (\text{junk}). \\ :\Gamma^\beta(w)\Gamma^\mu(z): &= e^\beta e^\mu (\text{same junk}). \\ \Gamma^\mu(z)\Gamma^\beta(w) - (-1)^{p(\mu)p(\beta)}\Gamma^\beta(w)\Gamma^\mu(z) & \end{aligned}$$

is local if and only if

$$e^\mu e^\beta = (-1)^{p(\mu)p(\beta)} (-1)^{(\mu, \beta)} e^\beta e^\mu.$$

(Here the $(-1)^{(\mu, \beta)}$ appeared because we changed the order of z and w in Equation ??.) That is,

$$(21.12.2) \quad C_{\mu, \beta} = (-1)^{p(\mu)p(\beta)} (-1)^{(\mu, \beta)} C_{\beta, \mu}.$$

In case $Q = \mathbb{Z}$, $(1, 1) = 1$, we have

$$C_{1,1} = -(-1)^{p(1)p(1)} C_{1,1},$$

so $p(1) = 1$ is forced on us! That is, $F^{\text{ch}} = \bigoplus_{m \in \mathbb{Z}} H^m$ can only be a vertex superalgebra, which we already knew (i.e. F^{ch} is a vertex superalgebra, but now we know it couldn't have been otherwise).

In fact, for all μ , we get

$$C_{\mu, \mu} = (-1)^{p(\mu)^2 + (\mu, \mu)} C_{\mu, \mu},$$

so we must assign a parity $(\mu, \mu) \bmod 2$ to H^μ .

In particular, if we take Q an even lattice, V will be an honest vertex algebra.

Exercise 21.13. Deduce (from Borchers identity, or otherwise) that $C : Q \times Q \rightarrow \mathbb{C}^\times$ must (in order for V to be a vertex superalgebra)

$$(21.13.1) \quad C(\alpha, \beta + \gamma) C(\alpha, \beta) C(\alpha, \beta) C(\alpha + \beta, \gamma)$$

for all $\alpha, \beta, \gamma \in Q$. (This is a cocycle condition.)

This apparently necessary condition on C was not very evident from the locality of $\Gamma^\mu, \Gamma^\beta, \dots$. Discuss.

There's a few other things we need apart from Equation 21.13.1. Namely, that $C(0, \alpha) = C(\alpha, 0) = 1$ (normalization condition) and Equation 21.12.2.

Equation 21.13.1 says C is a *group cocycle*. More precisely, consider the short exact sequence of (not necessarily abelian) groups

$$1 \longrightarrow \mathbb{C}^\times \longrightarrow \hat{Q} \longrightarrow Q \longrightarrow 1,$$

a central extension, i.e. $\hat{Q} = \mathbb{C}^\times \times Q$ as a set, $\mathbb{C}^\times$ central, and the product given by $(a, \alpha)(b, \beta) = (c(\alpha, \beta)ab, \alpha + \beta)$ which, to be associative, $C : Q \times Q \rightarrow \mathbb{C}^\times$ must satisfy Equation 21.13.1, that is, $C : Q \times Q \rightarrow \mathbb{C}^\times$ is a 2-cocycle (with coefficients in the trivial Q -module $\mathbb{C}^\times$). (Also, $\underbrace{(1, 0) \cdot (1, \alpha)}_{e \in \hat{Q}} = (1, \alpha) = (1, \alpha)(1, 0) \iff$

$$C(0, \alpha) = C(\alpha, 0) = 1.)$$

Remark 21.14. If we set up a linear isomorphism $\eta : V \rightarrow V$, of the form $\eta|_{H^\beta} = \eta(\beta) \in \mathbb{C}^\times$ for all β , then $\eta : Q \rightarrow \mathbb{C}^\times$ arbitrary function, conjugating C by η , i.e.

$$\tilde{C}(\alpha, \beta) = \eta(\alpha, \beta) C(\alpha, \beta) \eta^{-1}(\alpha) \eta^{-1}(\beta)$$

i.e.,

$$\tilde{C}(\alpha, \beta) = C(\alpha, \beta) \frac{\eta(\alpha + \beta)}{\eta(\alpha) \eta(\beta)}.$$

This corresponds to modifying C by the coboundary $d\eta$. It's the same vertex superalgebra, though.

So really vertex superalgebra structures on V are classified by

$$H^2(Q, \mathbb{C}^\times) = \frac{Z^2(Q, \mathbb{C}^\times)}{B^2(Q, \mathbb{C}^\times)}.$$

By the universal coefficient theorem, if G is any group and M any trivial G -module,

$$0 \longrightarrow \text{Ext}_{\mathbb{Z}}^1(H_{n-1}(G, \mathbb{Z}), \mathbb{Z}) \longrightarrow H^n(G, M) \longrightarrow \text{Hom}_{\mathbb{Z}}(H_n(G, \mathbb{Z}), M) \longrightarrow 0$$

In this case, $G = Q \simeq \mathbb{Z}^m$, and $H_k(Q, \mathbb{Z}) \simeq \Lambda^k Q$. So

$$0 \longrightarrow \text{Ext}_{\mathbb{Z}}^1(Q, \mathbb{C}^\times) \longrightarrow H^2(Q, \mathbb{C}^\times) \longrightarrow \text{Hom}_{\mathbb{Z}}(\Lambda^2 Q, \mathbb{C}^\times) \longrightarrow 0$$

Now since $\mathbb{C}^\times$ is a divisible group (algebra fact), hence injective as a $\mathbb{Z}$ -module, so $\text{Ext}_{\mathbb{Z}}^1(Q, \mathbb{C}^\times) = 0$. (See [Weibel, 6.1.5] for more on universal coefficients and group homology.) So we have an isomorphism

$$H^2(Q, \mathbb{C}^\times) \rightarrow \text{Hom}_{\mathbb{Z}}(\Lambda^2 Q, \mathbb{C}^\times).$$

Concretely, the map is $C \mapsto B$, where

$$B(\alpha, \beta) = \frac{C(\alpha, \beta)}{C(\beta, \alpha)}$$

Recall that our C was required to satisfy

$$C(\alpha, \beta) = (-1)^{\frac{(\alpha, \alpha)}{2} \frac{(\beta, \beta)}{2} + (\alpha, \beta)} C(\beta, \alpha)$$

hence the associated B is

$$B(\alpha, \beta) = (-1)^{\frac{\alpha, \alpha}{2} + (\alpha, \beta)}$$

In conclusion, on $V_Q = \bigoplus_{\beta \in Q} H^\beta$, there exists a unique conformal vertex superalgebra structure which extends that on H^0 .

$Q = \mathbb{Z}$ gave $V_Q = F^{\text{ch}}$. $Q = \mathbb{Z}\sqrt{2}$ gives $V_Q \simeq L_1(\mathfrak{sl}_2)$, simple quotient of $V^1(\mathfrak{sl}_2)$, affine $\mathfrak{sl}_2$.

Consider $\mathfrak{g} = B_2$, which was two different lengths of weight vectors, so as a lattice it's just $Q = \mathbb{Z}^2$. So $V_Q = V_{\mathbb{Z}} \otimes V_{\mathbb{Z}} = F_2^{\text{ch}}$, the charged fermions in two variables. Compare this with affine B_2 , $L_1(B_2)$. Here B_2 sits (horizontally) as the charge 1 piece. So they are different!

Finally, it turns out that the vertex algebra associated to the Leech lattice V_{Leech} is related to the moonshine module V^{natural} by an orbifold step. In a paper by [Jethro-?] it was shown that in rank 24 it's possible to pass from any vertex algebra to V_{Leech} by a single orbifold step.

22. REPRESENTATIONS OF VERTEX ALGEBRAS

In this section we aim at describing the irreducible representations of lattice vertex algebras V_Q . By restriction, M is a H -module, where $H = H^0(\mathfrak{h})$ is the Heisenberg vertex algebra (and also a Fock representation of the Heisenberg Lie algebra, which has a vertex algebra structure, unlike other Fock representations) associated with the commutative Lie algebra $\mathfrak{h} = \mathbb{C} \otimes_{\mathbb{Z}} Q$ with the induced bilinear form $(\cdot, \cdot) : \mathfrak{h} \times \mathfrak{h} \rightarrow \mathbb{C}$.

A few lectures ago, I stated (ref?)

Proposition 22.1. $H(\mathfrak{h})$ -modules are the same as smooth $\hat{\mathfrak{h}}$ -modules.

This is a special case of a construction that works for any $\mathfrak{h}$ abelian Lie algebra. Also recall

Proposition 22.2. $V^k(\mathfrak{g})$ -modules = {smooth level k $\hat{\mathfrak{g}}$ -modules}.

Let $\mathfrak{h} = \mathbb{C}$ now for simplicity (general case is similar) and let $\mathfrak{a} = \bigoplus_{n \neq 0} \mathbb{C}h_n \oplus \mathbb{C}K \subset \hat{\mathfrak{h}}$. (We have excluded 0!)

Proposition 22.3. Consider $\mathfrak{a}$ -modules with $\mathbb{Z}$ -grading $M = \bigoplus_{k \in \mathbb{Z}} M_k$ with degrees bounded from below, that is such that $h_n(M) \subset M_{k-n}$ and m_k for k sufficiently negative. We call this a positive energy module. (Will this work for any smooth??) $\mathfrak{a}$ -module is isomorphic to a direct sum of copies of the Fock space $F = \mathbb{C}[x_1, x_2, x_3, \dots]$ with

$$h_n = \begin{cases} n \frac{\partial}{\partial x_n} & n > 0 \\ x_{-n} & n < 0. \end{cases} \quad K = 1.$$

Proof. Let M be an $\mathfrak{a}$ -module. Define in general the space of vacuum vectors by

$$\Omega_M = \{x \in M : h_n x = 0 \text{ for } n \geq 1\}.$$

Let $\Omega = \Omega_M$ and define

$$\begin{aligned} \varphi : F \otimes \Omega &\longrightarrow M \\ p(\underline{x}) \otimes v &\longmapsto p(\underline{h})v. \end{aligned}$$

We claim that φ is injective. Indeed, if $v \in \text{Ker}(\varphi) \subset F \otimes \Omega$, then there exists $v' \in 1 \otimes \Omega \cap \text{Ker}$. To see why think of v as a finite sum of polynomials times elements in the module, take the highest degree of the polynomials, apply as many operators as you need to get down to the vacuum, this gives a sum of vacuum vectors. (We had used this sort of argument to show irreducibility of modules)

[Rest of proof made in class.] $\square$

Now let M be a V_Q -module which is positive energy as an $\mathfrak{a}$ -module. So $M \cong F \otimes \Omega$, as above. We have the action of quantum fields

$$h^M(z) = \sum_{n \in \mathbb{Z}} h_{(n)}^M z^{-n-1}$$

on M , for every $h \in \mathfrak{h}$. We also have

$$\Gamma_\alpha^M(z) = \sum_n \Gamma_{\alpha(n)}^m z^{-n-1} = Y^M(e^\alpha, z) \quad \forall \alpha \in Q.$$

We have

$$[h_{(m)}^M, \Gamma_M^\alpha(z)] = (h, \alpha) z^m \Gamma_M^\alpha(z), \quad \forall m \in \mathbb{Z}, h \in \mathfrak{h}, \alpha \in Q.$$

(We saw this relation in V_a itself a few lessons ago.)

From this we deduce

$$\Gamma_M^\alpha(z) = \exp \left(\sum_{j < 0} \frac{z^{-j}}{-j} \alpha_j \right) \exp \left(\sum_{j > 0} \frac{z^{-j}}{-j} \alpha_j \right) A_\alpha(z)$$

(exactly as previously), where the coefficients of $A_\alpha(z)$ commute with all h_n for $n \neq 0$. In other words, it commutes with $\mathfrak{a}$ as defined above.

Since $T = L_{-1}$ in V_Q , we have (as before) that

$$\partial_z \Gamma_M^\alpha(z) = : \alpha(z) \Gamma_M^\alpha(z) :$$

which implies (as before) that

$$A_\alpha(z) = C e_M^\alpha z^{\alpha_0}.$$

So

$$\Gamma_M^\alpha(z) = e^\alpha \exp(\sum \dots) \exp(\sum \dots) z^{\alpha_0},$$

where $e_M^\alpha : \Omega_M \rightarrow \Omega_M$.

In particular α_0 must act on Ω by integer eigenvalues for all $\alpha \in Q$, $\alpha_0|_{\Omega_\mu} = (\alpha, \mu)$. So

$$\Omega = \bigoplus_{\mu \in Q^\vee} \Omega_\mu.$$

Furthermore (as before), $e_M^\alpha : \Omega_\mu \rightarrow \Omega_{\mu+\alpha}$. From Borchers identity, we can deduce

$$(22.3.1) \quad e_M^\alpha e_M^\beta = c(\alpha, \beta) e_M^{\alpha+\beta} \quad \forall \alpha, \beta \in Q.$$

(Compare with Equation 21.13.1.)

In particular, if $x \in \Omega_\mu$ nonzero, then $V_Q \cdot x \subset M$ is of the form

$$\bigoplus_{\alpha \in Q} H^{\mu+\alpha} e^\alpha x.$$

If M is irreducible, then $M = \bigoplus_{\alpha \in Q} H^{\mu+\alpha} e^\alpha x$.

As before, we can picture this as a lattice (say, triangular) with a Fock module sitting in every point. V_Q is just the direct sum of these modules. We can also picture the dual lattice $Q^\vee$, labelled $Q + \mu$.

(I think the upshot is that the weight modules are indexed by the quotient $Q^\vee/Q$.)

Finally, for each $\mu \in Q^\vee/Q$, construct (or confirm it exists) $V_{\mu+Q}$. Recall the cocycle $Q \times Q \rightarrow \mathbb{C}^\times$, $C(\alpha, \beta) \in \mathbb{C}^\times$ had to satisfy Equation 21.12.2, namely:

$$C(\alpha + \beta, \gamma) C(\alpha, \beta) = C(\alpha, \beta + \gamma) C(\beta, \gamma),$$

etc, $\Lambda^2 Q \rightarrow \mathbb{C}^\times$,

$$\frac{C(\alpha, \beta)}{C(\beta, \alpha)} = (-1)^{(\alpha, \alpha)(\beta, \beta) + (\alpha, \beta)}.$$

For an explicit construction of $C(\alpha, \beta)$ see [?, Section 5].

The idea is to consider

$$V_{Q^\vee} = \bigoplus_{\mu \in Q^\vee} H^\mu \cdot e^\mu.$$

Define an action

$$V_Q \times V_{Q^\vee} \rightarrow V_{Q^\vee}[[z^{\pm 1}]]$$

by a similar formula, (extend $C : Q \times Q \rightarrow \mathbb{C}^\times$ to $C : Q \times Q^\vee \rightarrow \mathbb{C}^\times$) get a V_Q -module $V_{Q^\vee}$. Then

$$V_{Q^\vee} = \bigoplus_{\mu \in Q^\vee/Q} V_{\mu+Q}$$

where the $V_{\mu+Q}$ are irreducible modules.

Let $\mathfrak{g}$ be a Lie algebra and $(\cdot, \cdot)$ a symmetric invariant bilinear form. We have the affine Lie algebra $\hat{\mathfrak{g}} = \mathfrak{g}[t^{\pm 1}] \oplus \mathbb{C}K$. We defined a smooth $\hat{\mathfrak{g}}$ -module and level k

(which means that $K \mapsto k\text{Id } \hat{\mathfrak{g}}$ -modules, and the vertex algebra $V^k(\mathfrak{g})$). Again recall that $V^k(\mathfrak{g})$ -modules are in correspondence with smooth level k $\hat{\mathfrak{g}}$ -modules.

Consider U an associative algebra over $\mathbb{C}$ with $U_0 \supset U_1 \supset U_2 \supset \dots$ a decreasing filtration by subspaces U_p , such that

- (1) (Separating.) $\bigcap_p U_p = 0$. [The idea here is that the U_p may be thought of as a topological basis, and this condition means the algebra, as a topological algebra, separates points: there's an open set containing zero but not any given point.]
- (2) For any $u \in U$ and $p \geq 0$, there exists ℓ such that $U_\ell \cdot u \subset UU_p$. (think: for any p , no matter how big, there exists ℓ , maybe much bigger, such that $\dots$) [The idea here is that we want to turn the U_p into ideals, just by multiplying by U — indeed, UU_p is an ideal.]
- (3) (Exhaustive.) $U_0 = U$? Maybe not necessary.

We say a left U -module M is *smooth* if for all $x \in M$ there exists p such that $U_p \cdot x = 0$.

Given U and $\{U_p\}$ as above, we can form the *completion* U^c as a categorical limit of the diagram

$$U/UU_0 \longleftarrow U/UU_1 \longleftarrow U/UU_2 \longleftarrow \dots$$

Can think of U^c as

$$\left\{ \sum_i s_i : \sum_i s_i \bmod UU_p \text{ is finite for each } p \right\}.$$

Formally,

$$U^c = \prod_{p \in \mathbb{Z}_{\geq 0}} U/UU_p = \left\{ (u_0, u_1, u_2, \dots) : \begin{array}{l} u_i \in U/UU_i \\ \text{and } \overline{u_{i+1}} = u_i \end{array} \right\}.$$

One can then set $s_i = u_i - u_{i-1}$, then

$$(u_0, u_1, u_2, \dots) = (s_0, s_0 + s_1, s_0 + s_1 + s_2, \dots) \text{ "}" \sum_i s_i$$

and we recover the first description.

Clearly the action of U on a smooth U -module M extends to a well-defined action of U^c . Now let $a(w) = \sum_{n \in \mathbb{Z}} a_{(n)} w^{-n-1} \in U^c[[w^{\pm}]]$, we call $a(w)$ a *continuous distribution* if for every $p \geq 0$ there exists $N \geq 0$ such that $a_{(n)} \in UU_p$ for all $n \geq N$. We denote the vector space of continuous distributions by $U^c[[w]]^{\text{cont}}$.

Then it's clear that

Lemma 22.4. *If M is a smooth U -module (denote action of $a_{(n)}$ on M by $a_{(n)}^M \in \text{End}(M)$), and $a(w)$ a continuous distribution, then*

$$a^M(w) = \sum_n a_{(n)}^M w^{-n-1}$$

is a quantum field on M .

Proposition 22.5. *If $a(w), b(w) \in U^c[[w^{\pm 1}]]^{\text{cont}}$, the n -th product*

$$a(w)_{(n)} b(w) = \text{Res}_z (a(z) b(w) i_{z,w} - b(w) a(z) i_{w,z}) (z - w)^n$$

is well defined and lies in $U^c[[w^{\pm 1}]]^{\text{cont}}$.

Proof. Analogous to Exercise 9.10. □

Definition 22.6. Let $a(w), b(w) \in U^c[[w^{\pm 1}]]^{cont}$. We say the $a(w), b(w)$ is

- (1) *weakly local* if there exists N such that $a(w)_{(n)}b(w) = 0$ for all $n \geq N$.
- (2) *local* if there exists N such that $(z - w)^N[a(z), b(w)] = 0$.

Obviously local implies weakly local.

Let $\mathcal{F} \subset U^c[[w^{\pm 1}]]^{cont}$ be a subset such that all pairs of elements are local. Let $\overline{\mathcal{F}} \subset U^c[[w^{\pm 1}]]^{cont}$ be the closure of $\mathcal{F}$.

$$\mathcal{F}^0 = \{1_{U^c}\}, \quad \mathcal{F}^k = \left\{ a(w)_{(n)}b(w) : \begin{array}{l} a \in \mathcal{F}, \\ b \in \mathcal{F}^{k-1}, \\ n \in \mathbb{Z} \end{array} \right\}.$$

The analogue of Dong's lemma 11.3 guarantees all $a, b \in \overline{\mathcal{F}}$ are local.

Then we obtain a vertex algebra:

Theorem 22.7. *The vector space $\overline{\mathcal{F}}$, equipped with $|0\rangle = 1_{U^c}$, $Y(a(w), x) = \sum_{n \in \mathbb{Z}} a(w)_{(n)}x^{-n-1}$, $Ta(w) = \partial_w a(w) = a(w)_{(-2)}1_{U^c}$ is a vertex algebra.*

Proof. Exercise. □

Now let M be a smooth U -module (thought of as a U^c -module.)

Let's assign

$$Y^M(a(w), x) = a^M(x) = \sum_{n \in \mathbb{Z}} a_{(n)}^M x^{-n-1}.$$

Proposition 22.8. *(M, Y^M) is an $\overline{\mathcal{F}}$ -module.*

Proof.

$$Y(a(w)_{(n)}b(w), x) = Y(a(w), x)_{(n)}Y(b(w), x),$$

because $\overline{\mathcal{F}}$ is a vertex algebra.

So

$$\begin{aligned} Y^M(a(w)_{(n)}b(w), x) &= a^M(x)_{(n)}b^M(x) \quad \text{by definition} \\ &= Y^M(a(w), x)_{(n)}Y^M(b(w), x). \end{aligned}$$

□

[It looks as if this construction was a slight generalization of our earlier work for vertex algebras over vector spaces to vertex algebras over associative algebras; now we have to blend in this topological algebra notions - completion, etc.]

Now let's look at the case of $\hat{\mathfrak{g}}$ -modules and $V^k(\mathfrak{g})$. We have $\hat{\mathfrak{g}} = \mathfrak{g}[t^{\pm 1}] \oplus \mathbb{C}K$. Let

$$U = U(\hat{\mathfrak{g}})/(K - k_{\text{id}}).$$

Then U -modules are the same as level k $\hat{\mathfrak{g}}$ -modules. Now consider $U_p = \text{Im}(\mathfrak{g}t^{\geq p}) \subset U$. I claim U with U_p so defined satisfies the conditions (1) and (2). Indeed, (1) is given by PBW theorem since every monomial has a highest degree term. As for (2), e.g., for $u = at^{-5}$, $p = 10$, if I take $\ell = 15$, then $u_{\ell} \ni bt^{15}$, $[bt^{15}, at^{-5}] = [b, a]t^{10}$.

Now, in $U^c[[w^{\pm 1}]]^{cont}$ we have, for each $a \in \mathfrak{g}$, the continuous distribution

$$a(w) = \sum_{n \in \mathbb{Z}} at^n w^{-n-1}.$$

By the commutation relation $[at^m, bt^n] = [a, b]t^{m+n} + m\delta_{m, -n}(a, b)k1$ we have $(z - w)^2[a(z), b(w)] = 0$ so these are mutually local, $([a(z), b(w)] = [a, b](w)\delta(z, w) + (a, b)k1\partial_w\delta(z, w).)$

So we get a vertex algebra $\overline{\mathcal{F}}$ from $\mathcal{F} = \{a(w) : a \in \mathfrak{g}\}$ and any smooth $\hat{\mathfrak{g}}$ -module of level k gives an $\overline{\mathcal{F}}$ -module.

In particular $V^k(\mathfrak{g}) = U(\hat{\mathfrak{g}}) \otimes_{U(\hat{\mathfrak{g}}_+)} \mathbb{C}K$ (which happens to be a vertex algebra), being a smooth level k $\hat{\mathfrak{g}}$ -module is an $\overline{\mathcal{F}}$ -module.

In $M = V^k(\mathfrak{g})$, for $a \in \mathfrak{g}$, $a(w)$ (from $\overline{\mathcal{F}}$) acts via $Y^M(a(w), x) = a(x) = T(a_{(-1)}|0\rangle, x)$.

$$\begin{array}{ccc} a(w) & \longmapsto & Y(a(w), z) \\ & & \\ \hat{\mathcal{F}} & \begin{array}{c} \longrightarrow \{Y(a, z) : a \in V^k(\mathfrak{g})\} \\ \searrow \\ \downarrow \\ V^k(\mathfrak{g}) \end{array} & \begin{array}{c} a(z) \\ \downarrow \\ a(z)|0\rangle|_{z=0} \end{array} \end{array}$$

which is the state-field correspondence,

$$a(w)_{(n)}b(w) \longmapsto Y(a(w), z)_{(n)}Y(b(w), z)$$

$$\begin{array}{ccc} \hat{\mathcal{F}} & \begin{array}{c} \longrightarrow \{Y(a, z) : a \in V^k(\mathfrak{g})\} \\ \searrow \phi \\ \downarrow \\ V^k(\mathfrak{g}) \end{array} & \begin{array}{c} a(z)_{(n)}b(w) \\ \downarrow \\ a_{(n)}b \end{array} \end{array}$$

So ϕ is a homomorphism of vertex algebras

$$\begin{aligned} \phi(a_{(n)}b) &\mapsto \phi(a)_{(n)}\phi(b) \\ 1 &\mapsto |0\rangle. \end{aligned}$$

Clearly ϕ is surjective. Want to show $\text{Ker}(\phi) = 0$.

We can characterize $V^k(\mathfrak{g})$ as the universal highest weight level k $\hat{\mathfrak{g}}$ -module, i.e. if M is a $\hat{\mathfrak{g}}$ -module, with $x \in M$ such that

- (1) $Kx = kx$,
- (2) $at^{\geq 0}x = 0$

then there exists a unique $\hat{\mathfrak{g}}$ -module map

$$\begin{aligned} \psi : V^k(\mathfrak{g}) &\longrightarrow M \\ |0\rangle &\longmapsto x. \end{aligned}$$

Lemma 22.9. *I claim*

- (1) $\overline{\mathcal{F}}$ is a $\hat{\mathfrak{g}}$ -module, in which $x = 1_{U^c}$ satisfies
- (2) $Kx = k$
- (3) $\mathfrak{g}t^{\geq 0}x = 0$.

Proof. (1) $\overline{\mathcal{F}}$ is a $\hat{\mathfrak{g}}$ -module. For $c(w) \in \overline{\mathcal{F}}$, define $\rho(at^m)c(w) = a(w)_{(m)}c(w)$.
Then

$$\begin{aligned} [\rho(at^m), \rho(bt^n)] &= [a(w)_{(m)}, b(w)_{(n)}] \\ &= \sum_{j \geq 0} \binom{m}{j} (a(w)_{(j)}b(w))_{(m+n-j)} \\ &= (*). \end{aligned}$$

But for $a, b \in \mathfrak{g}$, we have $[a(z), b(w)] = \dots$ see above. So

$$\begin{aligned} a(w)_{(0)}b(w) &= [a, b](w) \\ a(w)_{(1)}b(w) &= k(a, b)1_{U^c} \\ a(w)_{\geq 2}b(w) &= 0. \end{aligned}$$

Then

$$\begin{aligned} (*) &= [a, b](w)_{(m+n)} + mk(a, b)1_{U^c}(w)_{(m+n-1)} \\ &= \rho([a, b]t^{m+n} + m\delta_{m,-n}(a, b)K). \end{aligned}$$

So ρ is a representation of $\hat{\mathfrak{g}}$ on $\overline{\mathcal{F}}$.

Obviously ρ is level k and obviously $x = 1_{U^c}(w) \in \overline{\mathcal{F}}$ is highest weight.
 $a(w)_{(j)}1(w) = 0$ for all $j \geq 0$.

So we have maps

$$\begin{array}{ccc} \overline{\mathcal{F}} & \xrightarrow{\phi} & V^k(\mathfrak{g}) \xrightarrow{\psi} \overline{\mathcal{F}}. \\ \text{VA homo, hence} & & \hat{\mathfrak{g}}\text{-mod map} \\ \hat{\mathfrak{g}}\text{-mod map} & & \end{array}$$

Can check $\psi \circ \phi = \text{id}$. So $\overline{\mathcal{F}} \simeq V^k(\mathfrak{g})$ as vertex algebras. □

23. CHARACTERS OF VERTEX ALGEBRAS AND THEIR MODULES

Disclaimer! There may be several computational mistakes in this section.

Prelude. Let $q = e^{2\pi i\tau}$, $\tau \in \mathbb{C}$, $\text{Im}(\tau) > 0$ (so $|q| < 1$). Consider

$$\begin{aligned} \Delta(\tau) &= q \prod_{n=1}^{\infty} (1 - q^n)^{24} \\ &= \sum_{n=1}^{\infty} a_n q^n \\ &= q - 24q^2 + 252q^3 + \dots \end{aligned}$$

Then $\sum_{d|n} d^{11} \equiv a_n \pmod{691}$.

Let V be a vertex algebra, and $L \in V$ a conformal vector. Let $L(Z) = \sum L_n z^{-n-2}$. Suppose furthermore that L_0 acts on V by integer eigenvalues with finite dimensional eigenspaces. Write $V = \bigoplus_{n \in \mathbb{Z}} V_n$, where $V_n = \{v \in V : L_0 v = nv\}$ and write $\chi_V = \sum_{n \in \mathbb{Z}} \dim(V_n) q^n$.

Example 23.1. Recall that $V = H = \mathbb{C}[x_1, x_2, \dots]$ with conformal vector $L = \frac{1}{2} : hh : .$ We saw that $L_0 1 = 0$, $[L_0, h_n] = nh_n$. So

$$L_0(h_n v) = h_n(L_0 v) - nh_n v.$$

This means: take a vector with eigenvalue Δ , then apply again, you'll get an eigenvector with eigenvalue $\Delta - n$. In symbols: $v \in V_\Delta$, then $L_0 v \in V_{\Delta-n}$.

So $\Delta(x_1^{m_1} x_2^{m_2} \dots x_k^{m_k} 1) = \sum i m_i$ and

$$\begin{aligned} \chi_H &= \prod_{m=1}^{\infty} (1 - q^m)^{-1} \\ &= \sum_{N=0}^{\infty} p(N) q^N, \end{aligned}$$

where $p(N)$ is the number of integer partitions of N .

Example 23.2. If $\mathfrak{h}$ is an n -dimensional vector space $(\cdot, \cdot)$ a nondegenerate symmetric bilinear form, $\mathfrak{h}$ abelian Lie algebra, $\hat{\mathfrak{h}}$ affine Lie algebra. Then $H = H(\mathfrak{h})$ is the *rank n Heisenberg algebra*. Given a basis $\{a^1, \dots, a^n\}$ of $\mathfrak{h}$ and a dual basis $\{b^1, \dots, b^n\}$ i.e. $(a^i, b^j) = \delta_{ij}$, (so we need $(\cdot, \cdot)$ for this), we compute Virasoro vector to be $L = \frac{1}{2} \sum_{i=1}^n :a^i b^i:$. Then

$$[L_\lambda L] = TL + 2\lambda L + \frac{\lambda^3}{12} c,$$

where the central charge $c = n$ is the rank. Then

$$\chi_{H(\mathfrak{h})} = \prod_{m=1}^{\infty} \frac{1}{(1 - q^m)^n}.$$

Example 23.3. Let $Q \subset \mathbb{R}^n$ be an even integral lattice. Consider $\mathfrak{h} = \mathbb{R}^n \otimes_{\mathbb{R}} \mathbb{C} = \mathbb{C}^n$ with $(\cdot, \cdot)$ inherited. We have $V = V_Q = \bigoplus_{\alpha \in Q} H^\alpha(\mathfrak{h})$.

As part of our construction of V_Q we computed that

$$L_0 |\alpha\rangle = \frac{(\alpha, \alpha)}{2} |\alpha\rangle.$$

Hence $|\alpha\rangle \in V_{\frac{(\alpha, \alpha)}{2}}$. (Also $[L_0, h_n] = -n h_n$ for all $h \in \mathfrak{h}$ and all $n \in \mathbb{Z}$.)

Also $h_{-n_1}^{i_1} h_{-n_2}^{i_2} \dots h_{-n_s}^{i_s} |\alpha\rangle$ lies in V_Δ , $\Delta = \frac{(\alpha, \alpha)}{2} + \sum_{j=1}^s n_j$. (Just as above: we apply the operator again and obtain another eigenvector, this time with eigenvalue given by that sum.) Therefore

$$\chi_{V_Q} = \sum_{\alpha \in Q} q^{\frac{(\alpha, \alpha)}{2}} \prod_{m=1}^{\infty} \frac{1}{(1 - q^m)^n}.$$

This makes sense, since (for now) Q is assumed to be positive-definite.

In fact

$$\chi_{V_Q} = \prod_{m=1}^{\infty} \frac{1}{(1 - q^m)^n} \cdot \underbrace{\sum_{\alpha \in Q} q^{\frac{(\alpha, \alpha)}{2}}}_{\text{theta function}}.$$

The series on RHS is called the *theta function* of the lattice Q , denoted $\theta_Q(\tau)$. (Think of it as a function that knows how many vectors of each norm the lattice has.) More precisely, for $q = e^{2\pi i \tau}$,

$$\theta_Q(\tau) = \sum_{\alpha \in Q} q^{\frac{(\alpha, \alpha)}{2}} = \sum_{\alpha \in Q} e^{\pi i (\alpha, \alpha) \tau}.$$

As detour, notice that if Q is integral, but not even, then some $\frac{(\alpha, \alpha)}{2} \notin \mathbb{Z}$, but are in $\frac{1}{2}\mathbb{Z}$. As an example of this, $Q = \mathbb{Z} \subset \mathbb{R}$ is an odd lattice,

$$\theta_Q(\tau) = \sum_{n \in \mathbb{Z}} q^{\frac{n^2}{2}} = 1 + 2q^{\frac{1}{2}} + 2q^2 + 2q^{\frac{9}{2}} + \dots$$

Remark 23.4. Note that for any positive definite Q (even or odd) the series $\theta_Q(\tau)$ converges for all $\text{Im}(\tau) > 0$. (Easy application of some test or other.)

If Q is even, then only integer powers of q appear, and we have

$$\theta_Q(\tau + 1) = \theta_Q(\tau) \quad \forall \tau \in \mathbb{H}$$

where $\mathbb{H} = \{\tau \in \mathbb{C} : \text{Im}(\tau) > 0\}$.

Can then also think of θ_Q as a holomorphic function (of q) on the disk $D = \{q : |q| < 1\}$.

If Q is odd, then $\theta_Q(\tau + 2) = \theta_Q(\tau)$.

Many lectures ago (see Section 8), I proved for $Q = \mathbb{Z}$ that $\theta_Q(-\frac{1}{\tau}) = \sqrt{\frac{\tau}{i}} \theta_Q(\tau)$ using the Poisson summation formula.

What about for other Q ?

Proposition 23.5. *If Q is self-dual of rank n (and even (?)), then*

$$\theta_Q\left(-\frac{1}{\tau}\right) = (\tau/i)^n \theta_Q(\tau).$$

Proof. This is also an application of Poisson summation. □

More generally,

Proposition 23.6. *If Q is integral with dual $Q^\vee$, then*

$$\theta_Q\left(-\frac{1}{\tau}\right) = (\tau/i)^n \sum_{\mu \in Q^\vee/Q} s_{0,\mu} \cdot \theta_{\mu+Q}(\tau)$$

where the $s_{0,\mu} \in \mathbb{C}^\times$ are nonzero complex numbers and $\tau_{\mu+Q}(\tau) = \sum_{\beta \in \mu+Q} q^{(\beta, \beta)/2}$.

As opposed to the usual case, this function has very fractional powers of q . However, it is well-defined as a function on $\mathbb{H}$.

Proposition 23.7. *If Q is integral, then for all $\lambda \in Q^\vee/Q$, $\theta_{\lambda+Q}(-\frac{1}{\tau}) = (\tau/i)^n \sum_{\mu \in Q^\vee/Q} s_{\lambda\mu} \cdot \theta_{\mu+Q}(\tau)$.*

Again proved by Poisson summation. In the process an explicit formula for $\{S_{\lambda\mu}\}$ emerges. I can't remember, but it's something like

$$S_{\lambda\mu} = \frac{1}{|Q^\vee/Q|^{1/2}} e^{2\pi i(\lambda, \mu)} \quad (?)$$

Thinking of $Q^\vee/Q$ as a finite list $\{1, 2, \dots, N\}$, $N = |Q^\vee/Q|$, then $\{S_{\lambda\mu}\}_{\lambda, \mu=1, \dots, N}$ becomes a matrix, the *S-matrix* of Q .

Since $-1\frac{1}{(-\frac{1}{\tau})} = \tau$, we (naively) should have $S^2 = I$ identity matrix.

There is a slight problem: $\theta_{\mu+Q}(\tau)$ might not all be distinct.

Example 23.8. Consider $Q = A_2$, the triangular lattice. We know that $Q^\vee/Q \simeq \mathbb{Z}/3$. Actually, $Q^\vee/Q = \{Q, \mu + Q, 2\mu + Q\}$. In fact, two of the theta-functions associated to these three cosets are actually the same:

$$\begin{aligned}\theta_Q &= 1 + 6q + \dots \\ Q_{\mu+2} &= 3q^{\frac{2}{3}} + \dots \\ Q_{2\mu+Q} &= 3q^{-\frac{2}{3}} + \dots\end{aligned}$$

We can refine the $\theta_{\mu+Q}(\tau)$ to make them linearly independent.

Let $z \in \mathfrak{h}$, and define

$$\theta_{\mu+Q}(\tau, z) = \sum_{\beta \in \mu+Q} q^{\frac{(\beta, \beta)}{2}} e^{2\pi i(z, \beta)}.$$

Now

$$\theta_{\mu+Q} : \mathbb{H} \times \mathfrak{h} \rightarrow \mathbb{C}$$

(convergent and holomorphic).

Proposition 23.9. For all $\lambda \in Q^\vee/Q$,

$$\theta_{\lambda+Q}\left(-\frac{1}{\tau}, \frac{z}{\tau}\right) = (\tau/i)^n e^{\frac{\pi i(z, z)}{\tau}} \sum_{\mu \in Q^\vee/Q} S_{\lambda, \mu} \theta_\mu(\tau, z)$$

Proof. Also by Poisson summation, same $s_{\lambda, \mu}$. But now $\{\theta_{\mu+Q}(\tau, z) | \mu \in Q^\vee/Q\}$ are linearly independent. So S makes sense (as a matrix). So it no longer holds that $S^2 = I$. We apply S -transformations as follows

$$(\tau, z) \xrightarrow{s} \left(-\frac{1}{\tau}, \frac{z}{\tau}\right) \mapsto \left(-\frac{1}{-\frac{1}{\tau}}, \frac{z/\tau}{-1/\tau}\right) = (\tau, -z) \xrightarrow{s} \left(-\frac{1}{\tau}, -\frac{z}{\tau}\right) \xrightarrow{s} (\tau, z).$$

So $S^4 = I$. □

Recall that

$$\chi_{V_Q}(\tau) = \frac{1}{\prod_{m=1}^{\infty} (1 - q^m)^n} \theta_Q(\tau).$$

Recall the definition of the Dedekind η function (Theorem 8.3). Why do we need to put the factor $q^{1/24}$? The reason is that without it, it is not a modular form, whereas with it, it is.

Lemma 23.10.

$$\eta\left(-\frac{1}{\tau}\right) = \sqrt{(\tau/i)} \eta(\tau)$$

(that's why we normalised by $q^{1/24}$).

So

$$\begin{aligned}\prod_{m=1}^{\infty} (1 - q^m)^n &= q^{-n/24} \eta(\tau)^n \\ \chi_{V_Q}(\tau) &= \frac{\theta_Q(\tau)}{\eta(\tau)^n} q^{n/24}.\end{aligned}$$

Let's correct the definition of $\chi_{V_Q}(\tau)$. Note that $n = \text{rank}(Q)$ is the central charge c of V_Q .

Definition 23.11. The *character* of a conformal vertex algebra V is

$$\chi_V(\tau) = q^{-\frac{c}{24}} \sum_{n \in \mathbb{Z}} \dim(V_n) q^n.$$

If M is a V -module, we also define $\chi_M(\tau) = q^{-\frac{c}{24}} \sum_{n \in \mathbb{Z}} \dim(M_n) q^n$, where $!!$ are the eigenvalues on M of L_0 (maybe not in $\mathbb{Z}$).

We write

$$M_\Delta = \{m \in M : L_0 m = \Delta m\}.$$

A more uniform notation is

$$\begin{aligned} \chi_V(\tau) &= \text{Tr}_V q^{L_0 - c/24}, \\ \chi_M(\tau) &= \text{Tr}_M q^{L_0 - c/24}. \end{aligned}$$

Lemma 23.12. For $\mu \in Q^\vee/Q$, we have the V_Q -module V_μ . Then

$$\text{Tr}_{V_\mu} q^{L_0 - c/24} = \frac{\theta_\mu(\tau)}{\eta(\tau)^n}$$

and

$$\boxed{\chi_M(\tau, z) = \text{Tr}_M e^{2\pi i z_0} q^{L_0 - c/24}.$$

Indeed in $V_\lambda = \bigoplus_{\beta \in \lambda + Q} H^\beta(\mathfrak{h})$, $z \in \mathfrak{h}$. In $H^\beta(\mathfrak{h})$, $z_0|_{H^\beta} = (\beta, z) I_{H^\beta}$. So $e^{2\pi i(z, \beta)} = e^{2\pi i z_0}$.

$$\chi_{V_\lambda} \left(0 \frac{1}{\tau}, \frac{z}{\tau} \right) = e^{\frac{\pi i(z, z)}{\tau}} \sum_{\mu \in Q^\vee/Q} S_{\lambda, \mu} \chi_\mu(\tau, z).$$

(New lecture.) Let's do a summary of what we have done in this section so far. For $Q \subset \mathbb{R}^n$ an integral lattice, we defined a function

$$\theta_Q : \underbrace{\mathbb{H}}_{\{\text{Im} > 0\}} \times \mathfrak{h} \rightarrow \mathbb{C}$$

by

$$\theta_Q(\tau, z) = \sum_{\alpha \in Q} q^{\frac{(\alpha, \alpha)}{2}} e^{2\pi i(\alpha, z)},$$

where $q = e^{2\pi i \tau}$, $z \in \mathfrak{h} = \mathbb{C}^n$. We explained that this series converges since $|q| < 1$ and it is raised to a quadratic term.

More generally, if $[\mu] \in Q^\vee/Q$, then

$$\theta_\mu(\tau, z) = \sum_{\alpha \in \mu + Q} q^{\frac{(\alpha, \alpha)}{2}} e^{2\pi i(\alpha, z)}$$

similarly.

If Q is self-dual,

$$\theta_Q \left(-\frac{1}{\tau}, \frac{z}{\tau} \right) = (\tau/i)^{n/2} e^{\frac{\pi i(z, z)}{\tau}} \theta_Q(\tau, z)$$

by the Poisson summation formula. More generally, for Q non self-dual,

$$\theta_\lambda \left(-\frac{1}{\tau}, \frac{z}{\tau} \right) = (\tau/i)^{n/2} e^{\frac{\pi i(z, z)}{\tau}} \sum_{\mu \in Q^\vee/Q} S_{\lambda\mu} \theta_\mu(\tau, z),$$

where $S = (S_{\lambda\mu})$ is some matrix, $S^4 = I$.

Remark 23.13. There is also the T -matrix, corresponding to $(\tau, z) \mapsto (\tau + 1, z)$.

$$\begin{aligned} S &\longleftrightarrow \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \in \mathrm{SL}_2\mathbb{Z} \\ T &\longleftrightarrow \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}). \end{aligned}$$

In fact, these generate $\mathrm{SL}_2(\mathbb{Z})$.

$$\begin{aligned} T : \underbrace{e^{2\pi i \tau}}_q &\rightarrow e^{2\pi i(\tau+1)} = e^{2\pi i \tau} = q \\ T : \underbrace{e^{2\pi i \tau a}}_{q^a} &\rightarrow e^{2\pi i a \tau + 2\pi i a} = q^a e^{2\pi i a}. \end{aligned}$$

In $\theta_{\mu+Q}(\tau, z)$, we have powers $q^{\frac{(\alpha, \alpha)}{2}}$ such that $\alpha \in \mu + Q$. Note that if $\mu, \mu' \in \mu + Q$, $\mu' - \mu \in Q$,

$$\frac{(\mu + \alpha, \mu + \alpha)}{2} = \frac{(\mu, \mu)}{2} + \underbrace{\frac{2(\mu, \alpha)}{2}}_{\in \mathbb{Z}} + \underbrace{\frac{(\alpha, \alpha)}{2}}_{\in \mathbb{Z}}$$

since Q is even.

So T acts on $\theta_{\mu}(\tau, z)$ by a constant. Indeed, by the constant $e^{2\pi i(\mu, \mu)/2} \in \mathbb{C}^\times$.

So we have a T -matrix, which is diagonal, with these numbers

$$\{e^{2\pi i(\mu, \mu)/2} | \mu \in Q^\vee / Q\}$$

on the diagonal.

In $\mathrm{SL}_2(\mathbb{Z})$, $S^4 = I$, $(ST)^3 = I$. These relations hold for our matrices S and T .

Proposition 23.14. *An even integral lattice Q gives a representation of $\mathrm{SL}_2(\mathbb{Z})$ of dimension $|Q^\vee / Q|$.*

Also earlier we noted that

$$\chi_{\mu+Q}(\tau, z) = \frac{\theta_{\mu+Q}(\tau, z)}{\eta(\tau)^n}$$

is the graded dimension of the V_Q -module $V_{\mu+Q}$. More precisely, $V = V_Q \curvearrowright M = V_{\mu+Q}$. $z \in \mathfrak{h} \subset H \subset V_Q$, so z_0 acts on $V_{\mu+Q}$, also L_0 acts on $V_{\mu+Q}$. And

$$\chi_{\mu+Q}(\tau, z) = \mathrm{Tr}_{V_{\mu+Q}} e^{2\pi i z_0} q^{L_0 - c/24}$$

where the central charge is $c = n = \mathrm{rank}(Q)$.

Recall (compare with Definition ??) that

Definition 23.15. A *modular form* of weight k is a holomorphic function $f(\tau)$ of $\tau \in \mathbb{H}$ such that

(1)

$$f\left(\frac{a\tau + b}{c\tau + d}\right) = (c\tau + d)^k f(\tau)$$

for all $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z})$.

(2) The Fourier series expansion of f (which exists since it must be periodic):

$$f(\tau) = \sum_{n=0}^{\infty} a_n q^n$$

has no $a_{<0}$ terms.

Now we explain a construction: let $\Lambda = \{m + n\tau \mid m, n \in \mathbb{Z}\} \subset \mathbb{C}$ a “lattice spanned by 1 and τ ”. (It’s not a lattice in the sense of Section 20.) Consider

$$\sum_{\lambda \in \Lambda \setminus 0} \frac{1}{\lambda^m} = G_m(\Lambda).$$

- (1) $G_m(\Lambda)$ can be thought of as $G_m(\tau)$.
- (2) The infinite series converges if $m \geq 3$. Inside the disk D_{2^N} , there are on the order of $(2^N)^2$ points. These points have

$$\left| \frac{1}{\lambda^m} \right| < \frac{1}{(2^{N-1})^m}.$$

$$\sum \left| \frac{1}{\lambda^m} \right| = \sum_N \sum_{N/2 < |\lambda| < N} \frac{1}{\lambda^m} < 2^m \sum_N \frac{2^{2N}}{2^{mN}}$$

converges for $m \geq 3$.

Think of $G_m(\tau)$ ($m \geq 3$). Consider $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$, a, b, c, d integers, and $a\tau + b \in \Lambda$, $c\tau + d \in \Lambda$. E.g. $\begin{pmatrix} 2 & 5 \\ 1 & 3 \end{pmatrix}$.

We claim that if $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = 1$, then

$$\{m(a\tau + b) + n(c\tau + d) \mid m, n \in \mathbb{Z}\} = \Lambda.$$

So, denoting $\sum' = \sum_{(m,n) \in \mathbb{Z}^2 \setminus (0,0)}$,

$$\begin{aligned} G_k(\tau) &= \sum' \frac{1}{(m\tau + n)^k} \\ &= \sum' \frac{1}{(m(a\tau + b) + n(c\tau + d))^k} \\ &= (c\tau + d)^{-k} \sum' \frac{1}{\left(m \frac{a\tau + b}{c\tau + d} + n\right)^k} \\ &= (c\tau + d)^{-k} G_k\left(\frac{a\tau + b}{c\tau + d}\right). \end{aligned}$$

So $G_k\left(\frac{a\tau + b}{c\tau + d}\right) G_k(\tau)$ for all $\begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{SL}_2(\mathbb{Z})$. This is called an *Eisenstein series* of weight k .

Remark 23.16. If k odd, then $\frac{1}{\lambda^m}$ and $\frac{1}{(-\lambda)^m}$ cancel, and $G_k(\tau) = 0$, nontrivial. So $G_4(\tau)$ is the first Eisenstein series.

Now let's talk about the Fourier series of $G_{2k}(\tau)$. Start with

$$(23.16.1) \quad (1-x^2)\left(1-\frac{x^2}{4}\right)\left(1-\frac{x^2}{9}\right)\cdots = \frac{\sin(\pi x)}{x},$$

which may be proved by applying Liouville's theorem to the quotient LHS/RHS.

In fact, this was Euler's method for proving the identity $\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$.

Do log derivative of Equation 23.16.1.

$$\pi \frac{\sin(\pi x)}{\cos(\pi x)} = \frac{d}{dx} \log \left(\prod_{n=1}^{\infty} \left(1 - \frac{x^2}{n^2}\right) \right) = \dots$$

Theorem 23.17.

$$G_{2k}(\tau) = (-1)^k \frac{\pi^{2k}}{(2k)!} B_{2k} \cdot E_{2k}(\tau),$$

$$E_{2k}(\tau) = 1 - \frac{4k}{B_{2k}} \sum_{m=1}^{\infty} \frac{m^{2k-1} q^m}{1 - q^m}.$$

Here B_{2k} are Bernoulli numbers,

$$\sum_{n=0}^{\infty} B_n \frac{t^n}{n!} = \frac{t}{e^t - 1}.$$

The first of them are

$$\begin{array}{cccccccc} 2k : & 0 & 2 & 4 & 6 & 8 & 10 & 12 & 14 \\ B_{2k} : & 1 & 1/6 & 1/30 & 1/42 & -1/30 & 5/66 & -691/2730 & 7/6 \end{array}$$

So

$$E_4(\tau) = 1 - 240 \sum_{m=1}^{\infty} \frac{m^3 q^m}{1 - q^m},$$

$$G_4(\tau) = \frac{\pi^4}{45} E_4(\tau).$$

Definition 23.18. Let $\mathcal{M}_{2k}^{\text{hol}}$ be the vector space of holomorphic modular forms of weight k .

Proposition 23.19. $\dim \mathcal{M}_{2k}^{\text{hol}}$ is

$$\begin{array}{cccccccccccc} 2k : & 0 & 2 & 4 & 6 & 8 & 10 & 12 & 14 & 16 \\ & 18 & & & & & & & & \\ \dim \mathcal{M}_{2k}^{\text{hol}} : & 0 & 0 & 1 & 1 & 1 & 1 & 2 & 1 & 2 \\ & 2 & & & & & & & & \end{array}$$

and $\dim \mathcal{M}_{2k+12}^{\text{hol}} = \dim \mathcal{M}_{2k}^{\text{hol}} + 1$.

Proof sketch. Let $f(\tau) \in \mathcal{M}^{\text{hol}_{2k}}$ and consider

$$\xi(\tau) = \frac{d}{d\tau} \log f(\tau) = \frac{f'(\tau)}{f(\tau)}.$$

Since

$$\begin{aligned} f(-1/\tau) &= \tau^{2k} f(\tau), \\ \xi(-1/\tau) &= \frac{2k}{\tau} + \xi(\tau), \end{aligned}$$

apply Cauchy-formula-type argument $\int_{\partial\mathcal{D}} \xi(\tau) d\tau = \text{num of zeroes of } f - \text{num of poles of } f \text{ in } \mathcal{D}$.

Then integrate along the boundary of the region above the unit circle, between the vertical lines $x = -1/2, 1/2$ and bounded above by some horizontal segment at height N . In conclusion, the vertical integrals cancel each other and $\frac{k}{12} \geq N(i\infty)$, where $N(i\infty) = \text{order of the zero of } f \text{ at } i\infty$. $\square$

One has, if $f \in \mathcal{M}_{2k}^{\text{hol}}$, $g \in \mathcal{M}_{2\ell}^{\text{hol}}$, $fg \in \mathcal{M}_{2(k+\ell)}^{\text{hol}}$. So $E_4^3 \in \mathcal{M}_{12}^{\text{hol}}$ and $E_{12} \in \mathcal{M}_{12}^{\text{hol}}$.

Then do

$$\begin{aligned} E_{12} - E_4^3 &= \frac{65520}{691}(q - 24q + 252q - \dots) \\ &= \frac{65520}{691}q \prod_{n=1}^{\infty} (1 - q^n)^{24}. \end{aligned}$$

Remark 23.20.

$$\begin{aligned} \Delta(\tau) &= q \prod_{n=1}^{\infty} (1 - q^n)^{24} \\ &= \eta(\tau)^{24}. \end{aligned}$$

In fact, $\bigoplus_{k=0}^{\infty} \mathcal{M}_{2k}^{\text{hol}} \simeq \mathbb{C}[E_4, E_{12}]$.

Next time. Let Q be unimodular of rank 24. So

$$P(\tau, z) = \frac{\theta_Q(\tau, z)}{\eta(\tau)^{24}}$$

satisfies

$$P\left(-\frac{1}{\tau}, \frac{z}{\tau}\right) = e^{\frac{\pi i(z, z)}{\tau}} P(\tau, z).$$

24. A NON-MODULAR FORM

$$\begin{aligned} G_2(\tau) &= \frac{\pi^2}{3} E_2(\tau) \\ E_2(\tau) &= 1 - 24 \sum_{n=0}^{\infty} \frac{nq^n}{1 - q^n}. \end{aligned}$$

The series $\sum_{\lambda \in \Lambda \setminus 0} \frac{1}{\lambda^2}$ does not converge absolutely, but converges conditionally.

We could define, i.e.,

$$G_2(\tau) = \sum_{n \neq 0} \frac{1}{n^2} + \sum_{m \neq 0} \sum_{n \in \mathbb{Z}} \frac{1}{(m\tau + n)^2}.$$

[Picture of dot arrangement in plane]

A modular transformation on Λ sends the rows here into some diagonal lines

[Picture of diagonal dor arrangement in plane]

and so the argument that proves modularity of $G_{2k}(\tau)$, $k \geq 2$, in this case shows

$$G_2\left(\frac{a\tau + b}{c\tau + d}\right) = (c\tau + d)^2 \left[\underbrace{\sum_{\lambda \in \Lambda \setminus 0} \frac{1}{\lambda^2}}_{\text{but summed in different order}} \right].$$

Proposition 24.1.

$$E_2(-1/\tau) = \tau^2 E_2(\tau) + \frac{12\tau}{2\pi i}.$$

Remark 24.2. This relation can be used to prove that $\eta(\tau) = q^{1/24} \prod_{n=1}^{\infty} (1 - q^n)$ is modular of weight 1/2. ($\eta(-1/\tau) = (\tau/i)^{1/2} \eta(\tau)$.) Idea: consider $\frac{\eta'(\tau)}{\eta(\tau)}$ and relate it to $E_2(\tau)$.

$$\bigoplus_{k=0}^{\infty} \mathcal{M}_{2k}^{\text{hol}} \cong \mathbb{C}[G_4, G_6].$$

Recall we were working with an even self-dual lattice Q of rank (say) 24. [This case is actually simpler than ranks 8 or 16 since we don't get fractional powers of q .] We had

$$\begin{aligned} \theta_Q(\tau, z) &= \sum_{\alpha \in Q} q^{\frac{(\alpha, \alpha)}{2}} e^{2\pi i(\alpha, z)}. \\ \theta_Q\left(-\frac{1}{\tau}, \frac{z}{\tau}\right) &= e^{\frac{\pi i(z, z)}{\tau}} (\tau/i)^{n/2} \theta_Q(\tau, z) \end{aligned}$$

Let's consider

$$P_Q(\tau, z) = e^{\frac{-2\pi i(z, z)}{24} E_2(\tau)} \theta_Q(\tau, z).$$

We can check that the non-modularity of $E_2(\tau)$ "cancels" the $e^{\frac{\pi i(z, z)}{\tau}}$ factors, and we get

$$P_Q\left(-\frac{1}{\tau}, \frac{z}{\tau}\right) = (\tau/i)^{n/2} P_Q(\tau, z).$$

Proof.

$$\begin{aligned} e^{-2\pi i \frac{(\frac{z}{\tau}, \frac{z}{\tau})}{24}} E_2\left(-\frac{1}{\tau}\right) &= e^{2\pi i \frac{(z, z)}{24\tau^2} (\tau^2 E_2(\tau) + \frac{12\tau}{2\pi i})} \\ &= e^{-2\pi i \frac{(z, z)}{24} E_2(\tau) - \frac{(z, z)}{2\tau}}, \end{aligned}$$

which is close enough. $\square$

Let's Taylor series expand P_Q (in $\mathbb{Z}$).

$$\begin{aligned} \theta_Q(\tau, z) &= \sum_{\alpha \in Q} q^{\frac{(\alpha, \alpha)}{2}} \left(1 + 2\pi i(\alpha, z) + \frac{(2\pi i)^2}{2} (\alpha, z)^2 + \frac{(2\pi i)^2}{6} (\alpha, z)^3 + \dots \right) \\ &= \sum_{\alpha \in Q} q^{\frac{(\alpha, \alpha)}{2}} + 2\pi i \underbrace{\sum_{\alpha \in Q} q^{\frac{(\alpha, \alpha)}{2}} (\alpha, z)}_{=0 \text{ since } \alpha, -\alpha \text{ cancel } \forall \alpha \in Q} + \frac{(2\pi i)^2}{2} \sum_{\alpha \in Q} q^{\frac{(\alpha, \alpha)}{2}} (\alpha, z)^2 + \dots \end{aligned}$$

Meanwhile,

$$e^{-2\pi i \frac{(z, z)}{24} E_2(\tau)} = 1 - \frac{2\pi i}{24} (z, z) E_2(\tau) + \frac{(2\pi i)^2}{24^2} (z, z)^2 E_2(\tau)^2 + \dots$$

Multiply to get

$$P_Q(\tau, z) = \theta_Q(\tau) + \left(\theta_Q(\tau) \frac{-2\pi i}{24} (z, z) E_2(\tau) + \frac{(2\pi i)^2}{2} \sum_{\alpha \in Q} q^{\frac{(\alpha, \alpha)}{2}} (\alpha, z)^2 \right) + \dots$$

Expanding further the z^2 term:

$$-\frac{2\pi i}{24} \underbrace{(1 - 24q + \dots)}_{E_2(\tau)} \sum_{\alpha \in Q} q^{\frac{(\alpha, \alpha)}{2}}(z, z) + \frac{(2\pi i)^2}{2} \sum q^{\frac{(\alpha, \alpha)}{2}}(\alpha, z)^2$$

write the q -series expansion: gives $(2\pi i)^2$ times

$$\begin{aligned} & -\frac{(z, z)}{24} \left(1 - 24q + \sum_{(\alpha, \alpha)=2} q + O(q^2) \right) \\ (24.2.1) \quad & + \frac{1}{2} \left(\sum_{(\alpha, \alpha)=2} q(\alpha, z)^2 + O(q^2) \right) \\ & = -\frac{(z, z)}{24} + q \left(\frac{1}{2} \sum_{(\alpha, \alpha)=2} (\alpha, z)^2 - \frac{(z, z)}{24}(N - 24) \right) + O(q^2) \end{aligned}$$

where $N = \#\{\alpha | (\alpha, \alpha) = 2\}$.

What good is this? Since $P_Q\left(-\frac{1}{\tau}, \frac{z}{\tau}\right) = \tau^{12} P_Q(\tau, z)$ (for $n = 24$). We have $(S$ and T -invariance, which generate the whole modular group)

$$P_Q\left(\frac{a\tau + b}{c\tau + d}, \frac{z}{c\tau + d}\right) = (c\tau + d)^{12} P_Q(\tau, z).$$

$$P_Q(\tau, z) = \sum_{i=0}^{\infty} P_Q^i(\tau) z^i,$$

one finds

$$P_Q^i\left(\frac{a\tau + b}{c\tau + d}\right) = (c\tau + d)^{12+i} P_Q^i(\tau).$$

In particular the quadratic (in z) term lies in $\mathcal{M}_{14}^{\text{hol}}$, which is 1-dimensional, spanned by $E_{14}(\tau) = 1 - 24q + \dots$

For us, this means that $\text{coef}(q) = -24\text{coef}(1)$ in the last equality of Equation ?? . I.e.,

$$(24.2.2) \quad \frac{1}{2} \sum_{(\alpha, \alpha)=2} (\alpha, z)^2 - \frac{(z, z)}{24}(N - 24) = (z, z).$$

$$\frac{1}{2} \sum_{(\alpha, \alpha)=2} (\alpha, z)^2 = (z, z) \frac{N}{24}.$$

Recall that

$$\Delta = \{\alpha \in Q | (\alpha, \alpha) = 2\}$$

is a simply laced root system. Δ is either A_n , D_n , E_6 , E_7 or E_8 (or an orthogonal sum of such components). A simple root system Δ has an associated invariant called the dual Coxeter number $h^\vee$. By definition, if $r = r_\alpha$ is the reflection in $\alpha \in \Delta$, then $(r_\alpha x, r_\alpha y) = (x, y)$ for all $x, y \in \mathbb{R}^n$ ambient space. In particular, $(\cdot, \cdot)$ is invariant for the action of the Weyl group $W = \langle r_\alpha | \alpha \in \Delta \rangle$, or rather, $W \subset O(\mathbb{R}^n)$. The following lemma claims that this form is unique.

Lemma 24.3. *If $\langle \cdot, \cdot \rangle : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ invariant under W , then $\langle \cdot, \cdot \rangle = \alpha(\cdot, \cdot)$ for some scalar α .*

OK, so consider

$$\langle x, y \rangle = \frac{1}{2} \sum_{\alpha \in \Delta} (\alpha, x)(\alpha, y).$$

This is W -invariant, and so is some multiple of $(\cdot, \cdot)$. The definition of $h^\vee$ is that

$$\langle x, y \rangle = h^\vee(x, y).$$

In particular,

$$\frac{1}{2} \sum_{\alpha \in \Delta} (\alpha, z)^2 = h^\vee(z, z),$$

which appears in Equation ??.

Example 24.4. For A_n , $h^\vee = n + 1$, for E_7 , 18, for D_n , $2n - 2$.

So, suppose

$$Q \supset \Delta = \{\alpha | (\alpha, \alpha) = 2\} = \Delta_1 \cup \Delta_2 \cup \dots \cup \Delta_k$$

where Δ_i is a simple root system.

Let $\beta \in \Delta_i$, our formula implies

$$\begin{aligned} h_i^\vee(\beta, \beta) &= (\beta, \beta) \frac{N}{24} \\ \implies h_i^\vee &= \frac{N}{24}. \end{aligned}$$

that is,

$$N = \sum_{i=1}^k |\Delta_i|,$$

and

$$h_i^\vee = \frac{N}{24}$$

for each component Δ_i .

Example 24.5. If $\Delta = \Delta_i$ is simple and type A_n , say A_{24} . We have $h^\vee = 25$,

$$\begin{aligned} \dim \mathfrak{sl}_{25} &= 25^2 - 1 = (25 + 1)(25 - 1) \\ N &= 25^2 - 1 - 24 = (24)(26) - 24 = 25(24). \end{aligned}$$

Remark 24.6. • The empty set is vacuously a solution.

- E_8^3 : $|\Delta_{E_8}| = 240$, $h^\vee = 30$, $h^\vee = \frac{3 \times 240}{24} = 30$, is a solution.
- A_2^{24} , $h^\vee = 2$ for all components, $\frac{|\Delta|}{24} = \frac{24(2)}{24} = 2$.

Consider $\mathfrak{g}$ a semisimple Lie algebra with root system Δ , $\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \dots \oplus \mathfrak{g}_k$, $\Delta(\mathfrak{g}_i) = \Delta_i$.

Note that if $\Delta \neq \emptyset$, then $\text{rank}(\Delta) = 24$. So we can rewrite $\#\Delta$ as $\dim \mathfrak{g} - 24$. so

$$(24.6.1) \quad h_i^\vee = \frac{\dim(\mathfrak{g})}{24} - 1.$$

There exist at most “partitions of 24 into 3 colours A, D and E” solutions to Equation ??.

Observe: ℓ : rank, d : dimension, $h^\vee$: $h^\vee$. For ADE, $h^\vee = O(\ell)$, $d = O(\ell^2)$. Use this to bound $\dim(\mathfrak{g})$.

Turns out there are 23 solutions (24 if we include the empty root system) A_{24} , E_8^3 , A_1^{24} , A_2^{12} , etc.

Theorem 24.7. *Each nonempty solution Δ is the root system of exactly one self-dual even lattice.*

Proof. Construct Q_Δ by orbifolding from E_8^3 . Compute $|\text{Aut}(Q_\Delta)|$,

$$RHS(\text{mass formula}) - \sum_{\Delta \neq \emptyset} \frac{1}{|\text{Aut}(Q_\Delta)|} = \frac{1}{10^{18}}.$$

□

The subalgebra $\mathfrak{g}_i \subset \mathfrak{g}$ induces a homomorphism of vertex algebras

$$L_1(\mathfrak{g}_i) \rightarrow V_Q.$$

Note that V_Q has one irreducible module, namely V_Q . Let $V = V_0 \oplus V_1 \oplus V_2 \oplus \dots$ be a vertex algebra with this parity,

$$V_0 = \mathbb{C}|0\rangle$$

$$V_1 = \mathfrak{g} \text{ some Lie algebra}$$

It turns out that $\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \dots \oplus \mathfrak{g}_k$ induces a set of embeddings

$$L_{k_i}(\mathfrak{g}_i) \rightarrow V, \quad k_i \in \mathbb{Z}_{\geq 1}.$$

One can prove that

$$\frac{h_i^\vee}{k_i} = \frac{\dim \mathfrak{g}}{24} - 1$$

now. If you allow $k_i \geq 2$, there are no longer 23 solutions but 221 solutions. 71 vertex algebra. This is called “Schellekens list”.

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